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Data Communication and Computer Networks

Block-1 Unit-1

Introduction to Internet

Topics to be Covered

- Introduction
- What is Internet?
  - ISP and Internet Backbone
  - Interconnect of ISPs
    - Taxonomy of network
    - Standard Internet Protocols
    - Public Network and Private network(Intranet)
    - Accessing the Internet
    - Internet Services

Topics to be Covered

- Network Topologies
- Important Topics
- Summary

Introduction

- In the era of computers, a worldwide network of computers allows people to share information electronically.
- Like a BIG book with many web pages on different topics.
- It can be accessed anywhere with an Internet connection.
- Every organization, small or big, doing business from a single site or multiple sites.
- Uses computers to handle various transactions but also communicate information among various sites located.
- So we can say, how information is exchanged among computers separated apart is known as the computer Internet.

Introduction

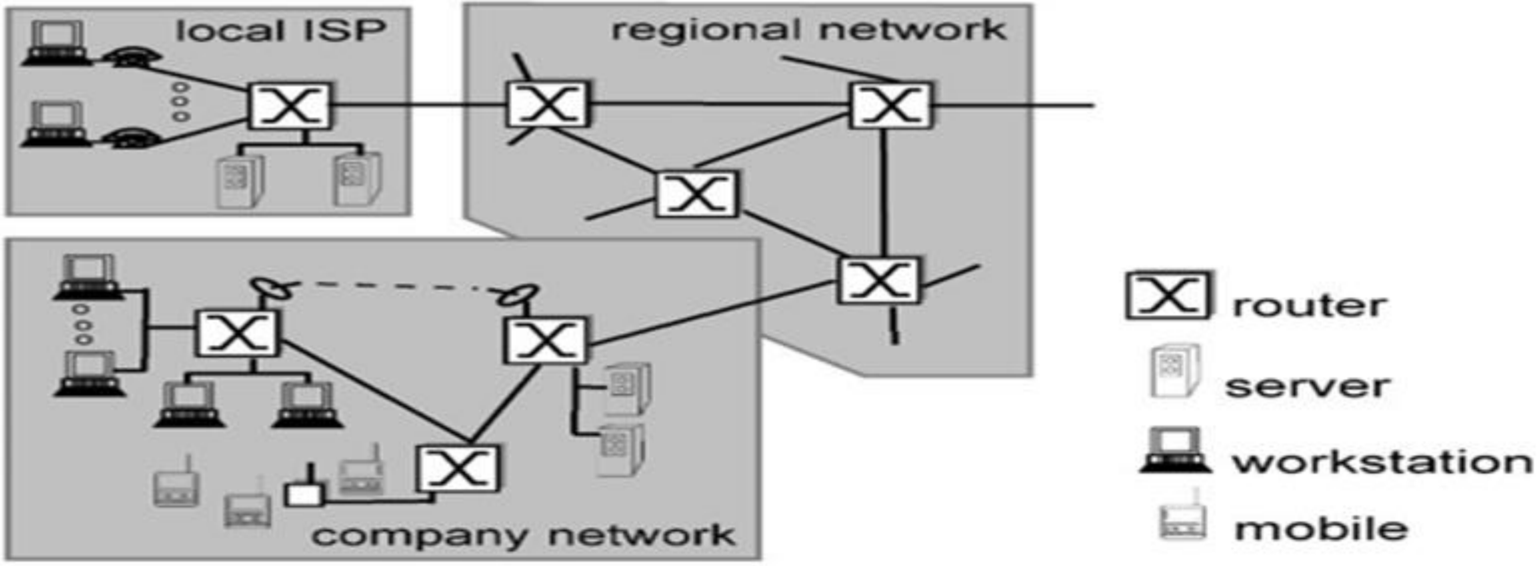
- A network is; an entity connected to exchange some information.
- When entities are replaced by computing devices, it is the computer network.
- Digital information is made available in just one click from one end of the world to another end within a fraction of a second.
- The productivity of every organization irrespective of their field has been increased enormously by sharing information in digital form through computer networks.

depositphotos.com/stock-photos/internet.html

What is Internet?

- The Internet is the biggest computer network with a worldwide scope. It is the network of networks.
- The Internet comprises lakhs of organizational networks of types: domestic or international, academic or business, and government, which altogether exchange various kinds of data/information.
- ARPANET, in full Advanced Research Projects Agency Network, an experimental computer Network that was the forerunner of the Internet.
- The ARPA, an arm of the U.S. Defense Department, funded the development of the ARPANET in the late 1960s.
- Bob Taylor is known as the father of the Internet, initiated the project in 1966 so that it can access remote computers.

What is Internet?



Overview of the Internet

ISP and Internet Backbone

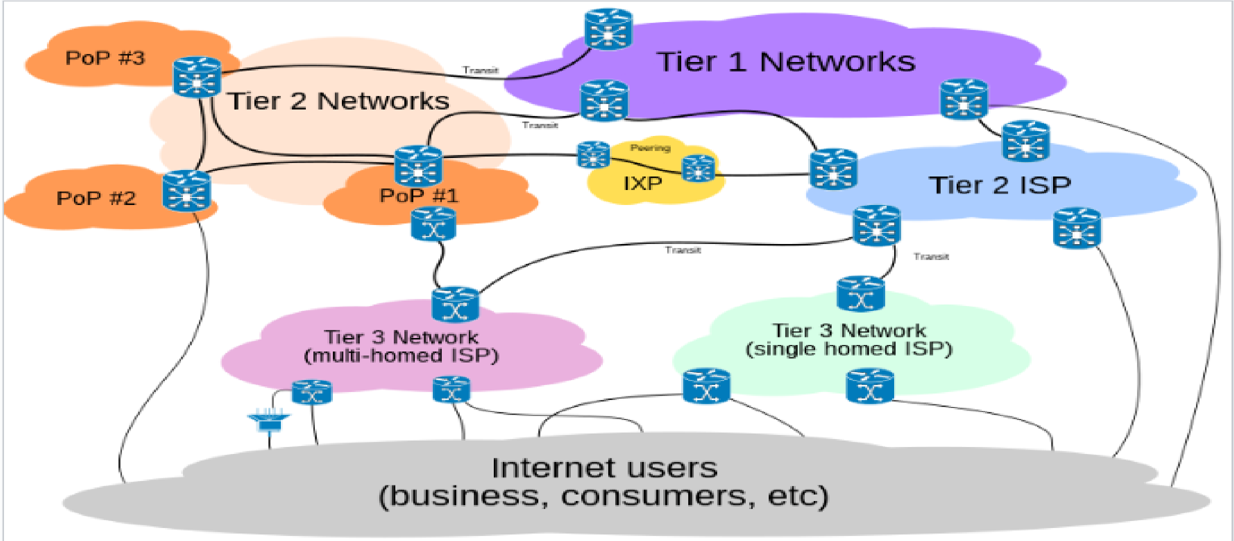
- ISPs (Internet Service Provides) are responsible for transporting traffic on the Internet.
- Internet is very vast in terms of traffic, number of end systems, number of intermediate devices, interconnecting links, and distribution of these over geographical locations.
- Hence a huge number of ISPs manage the Internet.
- To efficiently manage and control the services of the Internet, these ISPs are classified into three categories namely:
- Tier -1 ISPs, Tier-2 ISPs, and Tier-3 ISPs, and this hierarchical model is known as the 3-tier model.

Interconnection of ISPs

- End systems or hosts situated at the edge of the Internet are connected to the rest of the Internet through this 3-tier model of ISPs.
- The network through which end systems connect to the Internet is called the access network (i.e. local LAN or private network).
- Access network connects to the Internet via tier -3 ISPs also known as access ISPs and are at the bottom of 3 –the tier model.
- Access ISPs are the largest number of ISPs on the Internet. Tier-1 ISPs are relatively fewer in numbers and are at the top of the hierarchy.
- All three tiers of ISPs are the same in many ways as any network they all have connecting links and intermediate devices i.e. routers and are further connected to other networks/ISPs.

ISP - Tier -1 Model

- Tier -1 ISPs are also known as Internet backbone networks.
- Tier -1 ISPs are connected to a large number of tier-2 ISPs.

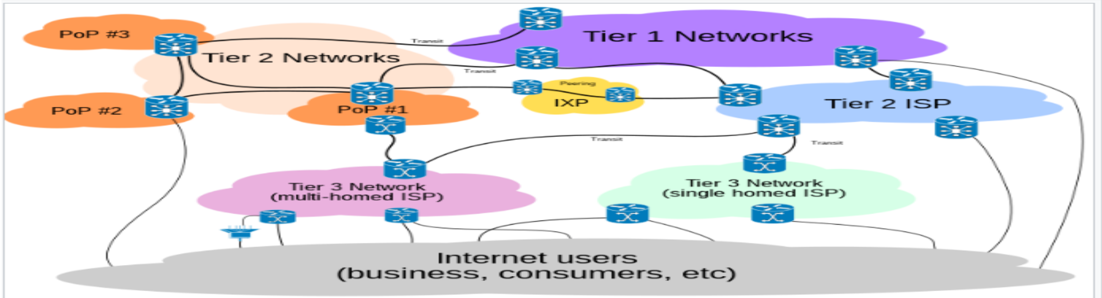


Relationship between the various tiers of Internet providers

en.wikipedia.org/wiki/Tier_1_network

ISP - Tier -2 Model

- Tier-2 ISPs have regional or national coverage and are connected to a few of the tier-1 ISPs.
- Thus, tier-2 ISPs need to route traffic through one of the tier-1 ISPs to reach a large portion of the global Internet.
- A tier-2 ISP is a customer of the tier-1 ISPs it is connected with, and their -1 ISPs are the service providers for their connected customers.



Relationship between the various tiers of Internet providers

Source : Wikipedia.org

Taxonomy of Network

- Computer networks are generally classified according to their structure, physical scales, transmission technology, etc. According to the network's physical area, it can be classified as:

Personal Area Network (PAN): PAN is for a person.

- It is a network established for a person for a small area of the range of a few meters.
- It can be wired (using a USB data cable) or wireless (using infrared, ZigBee, or Bluetooth).
- As an example, WPAN are: controlling the TV with a remote using infrared, mobile headsets with mobile using Bluetooth, wireless keyboards and mice, and wireless printers.

Taxonomy of Network

- **Local Area Network (LAN):** The physical scope of a Local area network is limited to a small geographical area, such as a single building or a campus.
- Within an organization or institution personal computers are connected among themselves and establish a local area network to share resources i.e. printers, scanners, etc., and to share information.
- **Metropolitan Area Network (MAN)**: The coverage of a MAN is spread over a city and is limited to 100KMs range.
- MAN can be wired or wireless. TV cable networks within a city and broadband networks (wired as well as wireless) are examples of MAN.

Taxonomy of Network

- **Wide Area Network (WAN):** The coverage scope of a WAN is spread over a large geographical area and covers many cities, states, and even countries.
- A WAN includes millions of computers connected and located over a large geographical scale. e.g. Internet is the largest Wide Area Network ever established.
- There are three categories of wide area networks as follows:
- **Enterprise network:** A network owned and managed by a single organization is named an enterprise network. It is the interconnected version of all the LANs of an Organization/Institution.
- **Global network:** Networks of several organizations over a wide area collectively known as a global network.

Taxonomy of Network

- Internet: A network of networks of broad area categories. It is the largest network worldwide.
- It includes many LANs, WANs, MANs.
- No single authority controls it; rather the local, national authority controls every segment of the Internet.

Networks by Scale

http://networking.layer-x.com/p050000-1.html

Taxonomy of Network

Architecture:

Based on the architecture, network can be classified as:

- Peer-to-peer network or
- Client - server architecture.

Transmission Technology:

According to the transmission technology networks can be classified as:

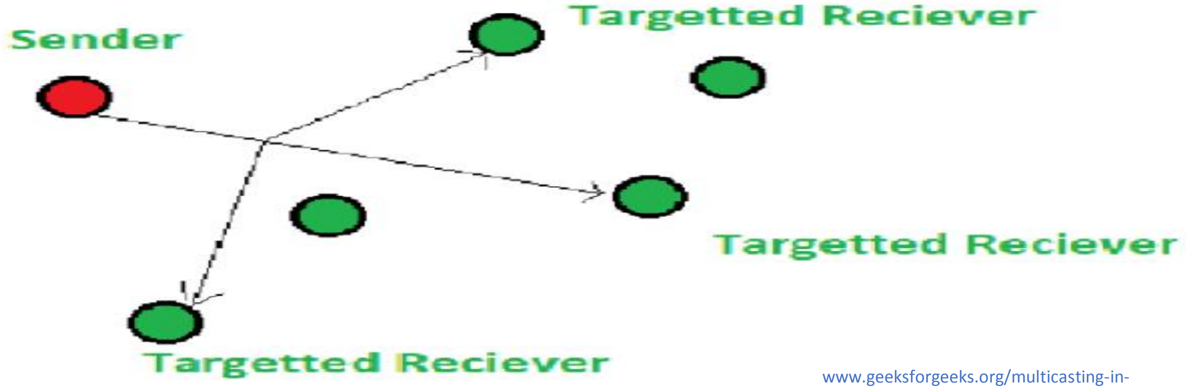
- Broadcast networks.
- Point-to-point or Switched networks.

Taxonomy of Network

- In a broadcast network a single communication channel is used which is shared by all the hosts connected to the network.
- In a broadcast network, a message sent by any member reaches all the members of the network.
- This mode of transmission is known as broadcasting. The sender sends the message to all the computers connected to the network, the receivers capable of and are interested to receive the message may get it.
- To send a packet to all the nodes (not known) broadcast address is used as the destination address in the packet and it is called ‘Broadcasting’.
- To send a packet to a group of nodes (all group members are known), ‘Multicasting’ is used.

Taxonomy of Network

- One possible way to achieve Multicasting is to reserve one bit to indicate multicasting and the remaining (n-1) address bits contain group number.
- Every machine can subscribe to any or all of the groups.
- Some of the common broadcast networks are the packet radio networks and satellite networks etc.



www.geeksforgeeks.org/multicasting-in-computer-network/

Standard Internet Protocols

- A protocol allows computers to talk with each other.
- A protocol defines the data type that may be exchanged, the set of commands used to exchange (send/ receive) data, and the way data is transferred are how it is confirmed that data is received.
- In the Internet various protocols are designed for networking applications.
- Like Ethernet protocol is used in wired networks, IEEE 802.11 protocol is defined for wireless communication.
- An Internet protocol suite is a set of standard protocols used for transmitting data over the Internet.

Categories of Standard Internet Protocols

- Standard Internet protocols may be classified into four major categories:
- **Application layer** – Hyper Text Transfer Protocol (HTTP), Simple Mail Transfer Protocol (SMTP).
- **Transport layer** – Transmission Control Protocol (TCP), User Datagram Protocol (UDP)
- **Network layer** – Internet Protocol (IP) version 4 and version 6
- **Data link layer** – Ethernet, IEEE 802.11 (Wi-Fi).

Public Network & Private Network (Intranet)

- A public network is an open network. Internet is the best example of a public network.
- Though both the public and networks there is no technical difference between a private and public network in terms of hardware and infrastructure, except for the security, addressing, and authentication systems in place.
- There are very few restrictions or access control rules are applicable to get connected and access resources of the network.
- In public networks least security measures are applicable therefore, it is the least secure network and one has to be very careful concerning the security aspects while connecting to the public.

Accessing the Internet

- Access to the Internet can be availed using either of the three methods namely: telephone network, cable network, and wireless network.
- **Telephone Network:** The PSTN (public switched telephone network) is the telephone network used for voice communication and spread throughout the world.
- A packet-switched network (PSN) is a kind of computer communications network that sends data in the form of small packets.
- Hence the telephone network system is an inseparable/integral part of the computer data transmission.

Dial-Up Lines

- Setting up data connection over existing analog telephone lines requires installing a modem on both sides (sender side and receiver side).
- Establishing a data connection over the telephone line using a dial-up setup is as similar as establishing a voice call over the same.
- Before sending the computer data, a connection is required to be established which is temporary between the sender and the receiver.
- To establish the connection modem on the sender side dials the telephone number of the receiver side’s modem.
- Once the connection is established data transmission can takes place.

Dial-Up Lines

- After the completion of the data transmission, either side of the communication can turn down the connection.
- During the communication no other connections can be established between end devices and even at a time either data or voice communication can be possible.
- The speed of the data transmission is very slow over the dial-up connection.
- The maximum speed that could be availed is 56kbps over a dial-up connection.

Dedicated Lines

- Connection established using dedicated lines does not require to dial the telephone number of the other end. A dedicated line establishes a separate connection for the data from the voice connection.
- It is always-on type of connection i.e. there is no need to reestablish a connection each time it is used. Dedicated lines provide a high quality connection for data transmission than a dial-up line.
- Dedicated lines are classified into two categories namely: analog lines and digital lines. In present scenario the digital lines are widely used to connect end users to Internet worldwide.
- The data transmission over digital dedicated lines is much faster than that over the analog lines.

Integrated Services Digital Network (ISDN)

- Data transmission over Integrated Services Digital Network (ISDN) is much faster as compared to the dial-up lines.
- ISDN also uses standard telephone lines for data transfer and was one of the popular technique for data communication.
- An ISDN modem is required to be installed at both the communicating ends.
- The end used should not be out of the range of 3.5 miles from ISDN modem of the telephone connection providing company.
- Thus, ISDN is suitable to urban areas only and is not an option for users located in rural areas far from the cities (where ISDN modems of telephone service providers are installed).

Cable Network

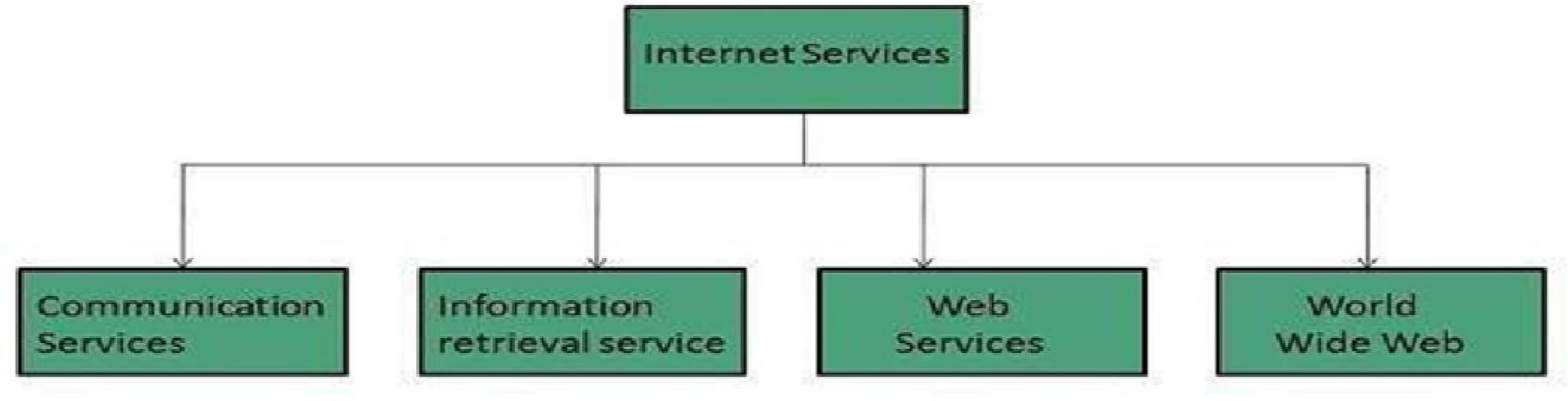
- In a cable network cables are used as a transmission medium in the network.
- Cable Internet uses the television cable infrastructure to provide Internet accessibility.
- This is also called broadband Internet access.
- Cable modem device is used for Internet access in the cable network.
- Coaxial cable is used in the cable network for connectivity. Cable Internet provides faster Internet speed of up to 1 Gbps than dial-up and DSL connections.
- Cable Internet access provides last-mile access. e.g the cable network of television infrastructure is used from Internet service provider to end user.
- In today’s scenario cable Internet access is suppressed by fiber optical networks, wireless networks, and mobile networks.

Wireless Network

- Wireless networks are much cheaper to install and maintain.
- Wireless network infrastructure does not require the setup of the wired network.
- A wireless modem/router/access point is required to establish a wireless network.
- Many wireless technologies are available to establish a wireless network.
- Bluetooth, Infrared, Wi-Fi, etc. are some of the wireless technologies commonly used to establish a wireless network.

Internet Services

- Using Internet services, one can access information available on the Internet.
- These Internet services can be classified into four major categories: Communication services, information retrieval services, web services, and



Internet Services

Communication Services

- Communication services are used to exchange information over the Internet.
- One of the commonly used communication services is Electronic mail.
- Electronic mail service is used to exchange messages in the form of electronic messages.
- Other services like Telnet – used to establish and exchanges messages between remote systems.
- Internet Relay Chat (IRC) – which is used to communicate in real-time.
- Internet Telephony (VoIP): the internet users are allowed to talk using Internet services by making a call.

Information Retrieval Services

- Information retrieval services are used to download or extract information from the Internet.
- One of the commonly used information retrieval services on the Internet is FTP (File Transfer Protocol): it enables Internet users to transfer files to or from the Internet.
- **Web Services**: Web applications use web services to interact with each other.
- **World Wide Web**: World wide web (WWW) provides a way to retrieve documents located on servers located in different geographical locations over the Internet.
- WWW documents may include audio, text, visuals, graphics, and links to other documents (hyperlinks).

Important Topics

- Computer Networks and its classification
- ISP
- Categories of Standard Internet protocols
- Taxonomy of Network
- Wireless
- Internet Services

Summary

- In this Session we discuss about Internet, Computer Networks, and its classification according to the Topologies.
- Then we discuss about ISDN.
- Taxonomy of network is discussed according to the area wise.
- Basically this session belongs to Client Server Architecture.

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Data Communication and Computer Networks

Block-1 Unit-2_Part1

Data Transmission Basics & Communication Transmission Media

Topics to be Covered

- Introduction
- Data Communication Terminology
    - Channel , Baud , Bandwidth , Frequency
- Modes of Data Transmission
    - Simplex, Half Duplex and Full Duplex Communication
    - Serial and Parallel Communication
    - Synchronous, Asynchronous and Isochronous Communication
- Analog and Digital Data Transmission
- Transmission Impairments

Topics to be Covered

- Attenuation, Delay Distortion, Noise, Signal To Noise Ratio, Concept Of Delays
- Transmission Media And Its Characteristics
  - Guided Media, Unguided Media
  - Wireless Transmission
  - Microwave Transmission, Radio Transmission, Infrared And Millimeter Wave
- Wireless Lan
- Important Topics
- Summary

Introduction

- Data communication not only involves the transfer of information but is successful only when the information is interpreted correctly by the receiver.
- For any communication to be successful, some rules has to be followed. A set of rules is called a protocol.
- As the communicating devices most of the time are heterogeneous i.e. may be different in hardware or software or operating system platform etc.
- So, certain rules have to be followed to bring them all on a common platform to make them capable of communicating and understand the information.

Data Communication Terminology

- Data communication is not just to transfer the data between sender and receiver but also the interpretation of the data should be the same and correct by both the sender and the receiver.
- Further, we will be looking after the terminologies commonly used in data communication here:
  - Channel
  - Baud
  - Bandwidth
  - Frequency

Data Communication Terminology

- **Channel:** A channel is the medium used to transmit the information between sender and receiver.
- In other words, the channel is the path through which the exchange of information between devices takes place. Channel is classified into two categories:
  - Wired and Wireless.
- In metal wired channel the electrons present within the metal carry the information. In fiber optics-based guided media concept of total internal reflection of light is used to transmit the information in a fiberglass cable.

Data Communication Terminology

- Wired channels are guided media i.e. as the electrons can move in the metal in a restricted way in a certain direction carrying the information.
- Whereas in wireless channels, the air particles are responsible for data transmission.
- Every channel has a channel capacity that limits its maximum data-carrying capacity over a certain period.
- Based on the type of information that could be propagated through the channel, there are two types of channels used in data communication namely: i.e. Analog and Digital.

Data Communication Terminology

- **Baud:** It is a unit of data transmission speed.
- The symbol rate or modulation rate of a signal is measured in bauds.
- Symbole rate is one of the elements considered to measure the data transmission speed of a data channel.
- In a second how many times a signal changes its state is known as the baud.
- For digital signals with two states, baud and bit per second are equal i.e.1 Bd = 1 bit/s.
- The symbol ‘Bd’ is used to represent the unit baud.

Data Communication Terminology

- **Bandwidth:** Bandwidth is the data carrying capacity of the channel/media per unit of time (It can be defined for the bus within a computer or the channel of a computer network).
- The bandwidth of a channel is the amount of data or signals transmitted in a time interval.
- Bandwidth of a channel depends on the length of the channel, the media considered, and the signaling method used. More is the bandwidth higher the throughput of the medium.

Data Communication Terminology

- **Bandwidth:**
- Bandwidth of the medium is the range of the frequencies used to transmit the data.
- The difference between the highest and the lowest frequencies the medium can carry.
- The bandwidth for digital devices is typically considered as the number of bits per second, expressed in Hertz (Hz) or the number of cycles per second for analog devices.

https://www.paessler.com/bandwidth-meter

Data Communication Terminology

- **Frequency:** Frequency is the rate of repetition of an event per second of duration. The unit of frequency is Hz which is the number of cycles per second.
- In data communication, the frequency of a signal is the number of cycles completed by the signal per unit time.
- Period and frequency are exactly opposite of each other.
- Period is defined as the minimum time in which a signal repeats itself and is denoted as seconds per cycle.
- Frequency is the number of periods a signal completes within a specified time duration and is denoted as cycles per second.

Modes of Data Transmission

- Data transmission mode means transmission of data from sender to receiver over a communication channel in the network.
- **Mode of transmission defines the direction of the flow of data.**
  - Simplex Mode
  - Half duplex mode
  - Full duplex mode

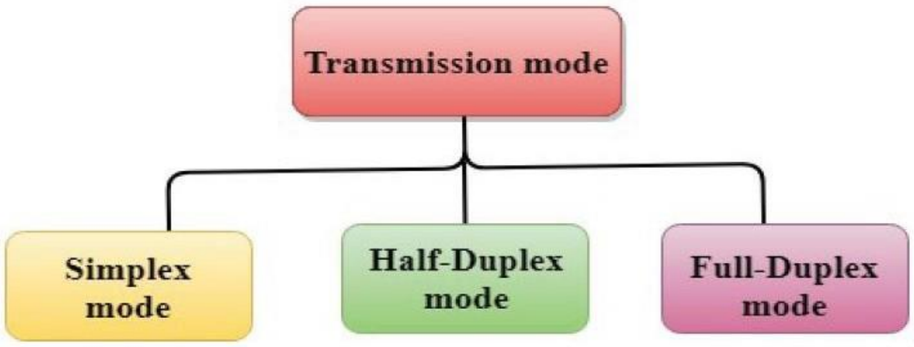


Figure: Data Transmission Modes

Data Transmission Modes

- In the Simplex mode of data transmission, the data through the channel can be transferred in one direction only.
- This mode of transmission is useful when one device always acts as the transmitter and another as the receiver

Half Duplex Mode

- In Half Duplex Mode of transmission the data can flow in both the direction but can be allowed in one direction at a time.
- At a time one of the communicating devices acts as a transmitter and another as receiver.

Figure: Half Duplex Mode

Full Duplex Modes

- In full duplex mode both the devices can act as transmitter and receiver simultaneously.
- Both the ends should be capable of transmitting and receiving the data simultaneously.

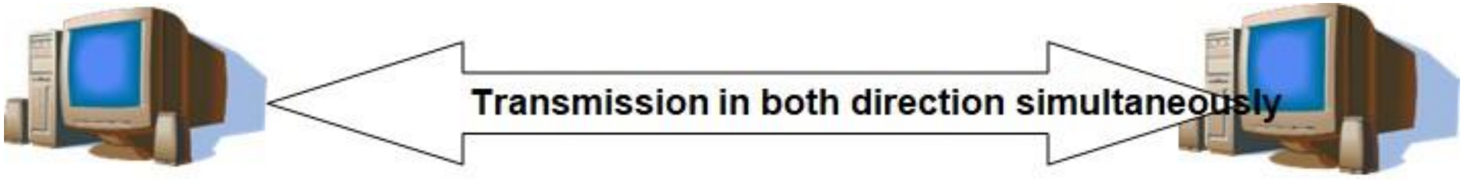


Figure: Full Duplex Mode

Serial And Parallel Communication

- Digital devices work on digital data in the form of binary bits. Data is transmitted among digital devices in the form of binary bits in two ways namely.
  - Serial transmission
  - Parallel transmission.

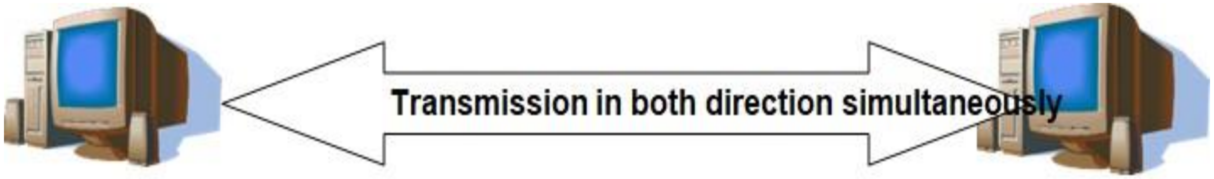
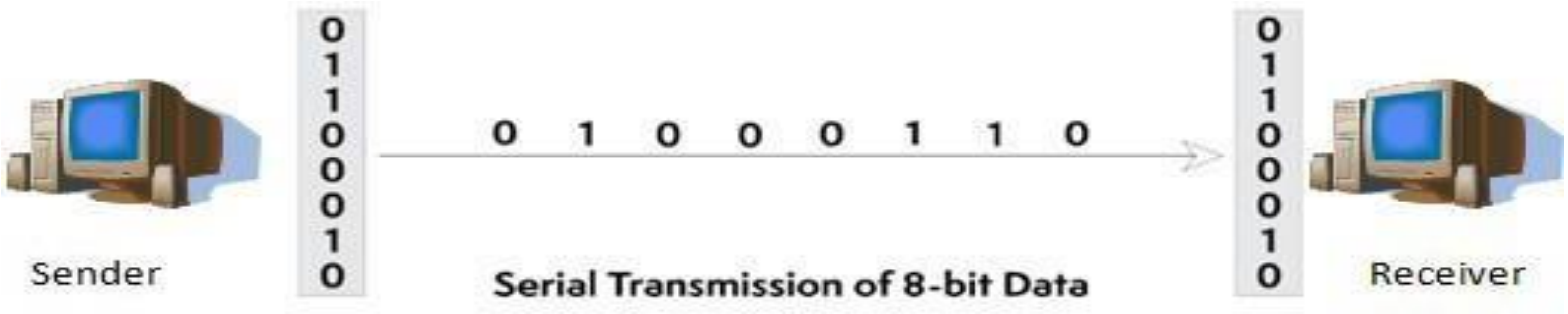


Figure: Full Duplex Mode

Serial Transmission

- In serial transmission bits are transmitted in a serial manner i.e. one after another through a channel.



Parallel Communication

- In parallel communication multiple bits are transmitted in parallel over multiple channels simultaneously.
- Data transmission through parallel data communication is much faster than serial communication methods.

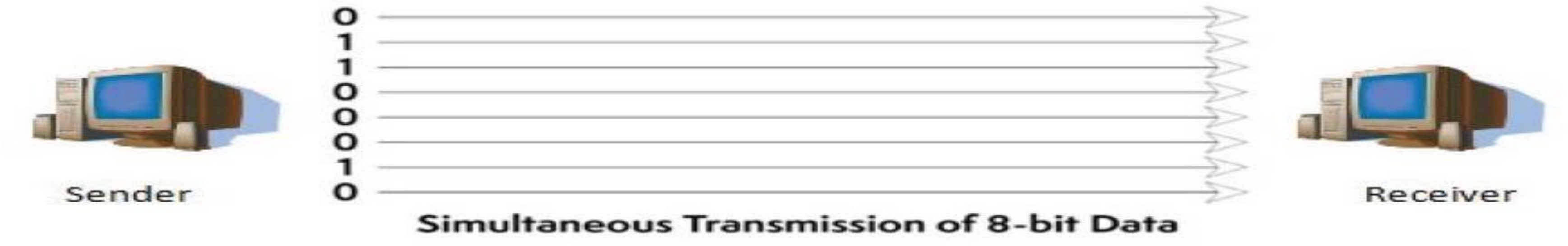


Figure: Parallel Data Transmission

Synchronous, Asynchronous and Isochronous Communication

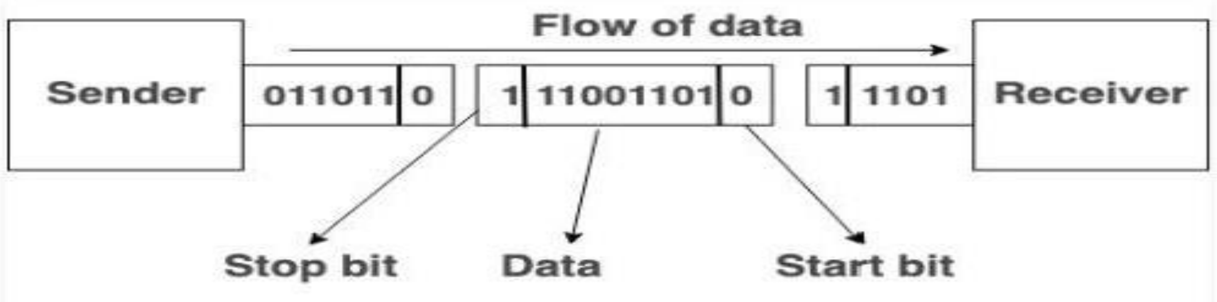
- In general serial data transmission technique is commonly used to transfer data on the Internet over long distances.
- The biggest issue of serial communication is the synchronization of the transmission speed of the sender and the receiving speed of the receiver.
- If the transmission speed and receiving speed of the sender and receiver mismatch, the receiver will not be able to interpret the received bit stream correctly.

Synchronous Communication

- In this communication, an external agent (a clock) regulates the data transmission.
- In the synchronous way of data communication, data is transmitted in the form of frames or chunks instead of characters.
- A special signal named: the synch signal is transmitted before sending the actual data notifying the receiver about the transmission of a new frame to get synchronized with the sender Sync signal.
- In general, is a unique bit pattern such that it cannot be a part of user data to avoid confusion between synch signal and user data.

Asynchronous Communication

- In asynchronous transmission, the size of frames should be kept as small as possible.
- It is commonly used to transmit characters generated at irregular intervals.
- When the user inputs characters with the keyboard to the computer is a typical example of asynchronous communication.
- As the size of the frame is small so, the parity bit is used to handle error control.



Frame Structure

- A frame is structured with:
- Start bit: it enables the receiver to synchronize itself with the frame (message).
- Data Bits: these are the actual user data to be transmitted.
- Parity Bits: it is optional to use and is used to handle errors encountered during transmission.
  Parity bit is capable of detecting single-bit error only.
- Stop bit: it enables the receiver to detect the need for a frame.

Synchronous vs. Asynchronous Transmission

Differentiating synchronous communication from asynchronous communication:

S.NO	Synchronous Transmission	Asynchronous Transmission
1.	Data is transmitted as frames or blocks.	Data is transmitted as characters or bytes.
2.	Transmission rate is faster.	Transmission rate is slower.
3.	It is expensive.	It is economical.
4.	The time interval between blocks is constant.	time interval is random.
5.	It is regulated by an external clock signal also shared with the receiver.	No need of synchronized clocks, as start and end bits are used.

Isochronous Communication

- Isochronous communication is the hybrid version of both synchronous and asynchronous communication.
- It inherits the start and stop bit from asynchronous communication i.e. start and stop bits are added to each character before transmission.
- Characters are transmitted at fixed intervals i.e. between two characters an exact multiple of one character time interval is added instead of a random time interval.
- As an example Let ‘t’ is the transmission time of a character frame with start bit, stop bits and parity bit, then the interval between any two character frames could be ‘n’, where ‘n’ belongs to positive integer.

Design Factors : for Transmission Media

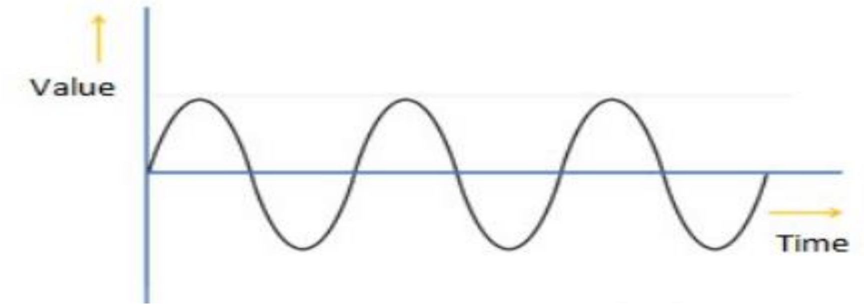
- **Bandwidth:** It is a measurement indicating the maximum capacity of a wired or wireless communications link to transmit data over a network connection.
- The greater the bandwidth of a signal, the higher the data rate that can be achieved, All other factors remaining constant.
- **Transmission impairments**. It limits the distance a signal can travel.
- **Interference:** When the wireless communication signals are interrupted or weakened by the presence of other wireless signals, can distort or wipe out a signal.
- **Number of receivers:** Each attachment introduces some attenuation and distortion, limiting distance and or data rate.

Analog and Digital Data Transmission

- Data can be transferred in the form of an analog signal or digital signal.
- In data communication, the data is represented in the form of electromagnetic signals like micro ware, radio wave, infrared for the wireless, electrical voltage for metallic wires, and light for fiber optical cable, radio wave, microwave, or infrared signal.
- Continuous data (like voice, data, image, signal, or video) is transmitted through analog signals and called analog transmission, and discrete data is transmitted through digital signals.

Analog Data Transmission

- Analog signals are generally represented in the form of sine wave passage way.
- Analog signals are continuous i.e. either amplitude or frequency are continuous and capable to represent continuous data.
- Video and audio data are transmitted as analog signals.
- The signal value is continuous function of time as shown in figure in PPT



Analog Data Transmission

- Over the distance, signal strength decreases so, to regain the signal strength amplifiers are installed along the transmission line.
- One of the major issue with analog signals is the accumulation of error during the journey.
- Signal is distorted during the journey due to external factors and amplifiers are dumb devices which simply increases the strength of input signal carrying noise also.
- So, along with the data signal, noise level is also regained which is very difficult to separate from the original signal.

Digital Data Transmission

- In contrast to the analog signal being a continuously variable waveform, the digital signal is a series of discrete pulses.
- Digital signals are used to transmit discontinuous data or events. All computing devices are based on digital data.
- Computers generate data in the form of digital signals.
- Digital signals are represented in the form of square waves.
- Digital signals also lose their signal strength or power over the distance.
- Repeater devices are used to regain the energy level of the signal.

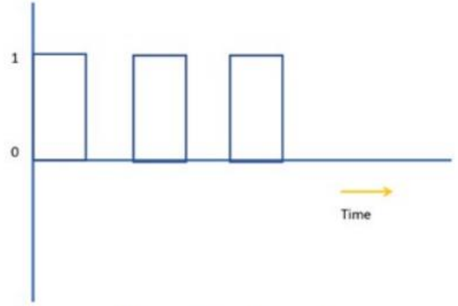
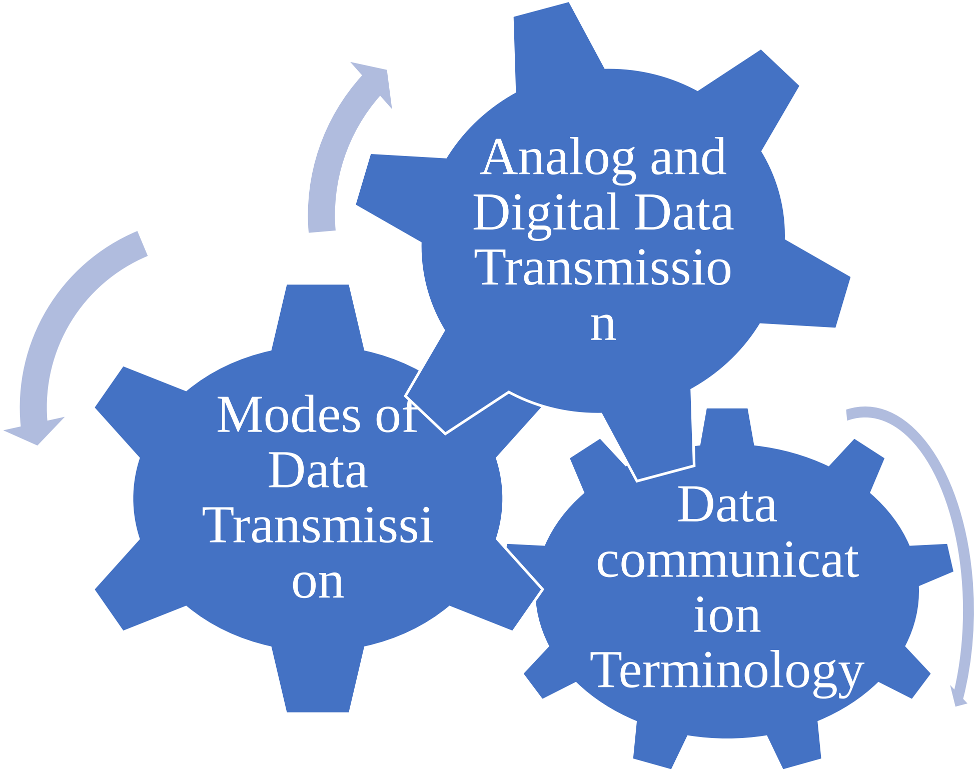


Figure: Digital Signal

Comparison of Analog and Digital signals

Analog	Digital
It is a continuous signal representing real time measurements.	It is discrete or time separated signalgenerated by digital modulation.
Sine wave is used to denote the analog signal.	Digital signals represented by square waves
Analog signal can represent continuous range of values.	Discrete values can be represented with digital signal.
Data produced by light sensors, FM radio, temperature sensors, etc are analog in nature and could be represented as analog signals.	Compact Drives, Digital video Disks and Computers etc. digital signal.
Bandwidth of analog signal is low	Bandwidth of digital signal is high.

Important Topics



Summary

- In this session, we have discussed about the data communication technologies.
- Starting from what is data communication, we have studied about the basic terminologies used in data communication i.e. channel, baud, bit rate, bandwidth, and frequency etc.
- Channel is the medium or the path carrying the data signals from source to destination.
- Baud is the number of symbols transmitted per unit time and the bit rate can be defined as [baud rate x bits in a baud].
- The frequency is measured in Hz and is the number of cycles completed by a signal in one second.

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Data Communication and Computer Networks

Block-1 Unit-3

Data Encoding and Multiplexing

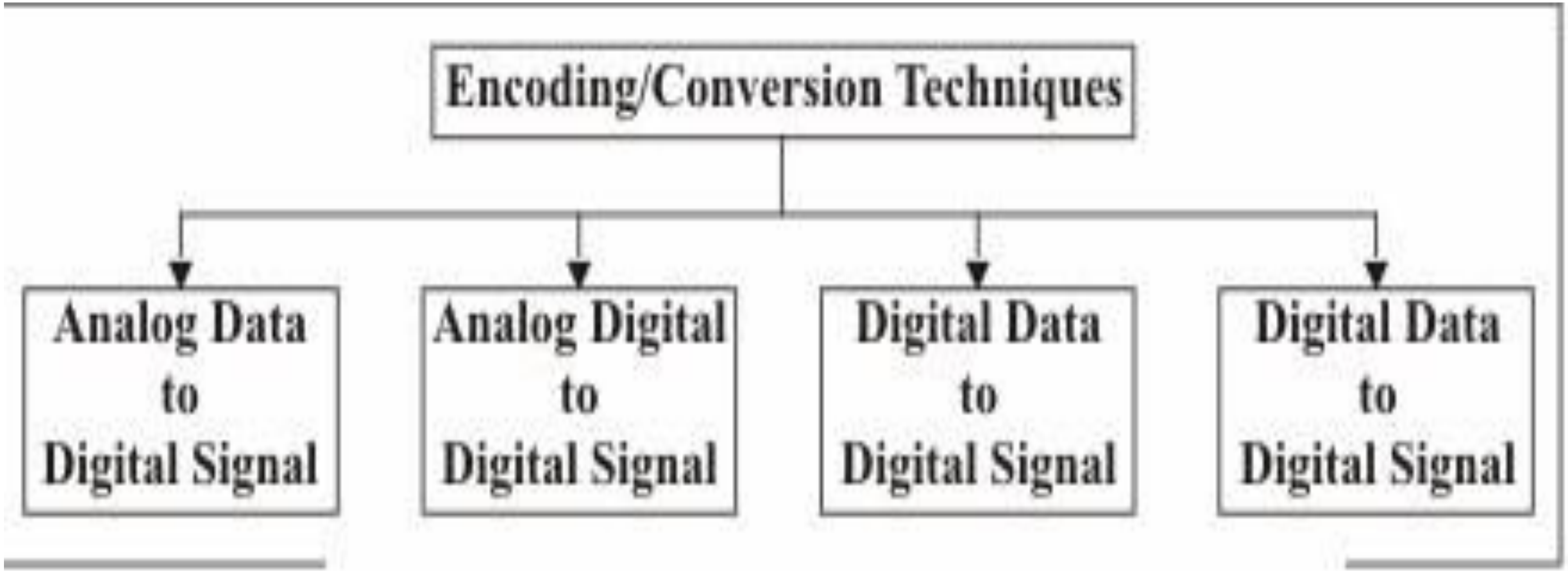
Topics to be Covered

- Introduction
- Encoding
- Analog to Analog Modulation
- Analog to Digital Modulation
- Digital to Analog Modulation
- Digital to Digital Encoding

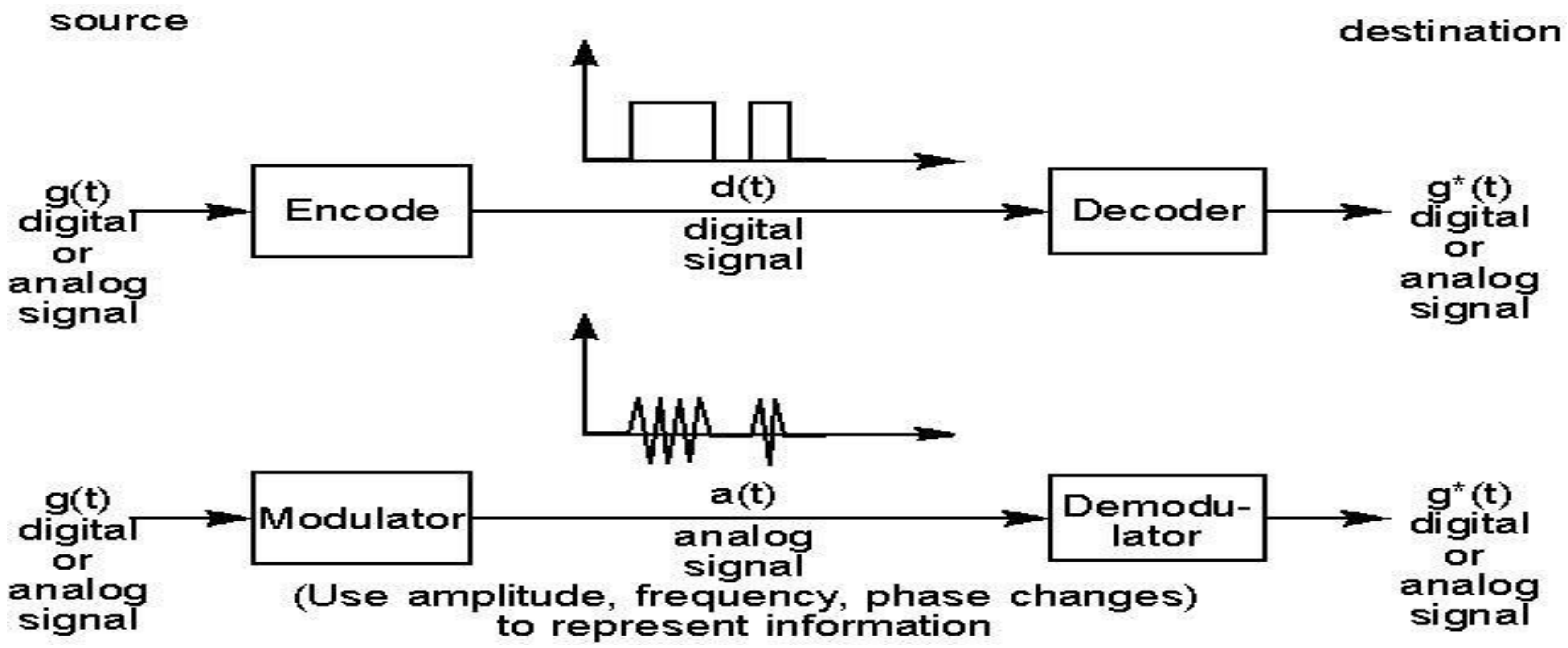
Introduction

- As we are aware about data is transmitted through a channel (wired or wireless) in the form of an analog or digital signal
- In this session, we will elaborate on the techniques to produce signals from analog or digital data.
- Usually, we cannot send a signal containing information directly over a transmission medium, at least not for long distances.
- We have to be able to transform our voice, to an electrical signal that can then, be sent that long distance.
- Then, we would also need to perform the reverse conversion at the other end.

Encoding Conversion Techniques



Encoding / Decoding



Analog to Analog Transmission

- The act of changing or encoding the information in the signal is known as modulation. An example of such encoding is radio broadcasting.
- When a low-frequency information signal is encoded over a higher-frequency signal, it is called modulation.
- Analog modulation, is the representation of analog information by an analog signal.
- Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us. An example is radio.

Ways of Encoding

- There are three different ways in which this encoding of the analog signal with analog information is performed.

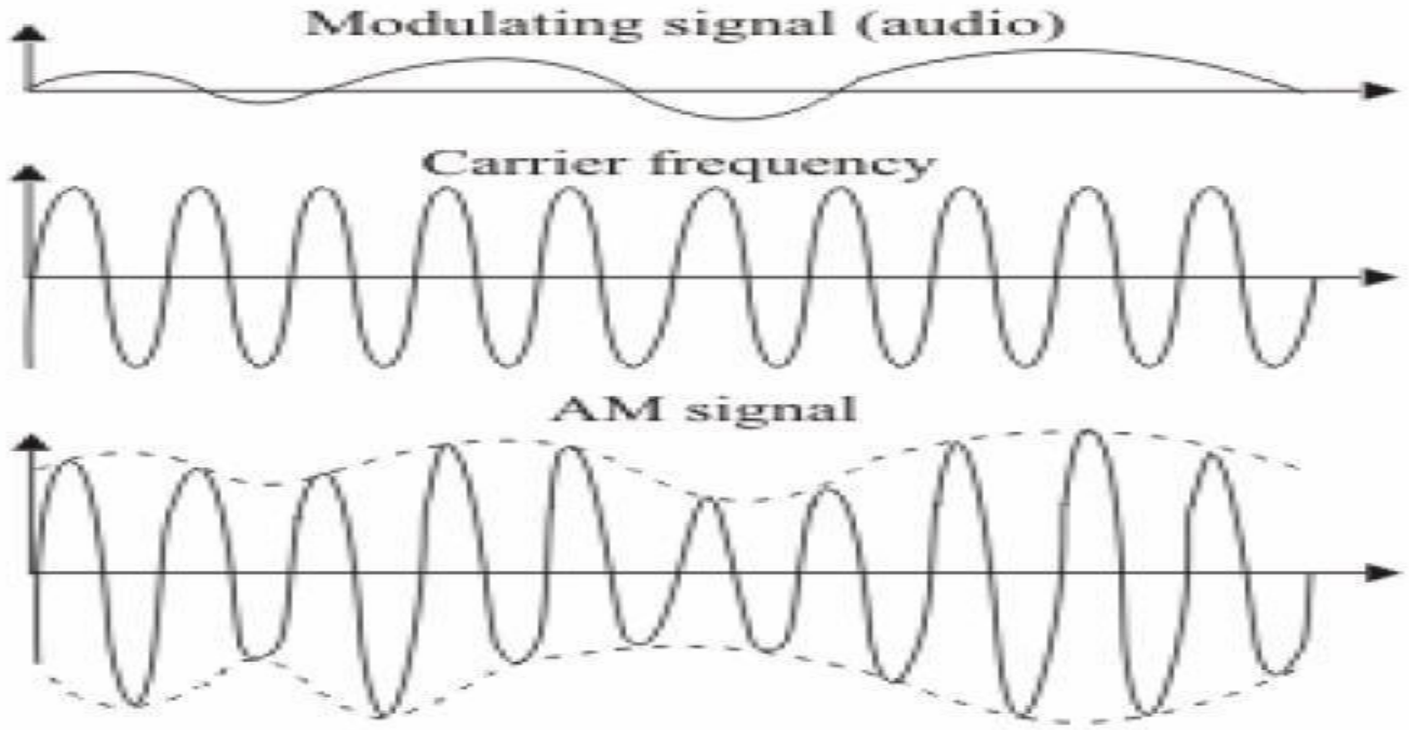
  These methods are:

- **Amplitude modulation (AM)**: where the amplitude of the signal is changed depending on the information to be sent
- **Frequency modulation:** where the frequency of the signal is changed depending on the information to be sent.
- **Phase modulation:** where it is the phase of the signal that is changed according to the information to be sent.

Amplitude Modulation (AM)

- In this modulation the frequency and phase of the carrier or base signal are not altered.
- The amplitude will only changes and we can see that the information is contained in the envelope of the carrier signal.
- It can be proud that the bandwidth of the composite signal is twice that of the highest frequency in the information signal that modulates the carrier.
- For radio transmission using AM, 9 KHz is allowed as the bandwidth.
- So, if a radio station transmits at 618 KHz, the next station can only transmit at 609 or 627 KHz.

Amplitude Modulation (AM)



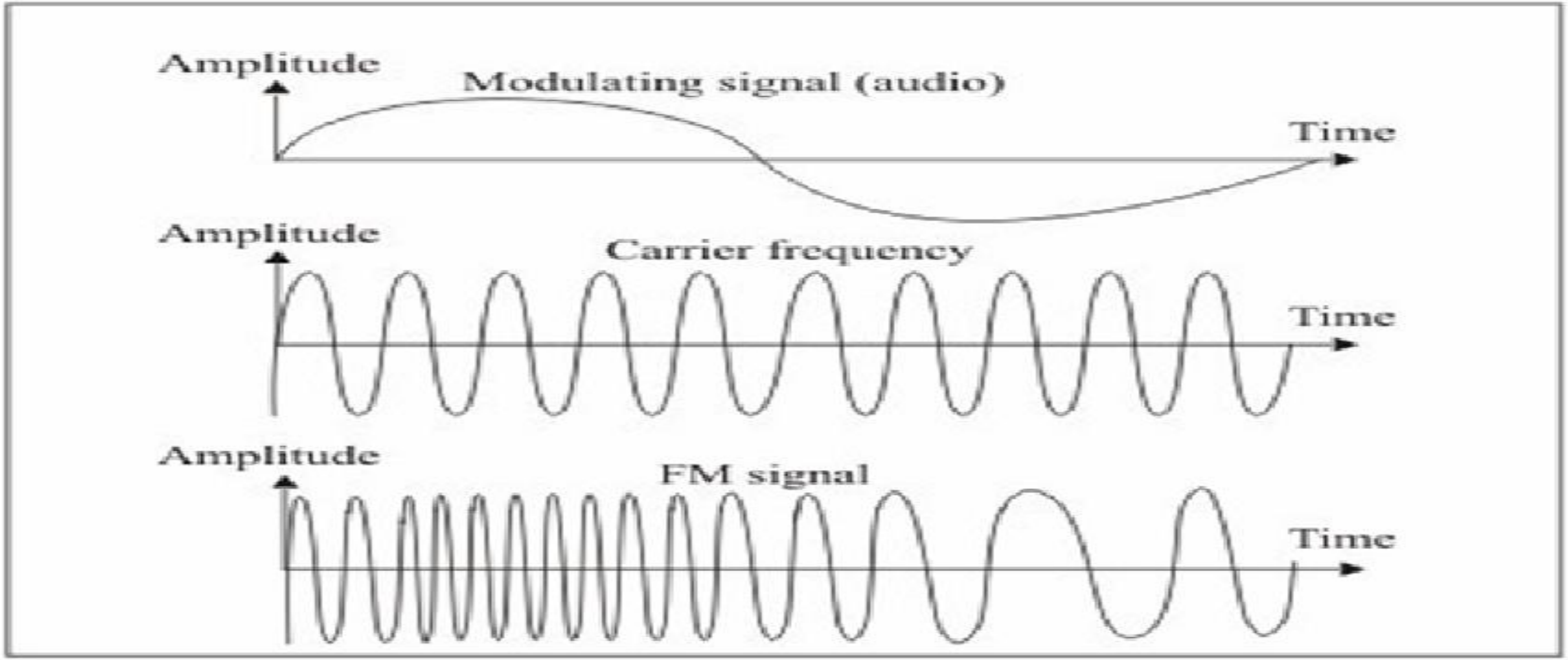
Frequency Modulation (FM)

- In contrast to Amplitude Modulation, it is the frequency of the base signal that is altered depending on the information that is to be sent.
- The amplitude and phase of the carrier signal are not changed at all.
- It can be shown that this results in a bandwidth requirement of 10 times the modulating signal, centered on the carrier frequency.
- FM radio transmission has been allocated the spectrum range 88 MHz to 108 MHz.
- As a good stereo sound signal needs 15 KHz of bandwidth, this means that FM transmission has a bandwidth of 150 KHz.

Frequency Modulation (FM)

- To be safe from interference from neighboring stations, 200 KHz is the minimum separation needed.
- As a further safety measure, in any given area, only alternate stations are allowed and the neighboring frequencies are kept empty.
- This is practical because FM transmission is a line of sight (because of the carrier frequency) and so, beyond a range of about 100 Km or so, the signal cannot be received directly (this distance depends on various factors).
- Therefore, we can have at most 50 FM stereo radio stations in a given geographical area.

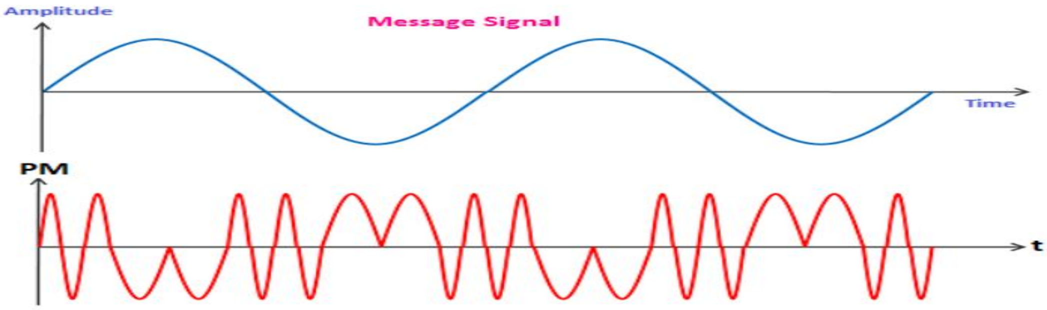
Frequency Modulation (FM)



Frequency modulation

Phase Modulation (PM)

- As you would have guessed by now, in Phase Modulation (PM) the modulating signal leaves the frequency and amplitude of the carrier signal unchanged but alters its phase.
- The characteristics of this encoding technique are similar to FM, but the advantage is that of reduced complexity of equipment.
- That is why, this method is not used in commercial broadcasting.



www.javatpoint.com/phase-modulation

Analog to Digital Modulation

- Natural phenomena are analog in nature and can take on any of the potentially infinite number of values.
- For example, the actual frequencies contained in a sound made by a human is an analog signal, as is the amplitude or loudness of the sound.
- One example of coding analog data in digital form is when, we want to record sound on digital media such as a DVD or in other forms such as MP3 or as a ".wav" file.
- Here, the analog signal, that is, the musical composition or the human voice is encoded in digital form. This is an example of analog to digital encoding.

Digital to Analog Modulation

- So far, we have looked at the codification of analog data, where the source information could take up any value.
- We also have need for the reverse type of encoding, where binary digital values have to be encoded into an analog signal.
- This may seem simple because after all, we only have to give two distinct values to the analog signal, but here you need to realize that, we have to be able to distinguish between successive digital values that are the same, say three 1’s in succession or two 0’s in succession.
- These should not get interpreted as a single 1 or a single 0! This sort of problem did not exist in the case of analog to analog encoding.

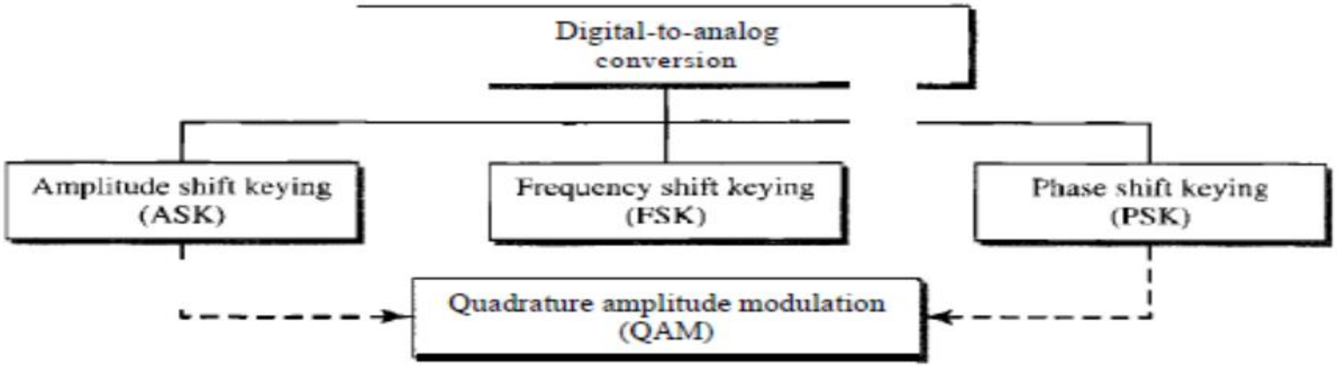
Digital to Analog Modulation

There are three types of digital to analog modulation:

Digital to analog modulation techniques

Digital to Analog Modulation

- Amplitude Shift Keying (ASK)
- Frequency Shift Keying (FSK)
- Phase Shift Keying (PSK)
- Quadrature Amplitude Modulation (QAM)



Digital to Analog Modulation

Amplitude Shift Keying (ASK)

- In this encoding, abbreviated as ASK, the amplitude of the Analog Signal is changed to represent the bits, while the phase and the frequency are kept the same.
- The bigger the difference between the two values, the lower the possibility of noise causing errors.
- There is no restriction, however, on what voltage should represent a 1 or a 0, and this is left to the designers.
- For the duration of a bit, the voltage should remain constant.

Digital to Analog Modulation

Frequency Shift Keying (FSK)

- As in the case of Analog-to-Analog encoding, we can encode a digital signal by changing the frequency of the analog carrier.
- This is frequency shift keying (FSK).
- When a bit is encoded, the frequency changes and remains constant for a particular duration.
- The phase and the amplitude of the carrier are unaltered by FSK.
- As in FM, noise in the line has little effect on the frequency and so, this method is less susceptible to noise.

Digital to Analog Modulation

Phase Shift Keying (PSK)

- We can also encode by varying the phase of the carrier signal.
- This is called phase shift keying (PSK) and here, the frequency and amplitude of the carrier are not altered.
- To send a 1 we could use a phase of 0 while we could change it to 180 degrees to represent a 0.
- Such an arrangement is not affected by the noise in the line, because that affects amplitude rather than the phase of the signal.

Digital to Analog Modulation

Quadrature Amplitude Modulation (QAM)

- To further improve the efficiency of the transmission, we can consider combining different kinds of encoding.
- Whereas at least one of them has to be held constant, there is no reason to alter only one of them.
- As of its bandwidth requirements, we would not bother to combine FSK with any other method.
- Yet it can be quite fruitful to combine ASK and PSK and reap the advantages. This technique is called Quadrature Amplitude Modulation (QAM).

Digital to Digital Modulation

- In Digital to Digital Modulation, Whenever while sending data from a computer to a printer, both of which understand digital signals.
- The transmission has to be for short distances over the printer cable and occurs as a series of digital pulses.
- Another example of such encoding is for transmission over local area networks such as an Ethernet, where the communication is between computer and computer.
- This kind of encoding is of three types
  - Unipolar, Polar and Bipolar techniques.

Digital to Digital Modulation

Kinds of encoding in Digital To Digital Modulation:

- **Unipolar Encoding:** This is a simple mechanism but is rarely used in practice because of the inherent problems of the method.
- As the name implies, there is only one polarity used in the transmission of the pulses.
- A positive pulse may be taken as a 1 and a zero voltage can be taken as 0, or the other way round depending on the convention we are using. The unipolar encoding uses less energy and also has a low cost.

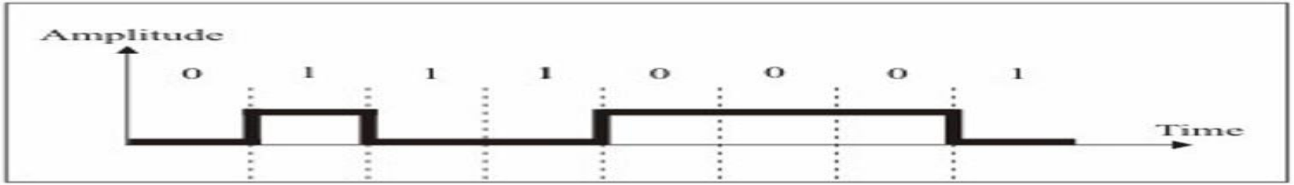


Figure 7: Unipolar encoding

Digital to Digital Modulation

- **Polar**: Unlike unipolar schemes, the polar methods use both a positive as well as a negative voltage level to represent the bits.
- So, a positive voltage may represent a 1 and a negative voltage may represent a 0, or the other way round. Because both are positive.
- Negative voltages are present, the average voltage is much lower than in the unipolar case, mitigating the problem of the DC component in the signal.
- There are three popular Polar Encoding Schemes.
  - Non-return to Zero (NRZ)
  - RZ Encoding.
  - Biphase Encoding.

Digital to Digital Modulation

- **Polar Encoding Schemes:**
- Non-return to Zero (NRZ): is a polar encoding scheme that uses a positive as well as a negative voltage to represent the bits.
- A zero voltage will indicate the absence of any communication at the time. Here again, there are two variants based on level and inversion.
- In NRZ-L (for level) encoding, it is the levels themselves, indicate the bits, e.g, a positive voltage for a 1 and a negative voltage for a 0.
- In NRZ-I (inversion) encoding, an inversion of the current level indicates a 1, while a continuation of the level indicates a 0.

Digital to Digital Modulation

- **Polar Encoding Schemes:**
- RZ Encoding: It allows for synchronization after each bit.
- This is achieved by going to the zero level midway through each bit.
- So a 1 bit could be represented by the positive to zero transition and a 0 bit by a negative to zero transition.
- At each transition to zero, the clocks of the transmitter and receiver can be synchronized.
- Of course, the price one has to pay for this is a higher bandwidth requirement.

Digital to Digital Modulation

Biphase Encoding:

- While in RZ, the signal goes to zero midway through each bit, in biphase it goes to the opposite polarity midway through each bit.
- These transitions help in synchronization, as you may have guessed. Again this has two flavors called Manchester encoding and Differential Manchester encoding.
- In the former, there is a transition in the middle of each bit that achieves synchronization.
- A 0 is represented by a negative to positive transition while a 1 is represented by the opposite.

Digital to Digital Modulation

Bipolar:

- This encoding method uses three levels, with a zero voltage representing a 0 and a 1 being represented by positive and negative voltages respectively.
- This way the DC component is done away with because every 1 cancels the DC component introduced by the previous 1.
- This kind of transmission also ensures that every 1 bit is synchronized.
- This simplest bipolar method is called Alternate Mark Inversion (AMI).
- The problem with Bipolar AMI is that synchronization can be lost in a long string of 0’s.

Important Topics

- Encoding
- Analog to Analog Modulation
- Analog to Digital Modulation
- Digital to Analog Modulation
- Digital to Digital Encoding

Summary

- In this Session, you have seen the need for data encoding and seen that there are four types of data encoding.
- These arise from the fact that there are 2 source signal types (Analog and Digital) and 2 transmission types (wired or wireless).
- We have seen the different kinds of techniques used for each type of data encoding – Analog to Analog, Analog to Digital, Digital to Analog and Digital to Digital.
- The bandwidth needed, noise tolerance, synchronization methods and other features of each different technique of encoding have been elaborated upon.

Thank you

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Data Communication and Computer Networks

Block-1 Unit-4

Multiplexing and Switching

Topics to be Covered

- Introduction
- Multiplexing
- Frequency Division Multiplexing
- Time Division Multiplexing
- Digital Subscriber Lines
- ADSL vs. Cable
- Switching
  - Circuit Switching
  - Packet Switching
- Important Topics
- Summary

Introduction

- The transmission of data over a network, whether wireless, requires to solve the problem of being able to send a large amount of data among different groups of recipients at the same time.
- If we think of the thousands of radio or television channels that are being broadcast throughout the world simultaneously, or of the millions of telephone conversations that are taking place every minute of the day.
- This feat is achieved by **multiplexing** the transmissions in some way.
- There are some differences in the characteristics of voice and data communication.
- Although, finally everything is really data.

MultiPlexing

- If one wanted to send data between a single source and a single destination, things would be comparatively easy.
- All it would need is a single channel between the two nodes, of a capacity sufficient to handle the rate of transmission based on Nyqist's theorem and other practical considerations.
- Granted that there would be no other claimants for the resources, there would be no need for sharing and no contention.

MultiPlexing

- The transmission channel would be available to the sole users in the world at all times, and any frequency they chose could be theirs.
- In a broadcasting case, we could have the single source transmitting at any time of its choosing and any agreed upon frequency.
- However, things are not so straightforward in the real world.
- We have a large number of nodes, each of which may be transmitting or receiving, from or to possibly different nodes each time.
- These transmissions could be happening at the same moment.
- So, when we need to transmit data on a large scale, we run into the problem of how to do this simultaneously.

MultiPlexing

- So, when we need to transmit data on a large scale, we run into the problem of how to do this simultaneously.
- As there can be many different sources and the intended recipients for each source are often different.
- The transmission resource, that is the available frequencies, is scarce compared to the large number of users.
- So, there will have to share the resource and consequently, the issue of preventing interference between them will arise.

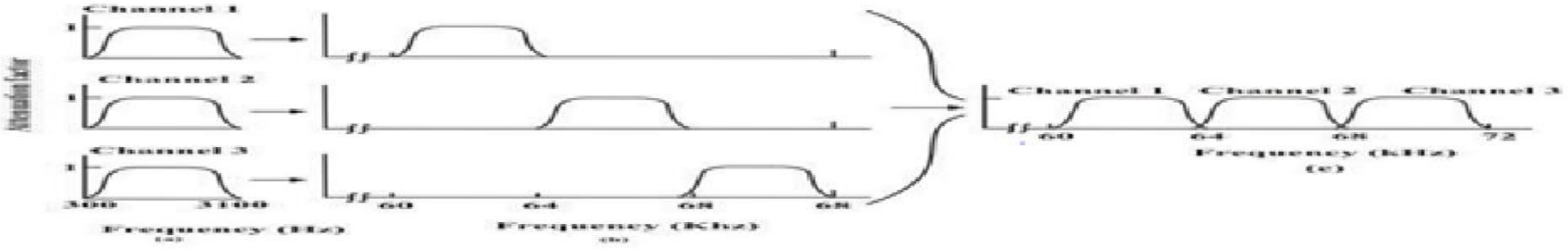
https://image1.slideserve.com/1910684/multiplexing1-l.jpg

Frequency Division MultiPlexing

- Suppose, it is a human voice that has to be transmitted, over a telephone.
- This has frequencies within the range of 300Hz to 3400 Hz.
- We can modulate this on a bearer or carrier channel, such as one at 300 kHz.
- Another transmission that has to be made can be modulated to a different frequency, such as 304 kHz, and yet another transmission could be made simultaneously at 308 kHz.
- We are thus, dividing up the channel from 300 kHz up to 312 kHz into different frequencies for sending data.
- As all the different transmissions are happening at the same time – it is only the frequencies that are divided up.

Frequency Division Multiplexing

- This is Frequency Division Multiplexing (FDM) because all the different transmissions are happening at the same time – it is only the frequencies that are divided up.
- The composite signal to be transmitted over the medium of our choice, is obtained by summing up the different signals to be multiplexed.
- The transmission is received at the other end, the destination, and there, it has to be separated into its original components, by DE multiplexing.



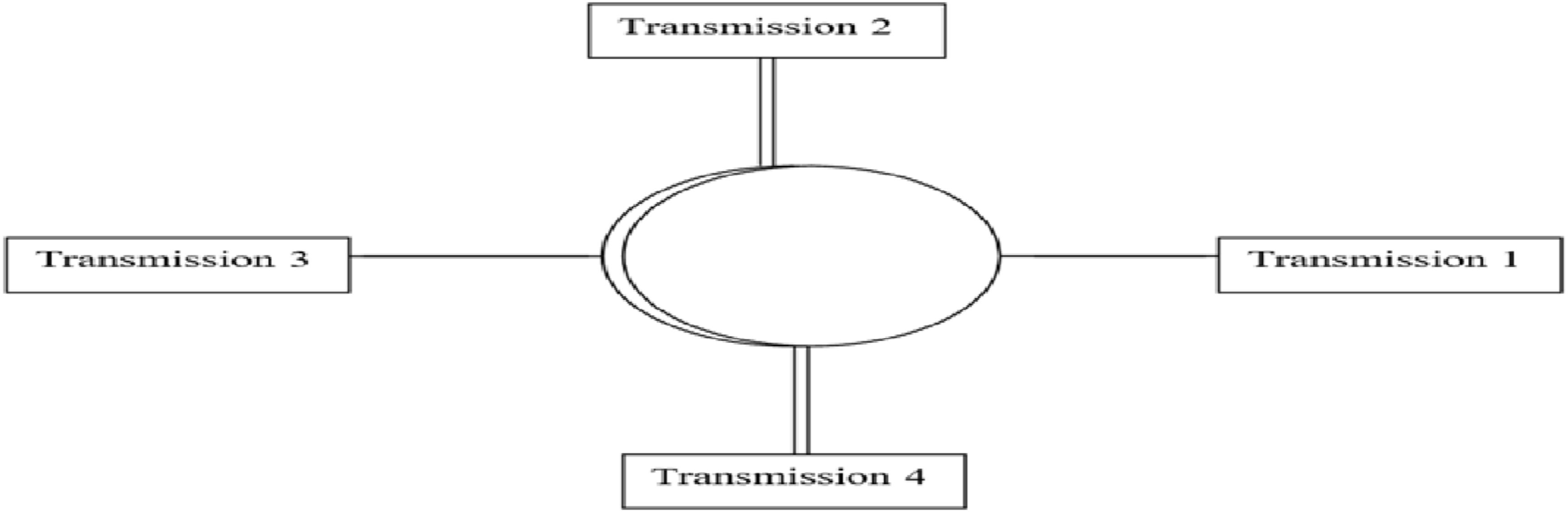
Frequency division multiplexing

Frequency Division Multiplexing

- In practice, a scheme like this may result in interference or pass speak among adjoining channels.
- As the bandpass filters which might be used to constrain the original statistics among the agreed upon frequencies (300 to 3400 kHz) aren't sharp.
- To minimize this, there are defend bands, or unused quantities of the spectrum between each two channels.
- Any other viable reason of interference should get up due to the fact that the device, together with amplifiers used to growth the strength of the signal, may not behave linearly over the whole set of frequencies that we are looking for to transmit.

Time Division Multiplexing

- Another multiplexing scheme is to use the entire bandwidth for each channel but to divide it into different time slots.



Time division multiplexing

Time Division Multiplexing

- In this Multiplexing, we have four different transmissions occurring with each being ¼ of the time slice.
- The slice should be small enough so that the slicing is not apparent to the application.
- So, for a voice transmission a cycle of 100ms could be sufficient as we would not be able to detect the fact that there are delays.
- In that case, each transmission could be allotted a slice of 25ms.
- At the receiving end, the transmission has to be reconstructed by dividing up the cycle into different slices, taking into account the transmission delays.

Synchronous Time Division Multiplexing

- In this kind of Time Division Multiplexing, the simpler situation is where the time slots are reserved for each transmission, irrespective of whether it has any data to transmit or not.
- Therefore, this method can be inefficient because many time slots may have only silence.
- But it is a simpler method to implement because the act of Multiplexing and DEMultiplexing is easier.
- This kind of TDM is known as Synchronous TDM.
- We usually transmit digital signals, although the actual transmission may be digital or analog.

Statistical Time Division Multiplexing

- We have seen that synchronous TDM can be quite wasteful.
- As an example, in a voice transmission, much of the bandwidth can be wasted because there are typically many periods of silence in a human conversation.
- Another example could be, that of a time-sharing system where many terminals are connected to a computer through the network.
- Here too, many of the time slots will not be used because there is no activity at the terminal.
- Despite being comparatively simpler, synchronous TDM is, therefore, often not an attractive option.

Digital Subscriber Lines

- Digital Subscriber Lines is an example of multiplexing in action. They are telephone lines that can support much higher data rates than are possible with an ordinary telephone connection.
- ADSL is comparatively simple for a service provider to offer, given an existing telephone network, and does not require much change to its already available equipment.
- It necessitates two modifications, one each, at the subscriber end and the end office.
- On the user's premises, a Network Interface Device has to be installed that incorporates a filter.

Digital Subscriber Lines

- This is called a splitter and it sends the non-voice portion of the signal to an ADSL (Asymmetric DSL) modem.
- The signal from the computer has to be sent to the ADSL modem at high speed, usually done these days by connecting them over a USB port.

www.britannica.com/technology/DSL

ADSL Vs. CABLE

- At first glance, a comparison between ADSL and cable may seem like a no contest.
- Cable television, sent over coaxial cables, has a bandwidth that is potentially hundreds of times that of the twisted pair Cat-3 cable used for telephone connections.
- As we go along, it turns out that there are considerations in favor of both sides.
- First, there are specific assurances regarding bandwidth that we get from telephone companies that provide ADSL connectivity.
- An ADSL link is a dedicated connection that is always available to the user, unlike television cable which is shared by scores or even hundreds of subscribers in the immediate neighborhood.

ADSL Vs. Cable

- So the kind of speeds that we can get over cable can vary from one moment to the next, depending on the number of users that are working at the time.
- There are security risks associated with the fact that cable is shared. Potentially other users can always tap in and read (even change) what you are sending or receiving.
- The problem does not exist on ADSL, because, each channel is separate and dedicated to the specific user.
- Though, cable traffic is usually encrypted by the provider, this situation is worse than ADSL where other users just do not get your traffic at all.

Switching

- There are many potential sources of data in the world and likewise, many potential recipients. Just think of the number of people who may like to reach one another over the telephone.
- How can one ensure that every data source can connect to the recipient? One cannot have a physical link between every pair of devices that might want to communicate!
- Therefore, we need a mechanism, to be able to connect devices that, need to transfer data between them.
- This is the problem of switching. There are two basic approaches to switching.
- Circuit switching and Packet switching.

Circuit Switching

- We create a circuit or link between devices, for the duration of time for which they wish to communicate.
- This requires a mechanism for, the initiating device to choose the device that it wants to send data to. All devices must also have an identifying, unique address that can be used to set up the circuit.
- Then, the transmission occurs, and when it is over, the circuit is dismantled so that the same physical resources can be used for another transmission.

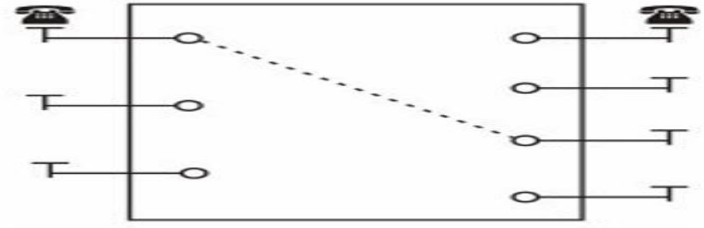


Figure 4: Circuit switching

Packet Switching

- The Packet switching can be used when we are using datagrams or packets to transmit data over the channel.
- Each datagram is a self-contained unit that includes within itself the needed addressing information.
- The datagrams can arrive at the destination by any route and may not come in sequence.

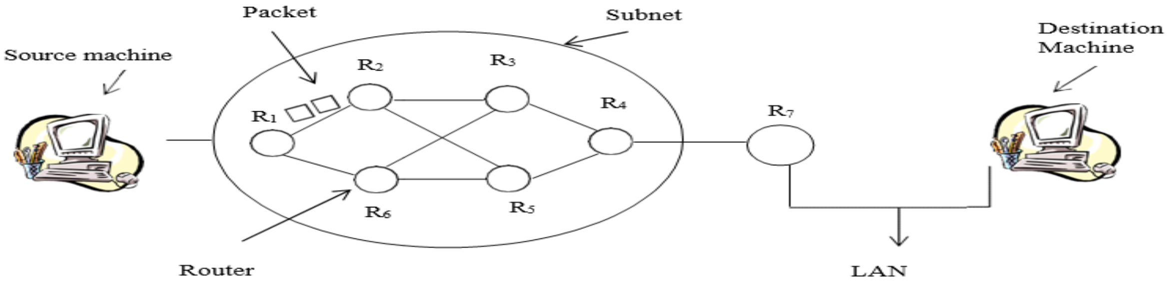


Figure 6: Packet switching

Important Topics

- Frequency Division Multiplexing.
- Time Division Multiplexing.
- Digital Subscriber Lines.
- Circuit Switching
- Packet Switching

Summary

- Multiplexing is needed in communication networks because of the scarcity of bandwidth compared to the large number of users.
- Frequency Division and Time Division Multiplexing are the two ways of multiplexing, of which Time Division multiplexing can be synchronous or asynchronous.
- The Asynchronous method is more complex but more efficient for data transmission.
- ADSL is a means of utilising the existing capacity of the local loop in the telephone system for providing subscribers with high speed data access.
- Switching is necessary to connect two nodes or devices over the network that intend to communicate for a limited duration.

Summary

- Circuit switching is more suitable for voice communication, and can be done using space division, time division or a combination of both kinds of switches.
- Multistage switching is needed to optimise the number of crosspoints needed.
- Packet switching is used for data transmission and allows for prioritising of data packets, alternate routing as needed and is also more efficient for the bursty traffic pattern of data communication.
- Datagrams are self contained packets of data that are routed by the intermediate nodes of the network. Switched or permanent virtual circuits can also be utilized.

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Data Communication and Computer Networks

Block-2 Unit-1_Part1

Data Link Layer Fundamentals

Topics to be Covered

- Introduction
- Framing
- Basics of Error Detection
- Forward Error Correction
- Cyclic Redundancy Check codes for Error detection
- Flow Control
- Important Topics
- Summary

Introduction

- Data Link Layer (DLL) is the second layer of the OSI model which takes services from the Physical layer and provides services to the network layer.
- DLL transfers the data from a host to a node or a node to another node.
- Data packets are encoded and decoded into bits. These are called frames.
- The Functions of the Data Link Layer are Framing, Frame synchronization, Error Handling Flow Regulation, Addressing and Access Control.
- The data link layer is divided into two sublayers:
    - The Media Access Control (MAC) layer and the Logical Link Control (LLC) layer.
- The MAC sublayer gains access to the link resources and LLC layer controls frame synchronization, flow control and error checking.

Framing

Flag	Control Information	Data Value	Control Information	Flag

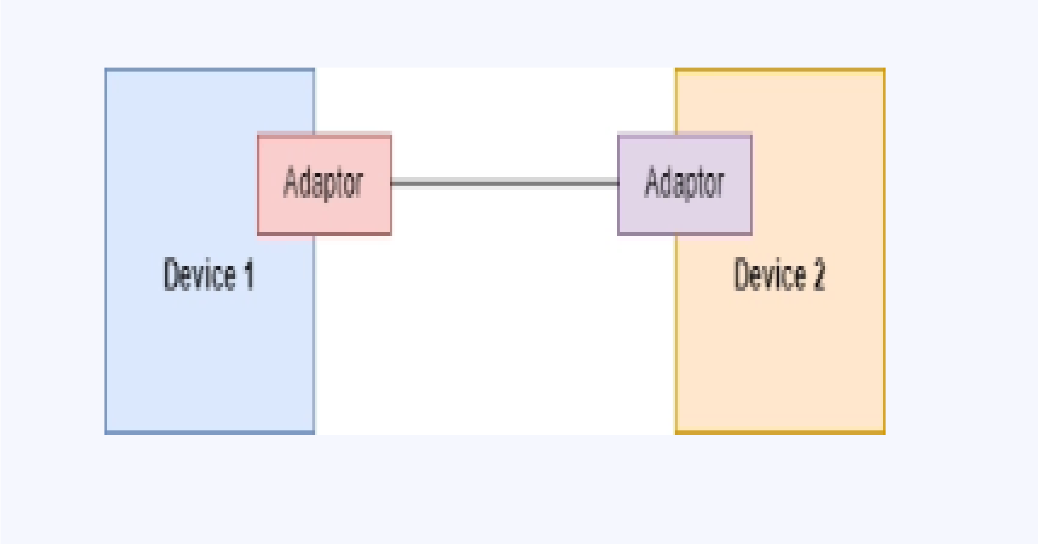
Figure 2: Frame format

- Frame synchronization
- Error Handling Flow Regulation
- Addressing and Accessing Control
- The data link layer is divided into two sub-layers:
- The Media Access Control layer and the Logical Link Control layer.
- The MAC sublayer controls how a computer on the network gains access to the link resources and grants permission to transmit it.
- The LLC layer controls frame synchronization, flow control, and error checking.

Methods of Framing

Methods of Framing are:

- Character Count
- Character Patterns
- Bit Patterns
- Framing by Illegal code
  (Code Violation)



Character Count

- Character count, uses a header field to specify the number of characters in the frame.
- The Data Link Layer at the destination checks the header field to know the size of the frame and hence, the end of the frame.
- However, problems may arise due to changes in character count value during transmission.

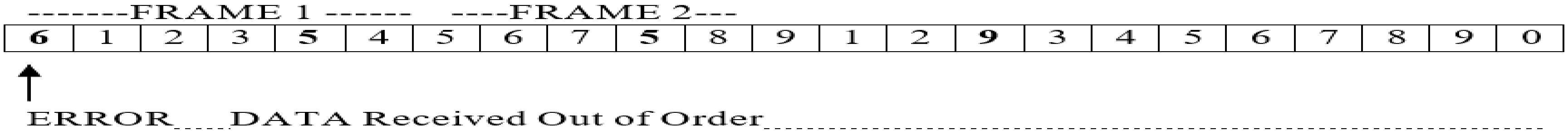
--FRAME 1-- ------FRAME 2---- -----FRAME 3--- -------------FRAME 4---------

4	1	2	3	**5**	4	5	6	7	**5**	8	9	1	2	**9**	3	4	5	6	7	8	9	0

Character count

Character Count

- However, problems may arise due to changes in character count value during transmission.
- As an example, in the first frame if the character counts 4 changes to 6, the destination will receive data out of synchronization and hence, it will not be able to identify the start of the next frame.
- In this example, Even, after a request from the destination to the source for retransmission comes, it does not solve the problem because the destination does not know from where to start retransmission.



Problems in character count

Character Patterns

- The second framing method, Character stuffing, solves the problem of out of synchronization.
- Here, each frame starts with special character set e.g., a special ASCII character sequence.
- The frame after adding the start sequence and the end sequence
- Begin each frame with DLE STX.
- After that End each frame with DLE ETX.
- Here, DLE stands for Data Link Escape, STX stands for Start of Text and
- ETX stands for End of Text

DLE	STX	....	DLE	ETX

Character patterns (Character stuffing)

Bit Patterns

- This method is similar to the one discussed earlier, except that, the method of bit stuffing allows the insertion of bits instead of the entire character (8 bits).
- Bit pattern framing uses a particular sequence of bits called a flag for framing.
- The flag is set as the start and the end of the frame.
- Use of bit patterns is to keep the sequence of data in the same order. Flag = 0 1 1 1 1 1 1 0 (begins and ends frame.)
- In this case the transmitter automatically inserts a 0 after 5 consecutive 1's in the data. This is called Bit stuffing.
- The receiver discards these stuffed 0's as soon as it encounters 5 consecutive 1's in the received data.

Bit Patterns

11111111110010011

Original data ready to be sent

011111101111101111100010011 01111110

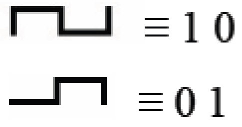
Data after adding Flag and stuffing

11111111110010011

Data after Destuffing

Framing by Illegal Code (Code violation)

- A fourth method is based on any redundancy in the coding scheme.
- In this method, we simply identify an illegal bit pattern and use it as a beginning or end marker, i.e., certain physical layers use a line code for timing reasons.
- As an example In Manchester Encoding,
- 1 –It can be coded into two parts i.e., high to low
- 0 – It can be coded into two parts i.e., low to high



- Codes of all low (000) or all high (111) aren’t used for the data and therefore, can used for framing.

Basics of Error Detection

- The Network have to ensure complete and accurate delivery of data from the source to the destination.
- But many times data gets corrupted during transmission. because many factors can corrupt or alter the data which leads to an error.
- A reliable system should have methods to detect and correct errors.
- Firstly, we will discuss what the error could be then, in the later section we will discuss the process of detecting and correcting them.

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Types of Error

- The Network should ensure complete and accurate delivery of data from the source node to destination node. But many times data gets corrupted during transmission.

- As already discussed, many factors can corrupt or alter the data that leads to an error. A reliable system should have methods to detect and correct the errors.

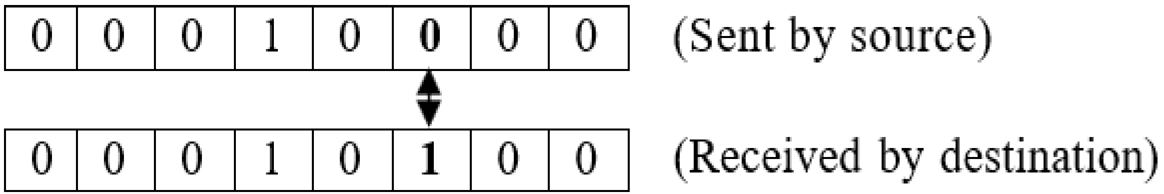
- **Types of Error**
- 1-bit error
- burst error
- lost message (frame).

Flag	Control Information	Data Value	Control Information	Flag

Figure 2: Frame format

Types of Error

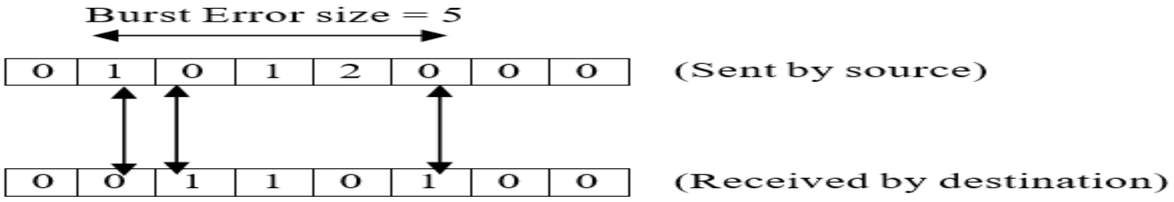
- **1-bit error: 1**-bit error/Single bit error means that only one bit is changed in the data during transmission from the source to the destination node i.e., either 0 is changed to 1 or 1 is changed to 0.
- This error will not appear generally in case of serial transmission. But it might appear in case of parallel transmission.



1 bit error

Types of Error

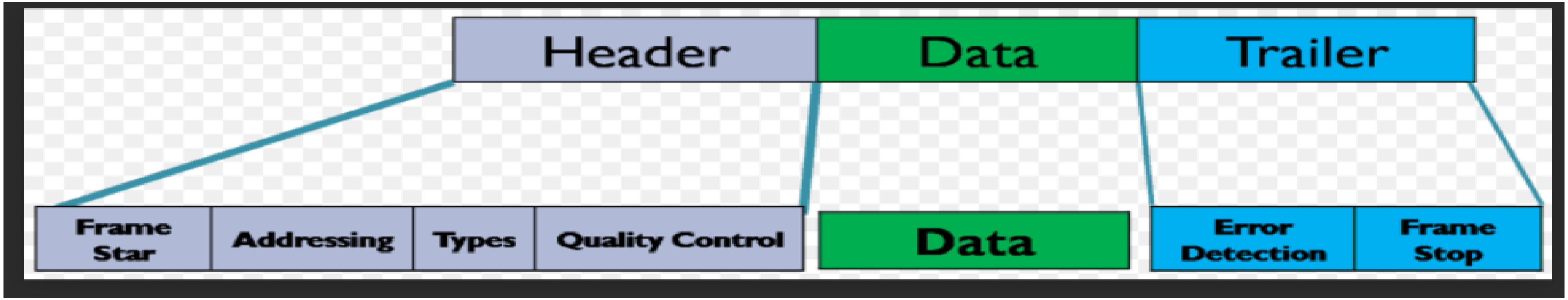
- **Burst error:** Burst error means that 2 or more bits of data are altered during transmission from the source to the destination node.
- It is not necessary that error will appear in consecutive bits.
- Size of burst error is from the first corrupted bit to the last corrupted bit.
- An *n*-bit burst error is a string of bits inverted during transmission.
- This error will hardly occur in case of parallel transmission. But, it is difficult to deal with all corrupted bits at one instance.



Burst error

Types of Error

- **Lost Message (Frame):** The sender has sent the frame but that is not received properly, this is known as loss of frame during transmission.
- To deal with this type of error, a retransmission of the sent frame is required by the sender.

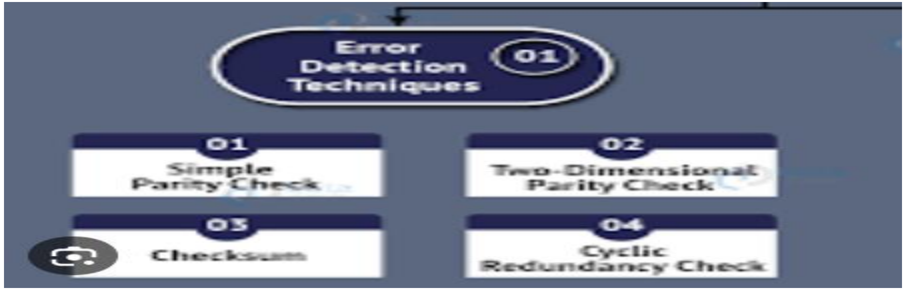


https://i0.wp.com/networkustad.com/wp-content/uploads/2019/06/Datal-link-layer-frame-fields.png?resize=768%2C194&ssl=1

Message (Frame)

- **Error Detection**: One of the important goals of this layer is, accurate delivery of data at the receiver's site.
- This implies that the receivers should get the data that is error free.
- However, due to some factors if, the data gets corrupted, we need to correct it using various techniques.
- So, we require error detection methods first to detect the errors in the data before correcting it.
- Error detection is an easy process.

data-flair.training/blogs/error-detection-and-correction-in-computer-network/



Error Detection

- For error detection the sender can send every data unit twice and the receiver will do bit by bit comparison between the two sets of information.
- Any alteration found after the comparison will indicate an error and a suitable method can be applied to correct the error.
- Hence, the basic strategy for dealing with errors is to include groups of bits as additional information in each transmitted frame, so that, the receiver can detect the presence of errors.
- This method is called Redundancy as extra bits appended in each frame are redundant.
- At the receiver end these extra bits will be discarded when the accuracy of data is confirmed.

Error Detection

Source node (Sender)

0	0	0	1	0	0	0	0

0	0	0	1	0	0	0	0	1	1	1

$\downarrow$

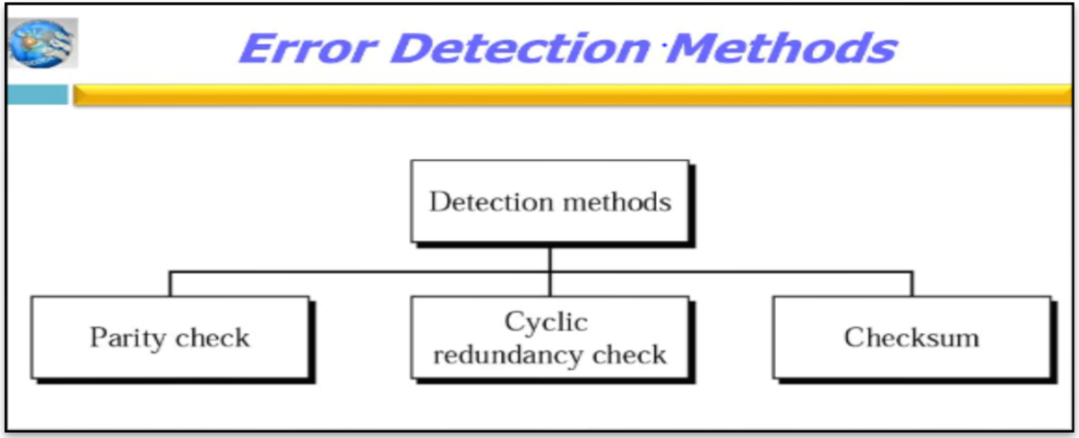
Destination node (Receiver)

0	0	0	1	0	0	0	0	1	1	1

0	0	0	1	0	0	0	0

Error Detection

- Redundancy check methods commonly used in data transmission are:
- Media Access Control
- and Data Link Layer
- Parity check
- CRC
- Checksum



https://image1.slideserve.com/2100892/slide17-l.jpg

Parity Check

- The most common method used for detecting errors when the number of bits in the data is small, is the use of the parity bit.
- A parity bit is an extra binary digit added to the group of data bits, so that, the total number of ones in the group is even or odd.
- Data bits in each frame are inspected before transmission and an extra bit (the parity bit) is computed and appended to the bit string to ensure even or odd parity, depending on the protocol being used.
- If odd parity is being used, the receiver expects to receive a block of data with an odd number of 1s.
- For even parity, the number of 1's should be even.

Parity Check

- In the example below, even parity is used. The ninth column contains the parity bit.
- 0 1 0 1 0 1 0 1 **0**
- 0 1 1 1 1 0 0 1 **1**
- 1 1 1 1 0 0 1 1 **0**
- Use of parity bits a rather weak mechanism for detecting errors.
- A single parity bit can only detect single-bit errors.

Block Sum Check Method

- This is an extension to the single parity bit method.
- This can be used to detect up to two-bit errors in a block of characters.
- Each byte in the frame is assigned a parity bit (row parity).
- An extra bit is computed for each bit position (column parity).
- The resulting set of parity bits for each column is called the block sum check.
- Each bit that makes up the character is the modulo-2 sum of all the bits in the corresponding column.

T T

Block Sum Check

- Block Sum Check,
- Sender’s data: 0000100 0010101 0101011
- 1 0 0 0 0 1 0 0
- 1 0 0 1 0 1 0 1
- 0 0 1 0 1 0 1 1
- 0 0 1 1 1 0 1 0
- Data after adding parity bits:
- 00001000 00101010 01010110 01110100
- This method detects single bit errors as well as increases the probability of finding burst error.

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Important Topics

The Framing

Error Detection

Error Codes

Summary

- In this session, we discussed about Data Link Layer.
- The concept of framing and different methods of framing like Character Count, Character Stuffing, Bit stuffing and Framing by Illegal code has been focused up on.
- In the Data Link layer, data flow and error control is discussed. This error and flow control is required in the system for reliable delivery of data in the network.
- For the same, different types of error, different methods for error detection and correction are discussed.
- Among various methods, Block sum check method detects burst of error with high probability.

Thank you

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Block-2 Unit-2

Retransmission Strategies

Topics to be Covered

- Introduction
- Stop and Wait ARQ
- Go-Back N ARQ
- Selective Repeat ARQ
- Pipelining
- Piggybacking
- Important Topics
- Summary

Introduction

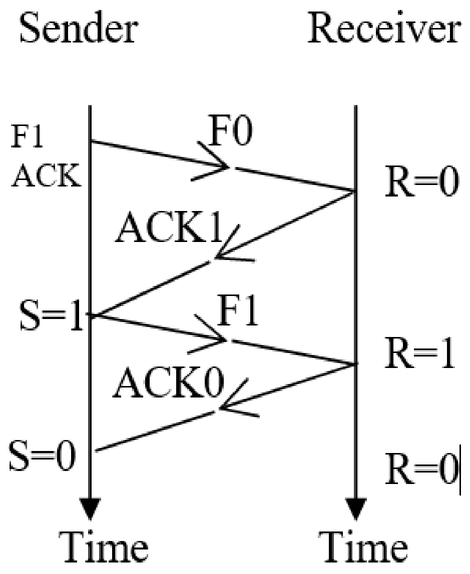
- Flow control deals with when the sender should send the next frame and for how long the sender should wait for an acknowledgment.
- Data link protocol takes care of the amount of data that a sender can send and that a receiver can process as the receiver has its limitation in terms of speed for processing the frames.
- It also sees the compatibility of speed of both the sender and the receiver We can send up to W frames before worrying about ACKs.
- We keep a copy of these frames until the ACKs arrive.
- This procedure requires additional features to be added to Stop-and-Wait ARQ.

Stop & Wait ARQ

- This is the simplest method for flow and error control.
- This protocol is based on the concept that, the sender will send a frame and wait for its acknowledgment.
- Until it receives an acknowledgment, the sender cannot send the next frame to the receiver.
- During the transmission of the frame over the network an error can appear.
- Normal Operation
    - When ACK is lost
    - When the frame is lost
    - When ACK time-out occurs

Normal Operation

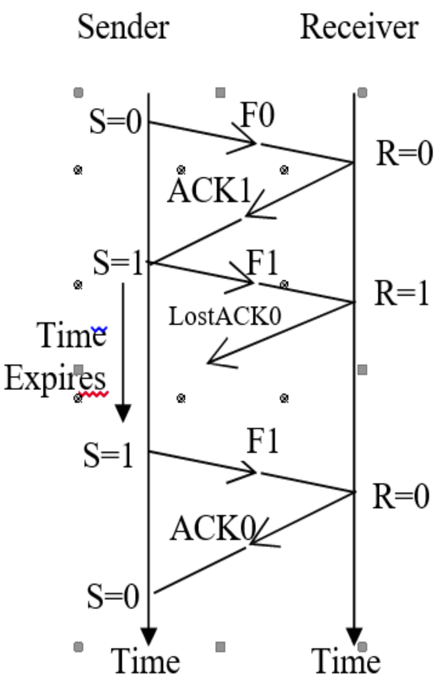
- If the sender is sending frame 0, then it will wait for Ack 1 which will be transmitted by the receiver with the expectation of the next frame numbered frame1.
- As it receives ACK1 in time (allotted time) it will send frame1.
- This process will be continuous till complete data transmission takes place.
- This will be successful transmission if Ack for all frames sent is received in time.



Stop and wait protocol

When ACK is lost

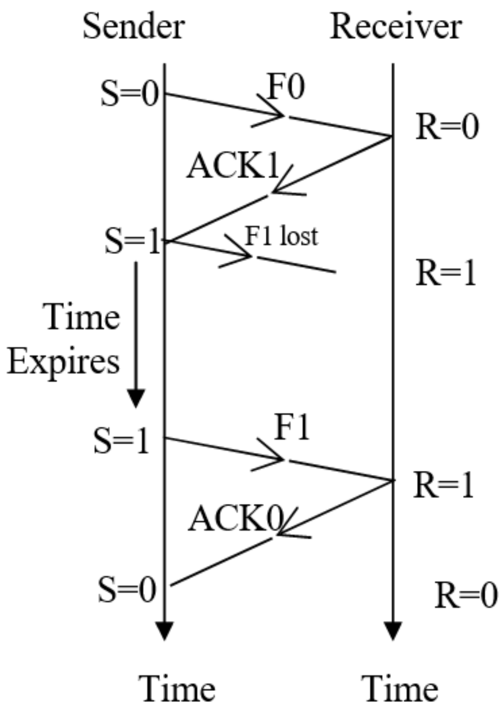
- Here, the sender will receive corrupted ACK1 for frame sent frame 0.
- It will simply discard corrupted ACK1 and as the time expires for this ACK it will retransmit frame 0. The receiver has already received frame 0 and is expecting frame 1, hence, it will discard a duplicate copy of frame 0.
- In this way, the numbering mechanism solves the problem of duplicate copies of frames. Finally, the receiver has only one correct copy of one frame.



Loss of ACK

When Frame is lost

- If the receiver receives corrupted/damaged frame1, it will simply discard it and assumes that the frame was lost on the way.
- And correspondingly, the sender will not get ACK0 as the frame has not been received by the receiver.
- The sender will be in the waiting stage for ACK0 till its time out occurs in the system.
- As soon as time out occurs in the system, the sender will retransmit the same frame i.e. frame1(F1) and the receiver will send ACK0 in reply.



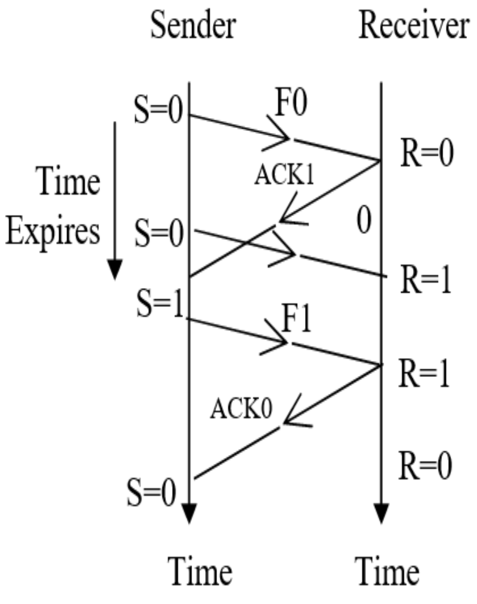
Loss of a frame

When ACK time out occurs

- In this operation, the receiver isn't suitable to send ACK1 for received frame0 in time, due to some problem at the receiver’s end or network communication.
- The sender retransmits frame0 as ACK1 isn't received in time. also, the sender retransmits frame0 as ACK1 time expires.
- At the receiver end, the receiver discards this frame0 as the duplicate copy is awaiting frame1 but sends the ACK1 formerly again corresponding to the copy received for frame0.
- At the sender’s point, the duplicate copy of ACK1 is discarded as the sender has received ACK1 before as explained with the help.

When ACK time out occurs

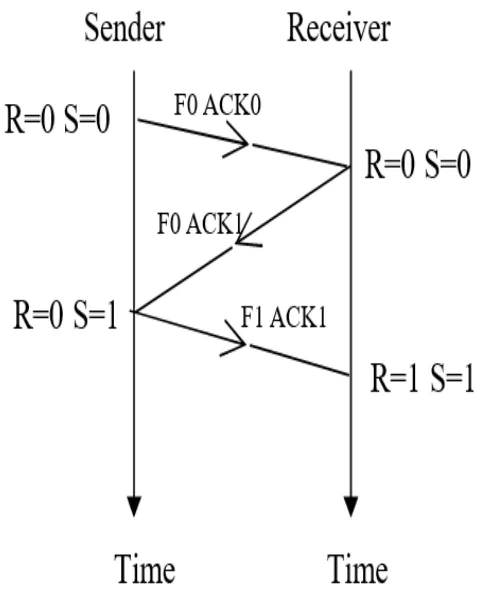
- These operations indicate the importance of the numbering mechanism while, transmitting frames over the network.
- The method above discussed has low efficiency due to improper use of the communication channel.
- So, now, we will discuss the case if flow of data is bidirectional.
- In Bidirectional transmission both the sender and the receiver can send frames as well as acknowledgement.



ACK time out

Piggy Backing

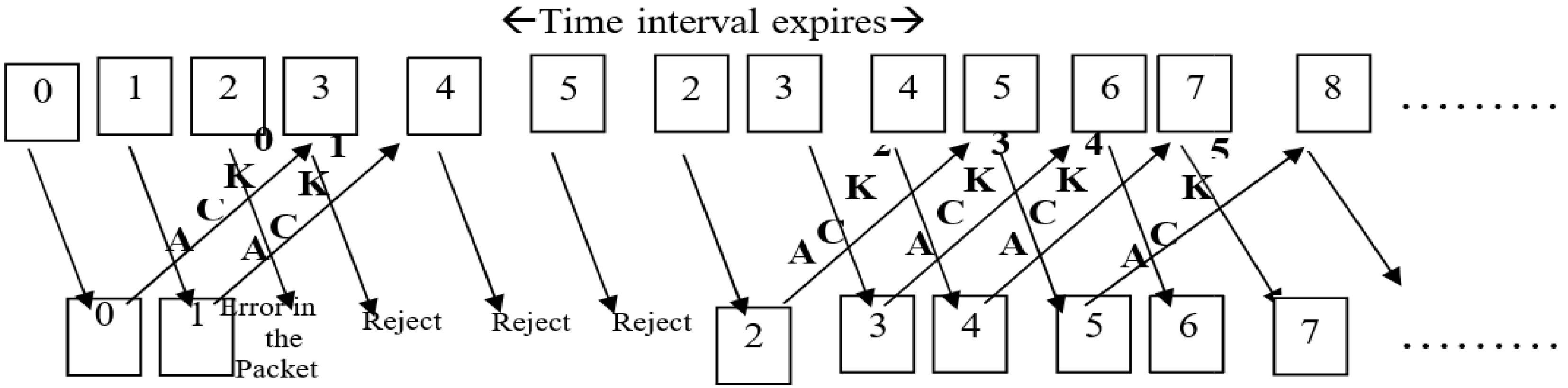
- Hence, both will maintain S and R variables.
  •To have an efficient use of bandwidth the ACK can be appended with the data frame during transmission.
  •The process of combining ACK with data is known as Piggybacking.
  •This reduces transmission overhead and increases the overall efficiency of data transmission.



Piggybacking 1

Go-Back-N ARQ

- In this, we can send n frames without making the sender wait for acknowledgements.
- At the same time, the sender will maintain a copy of each sent frame till acknowledgement reaches it safely.

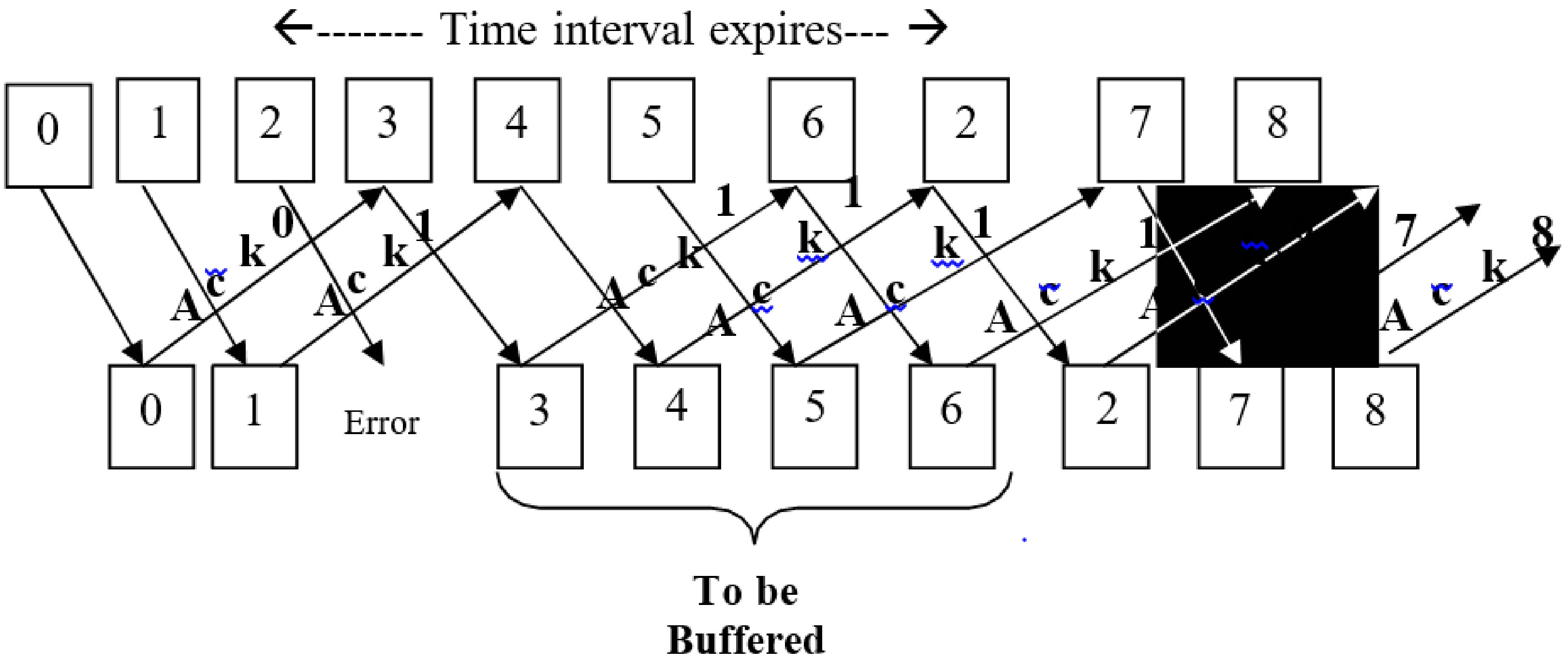


Go-Back-N

Selective Repeat

- In this method, the receiver has a window with a buffer that can hold multiple frames sent by the sender.
- The sender will retransmit only that frame that has some error and not all the frames as in Go- Back-N ARQ.
- Hence, it is named a selective repeat ARQ. Here the size of the sender and the receiver window will be the same.
- The receiver will not look forward only to one frame as in Go- Back-N ARQ but it will look forward to a continuous range of frames.
- The receiver also sends NAK for the frame which had the error and required to be retransmitted by the sender before the time-out event fires.

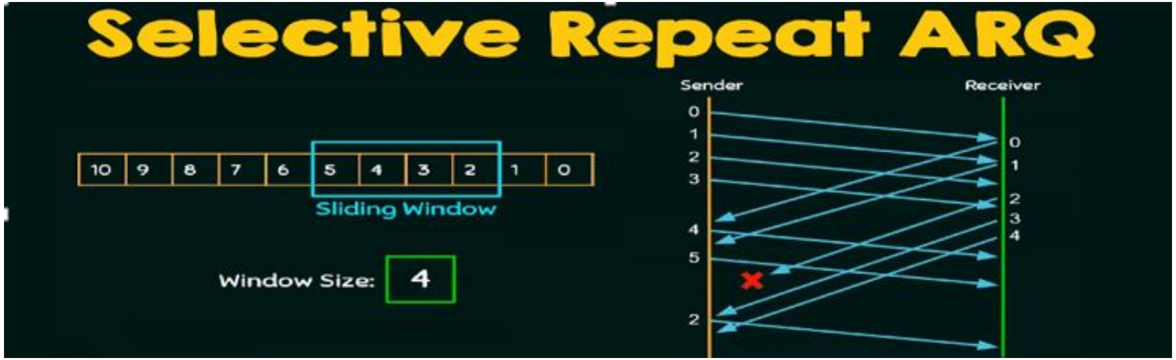
Selective Repeat



Selective Repeat

TCP uses Selective Repeat ARQ Strategy

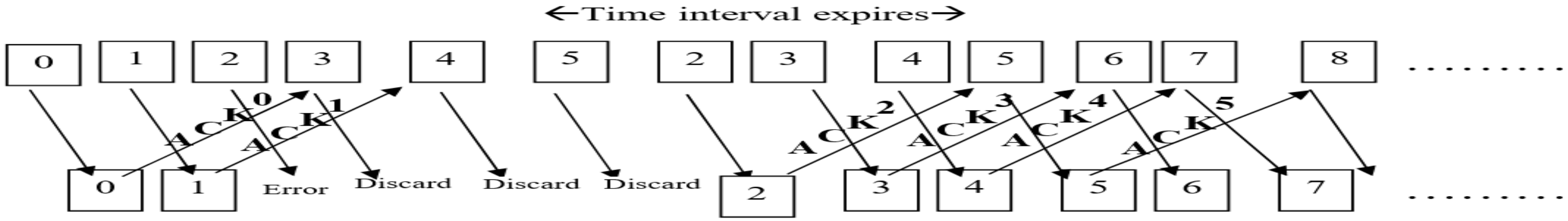
- If, we consider bidirectional transmission i.e., data and acknowledgement flow from both sender and receiver.
- Then, the concept of Piggybacking can be used in a similar fashion as already discussed in Stop & Wait ARQ method, in order to better utilize bandwidth.



www.youtube.com/watch?v=WfIhQ3o2xow

Pipelining

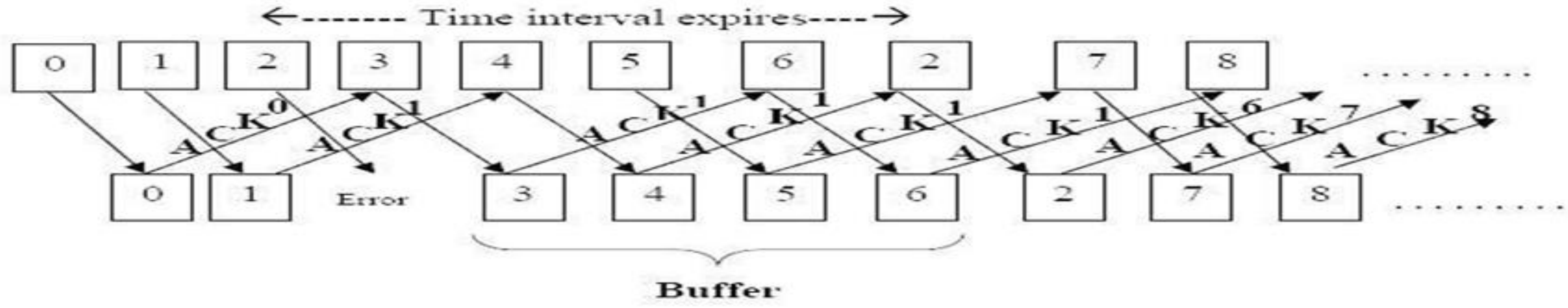
- Pipelining in the network is one task that starts before the previous one is completed.
- We might also say that the number of tasks is buffered in line, to be processed and this is called pipelining. For example, while printing, through the printer before one task printing is over we can give commands for printing second task. Stop & Wait ARQ does not use pipelining.



Pipelining in Go-Back-N

Selective Repeat Pipelining

- Here, we can show how pipelining increases the overall utilization of bandwidth.
- Frame0 and frame1 are sent in continuous order without making the sender wait for acknowledgment for frame0 first.



Pipelining in selective repeat

Piggybacking

- Piggybacking is the process that appends acknowledgment of frame with the data frame.
- The piggybacking process can be used if Sender and Receiver both have some data to transmit.
- This will increase the overall efficiency of transmission.
- While data is to be transmitted by both sender and receiver, both will maintain control variables S and R.
- We will explain the process of piggybacking when the sender and the receiver both are transmitting data.

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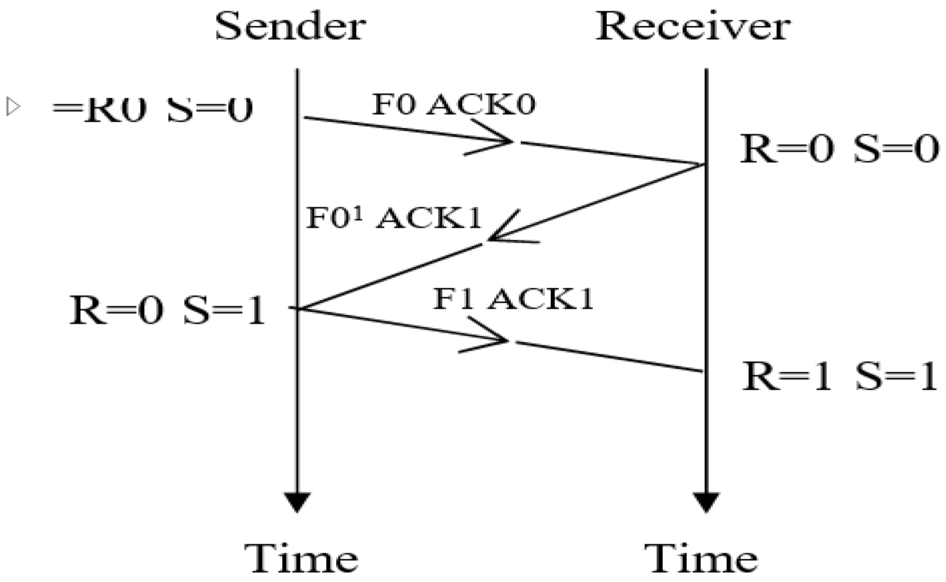
Piggybacking



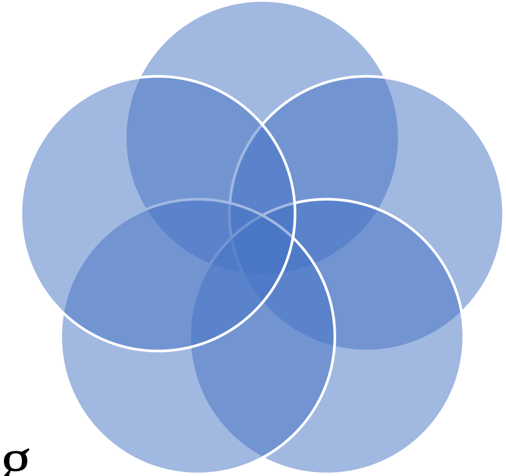
Piggybacking

Important Topics

Stop and Wait ARQ.

Piggybacking

Go-Back N ARQ



Pipelining

Selective Repeat ARQ

Summary

- This session focuses on one prime function of the Data link layer that is flow and error control for achieving the goal of reliable data transmission over the network.
- For flow and error control retransmission strategies are considered.
- Flow control specifically talks about the speed of sending the frame by the sender and processing the received frame by the receiver.
- The speed for the sender and the receiver must be compatible. So, that all frames can be received in order and processed in time.
- Error control technique combines two processes error detection and error correction in the data frame.

Summary

- In Stop & wait ARQ protocol sender waits for acknowledgment for the last frame sent.
- After the acknowledgment is received by the sender then only the next frame can be sent.
- In Go-Back-N ARQ frames can be sent continuously without waiting for the sender to send the acknowledgement.
- If an error is found in any frame then frames received after that will be discarded by the receiver.
- Retransmission of frames will start from the error frame itself. In selective repeat Process frames can be sent continuously.
- But here, the receiver has a buffer window that can hold the frames received after the error frame.

Thank you

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Block-1 Unit-3

Congestion Based Media Access Protocols

Topics to be Covered

- Introduction
- Advantages of Multiple Access Sharing of Channel Resources
- Pure Aloha
- Slotted Aloha
- Carrier Sense Multiple Access (CSMA)
- CSMA with Collision Detection (CSMA/CD)
- Important Topics
- Summary

Introduction

- Data Link Layer (DLL) is divided into two sub layers i.e.,
  - MAC layer and LLC layer.
- In a network the connection of nodes, a network can be divided into two categories, that is, point-to-point link and broadcast link.
- The protocol which decides who will get access to the channel and who will go next on the channel belongs to MAC sub-layer of DLL.
- Channel allocation is categorized into two, based on the allocation of broadcast among competing users that is Static channel allocation problem and Dynamic Channel allocation problem.

Channel Allocation Techniques

- Based on the connection of nodes, a network can be divided into two categories, Point-to-Point link and Broadcast link.
- In broadcast network, a control process for solving the problem of accessing a multi access channel is require.

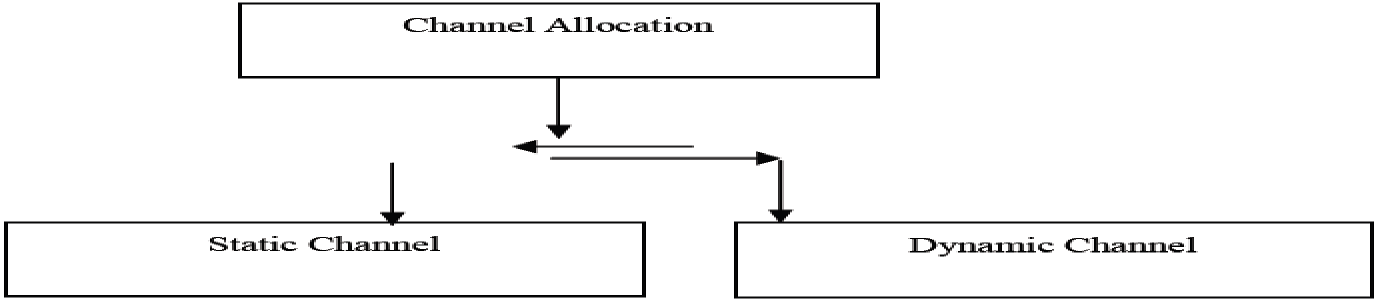


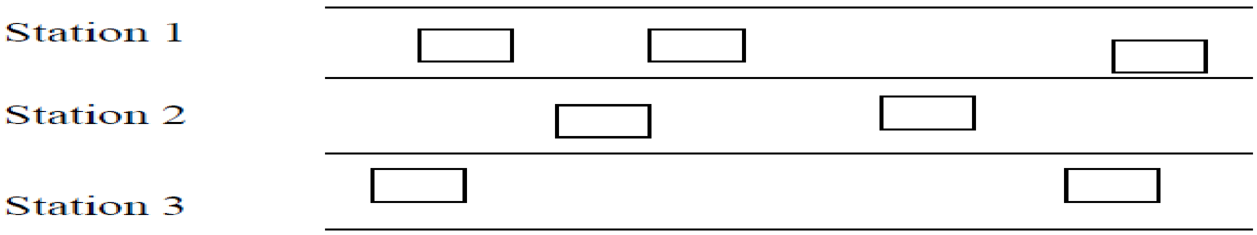
Figure 2: Channel allocation technique

Advantages of Multiple Access Sharing of Channel Resources

- MAC sub layer's primary function is to manage the allocation of one broadcast channel among N competing users.
- For the same, many methods are available such as static, and dynamic allocation methods.
- In the static channel allocation method, allocating a single channel among N competing users can be either
- FDM (Frequency division multiplexing)
- TDM (Time-division multiplexing).
- In FDM the total bandwidth will be divided into N equal parts for N users.
- In the dynamic channel allocation the important issues to be considered are whether the time is continuous or discrete

Pure Aloha

- In case of allocating a single channel among N uncoordinated competing users, then the probability of collision will be high.
- Station accesses the channel and when their frames are ready. This is called random access.
- In an ALOHA network one station will work as the central controller and the other station will be connected to the central station.

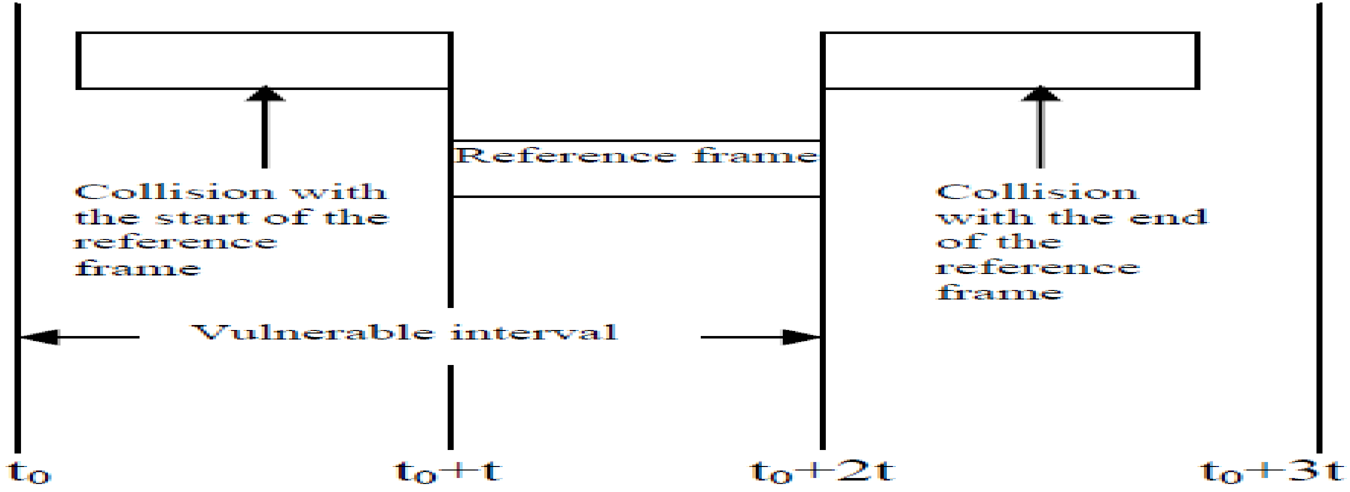


Pure Aloha

- Pure ALOHA protocol frames can be sent any time so, the probability of collision will be very high.
- Hence, to present a frame from colliding, no other frame should be sent within its transmission time.
- Let a frame is that transmitted at time t^0 and t be the time required for its transmission. I
- f, any other station sends a frame between t^0 and t^0+t then the end of the frame will collide with that earlier sent frame.
- Similarly, if any other station transmits a frame between the time interval t^0+t and t^0+2t again.
- It will result in a garbage frame due to collision with the reference frame.

Pure Aloha

- 2t is the Vulnerable interval for the frame. In case a frame meets with collision that frame is retransmitted after a random delay.
- For the probability of successful transmission, no additional frame should be transmitted in the vulnerable interval 2t.



Pure Aloha

- Pure Aloha is a Continuous time system whereas Slotted Aloha is discrete time system.
- Pure ALOHA doesn't check whether the channel is busy before transmission.
- Slotted ALOHA send the data at the beginning of timeslot.
- Pure aloha not divided in to time .
- Slotted aloha divided in to time.
- In slotted ALOHA, a frame can be sent only at fixed times, whereas in pure ALOHA, you can send any time.

Slotted Aloha

- Slotted ALOHA is a refinement over the pure ALOHA.
- The Slotted ALOHA requires that time be segmented into slots of a fixed length exactly equal to the packet transmission time.
- Every packet transmitted must fit into one of these slots by beginning and ending in precise synchronization with the slot segments.
- A packet arriving to be transmitted at any given station must be delayed until the beginning of the next slot.
- In contrast, for pure ALOHA, a packet transmission can begin at any time.

Carrier Sense Multiple Access

- Protocols in which the station senses the channel before starting transmission are in the category of CSMA protocols (also known as listen-before-talk protocols).
- CSMA has many variants available that are to be adapted according to the behavior of the station that has frames to be transmitted when the channel is busy or when some transmission is going on.
- Following are some versions of CSMA protocols:
    - 1-Persistent CSMA
    - Non-Persistent CSMA
    - p-Persistent CSMA

1-Persistent CSMA

- 1-persistent CSMA is a greedy protocol to capture the channel as soon as it finds it idle.
- Hence, it has a high frequency of collision in the system.
- In case of collision, the station senses the channel again after a random delay.
- In this protocol a station i.e., who wants to transmit some frame will sense the channel first, if it is found busy that some transmission is going on the medium, then, this station will continuously keep sensing that the channel.

Non-Persistent CSMA

- To reduce the frequency of the occurrence of collision in the system then, another version of CSMA that is non-persistent CSMA can be used.
- The station who has frames to transmit first sense whether the channel is busy or free.
- If the station finds that channel to be free it simply transmits its frame.
- Otherwise, it will wait for a random amount of time and repeat the process after that time span is over.
- As it does not continuously senses the channel to be, it is less greedy in comparison of 1-Persistent CSMA.

p-Persistent CSMA

- This category of CSMA combines features of the above versions of CSMA that is 1-persistent CSMA and non-persistent CSMA.
- This version is applicable for the slotted channel.
- The station that has frames to transmit, senses the channel and if found free then simply transmits the frame with p probability and with probability 1-p it, defers the process.
- If the channel is found busy then, the station persists sensing the channel until it became idle.
- Here value of p is the controlling parameter.

Csma/cd

- CSMA /CD or Carrier Sense Multiple Access/Collision Detect.
- In CSMA/CD, stations ensure that the transmission will start only when it finds that the channel is idle.
- In CSMA/CD the station aborts the process of transmission as soon as they detect some collision in the system.
- If two stations sense that the channel is free at the same time, then, both start transmission process immediately.
- And after that, both stations get information that collision has occurred in the system.
- This protocol is known as CSMA/CD and, this scheme is commonly used in LANs.

Csma/cd

- Let, t be the maximum transmission time between two extreme ends of a network system (LAN).
- At t^0 station A, at one extreme end of the LAN begins the process of transmitting a frame F^A.
- This frame reaches the station E which at another extreme end of the same network system in t propagation delay away.
- If no other station in between has started its frame transmission, it implies that A has captured the channel successfully.
- But, in case E_F station E starts its frame transmission just before the arrival of frame from station A frame then, collision will take place.
- Station A will get the signal of collision after 2t time. Hence, 2t time is required to ensure that station A has captured the channel successfully.

Csma/cd

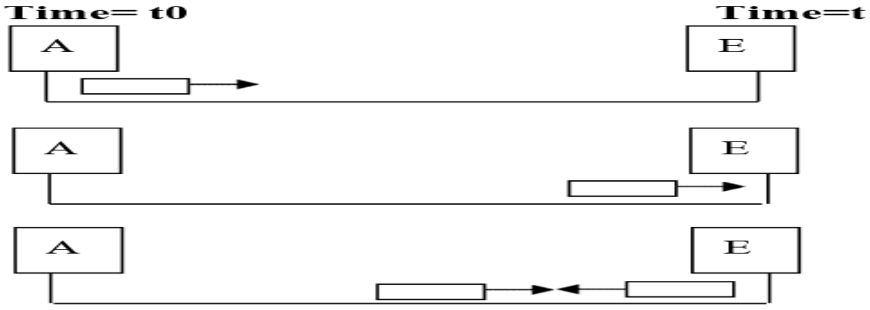


Figure 8: Collision detection

- Collision of frames will be detected by looking at the strength of electric pulse or signal received after collision.
- After a station detects a collision, it aborts the transmission process and waits for some random amount of time and tries the transmission again with the assumption that no other station has started its transmission in the interval of propagation time.

Csma/cd

- The channel in CSMA/CD can be any of the following three states.
- Transmission of frame is going on.
- Idle slot.
- Contention period/slot.

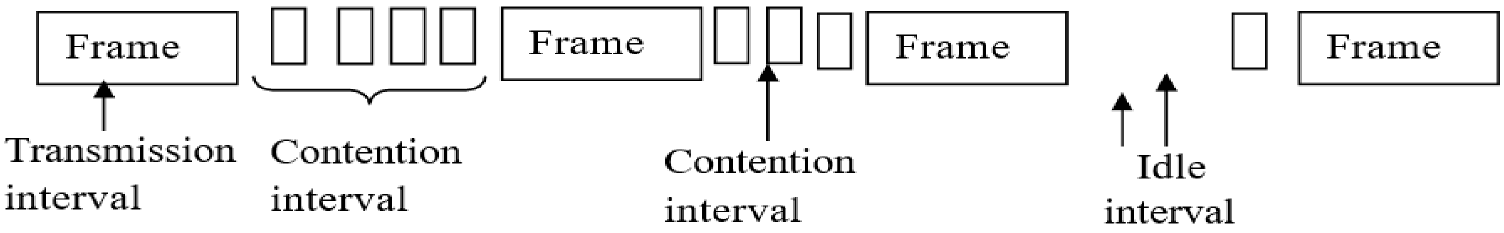
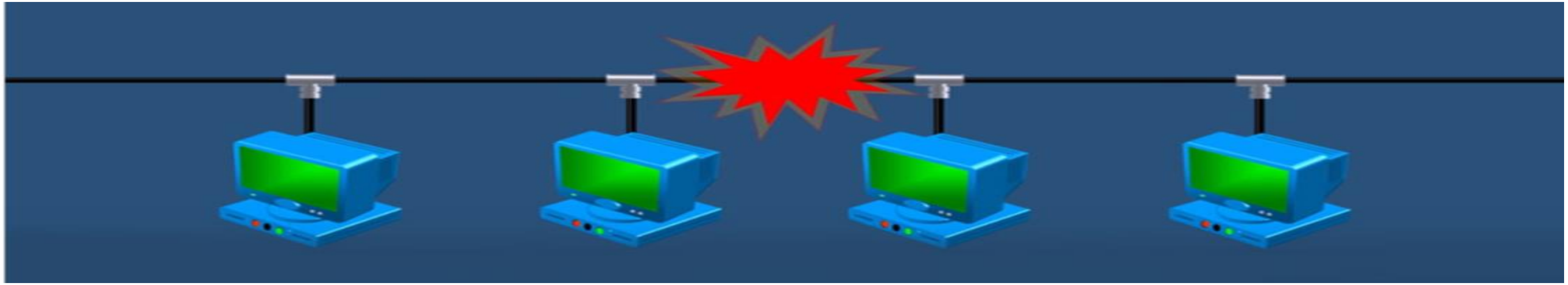


Figure 9: Transmission states

Csma/cd

- In CSMA/CD a station with a frame ready to begin transmission senses the channel and starts transmission if it finds that the channel is idle.
- Otherwise, if it finds it busy, the station can persist and use backoff algorithm which will be discussed in the next paragraph



www.propatel.com/difference-csma-cd-csma-ca/

Backoff Algorithm

- Time is divided into discrete slots with the length of the worst-case propagation time (propagation time between two extreme ends of LAN) 2t.
- After the first collision in the system, each station waits for 0 or 1 slot time before trying transmission for the next time.
- If two stations that collide are selected with the same random number then the collision will be repeated.
- After the second collision, the station will select 0,1,2, or 3 randomly and wait for these many slots.
- If, the process of collision will occur repeatedly, then, the random number interval would be between 0 and 2^i-1 for i^{th} collision and this number of slots will be the waiting time for the station.

Ethernet Frame Format (IEEE 802.3)

- Ethernet protocol is a MAC sublayer protocol.
- Ethernet stands for cable and IEEE 802.3 Ethernet protocol was designed to operate at 10 Mbps.
- Here, is the Ethernet cabling used for baseband 802.3 LANs.

Characteristic\Cable	10Base5	10Base2	10BaseT	10BaseF
Medium	Thick Coaxial Cable	Thin Coaxial Cable	Twisted Pair	Optical Fiber
Maximum Length of segment	500 m	200 m	100 m	2 Km
Topology	Bus	Bus	Star	Star
Advantages	Used for connecting workstation with tap on the cable	Low cost	Existing environment can use Hub and connect the stations	Good noise immunity and good to use

Characteristics Ethernet cable

MAC Frame Structure for IEEE 802.3

- Each frame has seven fields explained as:
- Preamble, Starting Delimiter (SD), Destination Address (DA),
- Source Address (SA), Length of Data Field,
- Data, Pad, Frame Checksum (FCS).

Preamble	Start Delimiter of frame	Destination Address	Source Address	Length of Data Field	Data	Pad	Frame Check Sum

Figure 10: Ethernet Frame Format

MAC Frame Structure for IEEE 802.3

- **Preamble:** The first field of 802.3 frame is 7 byte (56 bits) long with a sequence of alternate 1 and 0 i.e., 10101010.
- This pattern helps the receiver to synchronize and get the beginning of the frame.
- **Starting Delimiter (SD):** The second field start delimiter is 1 byte (8 bit) long. It has pattern 10101011.
- Again, it is to indicate the beginning of the frame and ensure that the next field will be a destination address.
- **Destination Address (DA):** This field is 6 byte (48 bit) long. It contains the physical address of the receiver. T T

MAC Frame Structure for IEEE 802.3

- **Source Address (SA):** This filed is also 6 byte (48 bit) long. It contains the physical address of the sender.
- **Length of Data Field:** It is 2 byte (16 bit) long. It indicates the number of bytes in the information field. The longest allowable value can be 1518 bytes.
- **Data:** This field size will be a minimum of 46 bytes long and a maximum of 1500 bytes as will be explained later.
- **Pad:** This field size can be 0 to 46 bytes long. This is required if, the data size is less than 46 bytes as a 802.3 frame must be at least 64 bytes long.
- **Frame Checksum (FCS):** This field is 4 bytes (32 bit) long. It contains information about error detection. Here it is CRC-32.

Minimum and Maximum Length of Frame

- Minimum frame length = 64 bytes = 512 bits.
- Maximum frame length= 1518 bytes = 12144 bits.
- Minimum length or lower limit for frame length is defined for normal operation of CSMA/CD.
- This is required so that, the entire frame is not transmitted completely before its first bit has been received by the receiver.
- If, this happens then the probability of the occurrence of collision will be high (the same has been explained earlier in the previous section CSMA/CD).

Minimum and Maximum Length of Frame

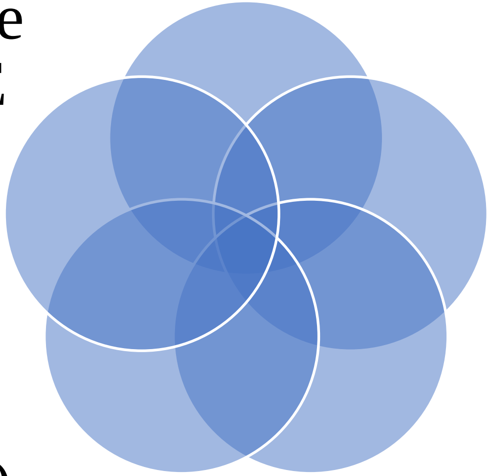
- Hence, Ethernet frame must be of 64 bytes long.
- Some of the bytes are header and trailer parts of the frame.
- If, we consider 6 bytes destination address, 6 bytes source address, 2 bytes length and 4 bytes FCS (6+6+2+4=18) then, the minimum length of data will be 64-18= 46 bytes.
- If, frame is less than 46 bytes then, padding bits fill up this difference.
- As per 802.3 standard, the frames maximum length or upper limit of frame

  is = 1518 bytes (excluding preamble and SD).
- If we subtract 18 bytes of header and trailer then, the maximum length will be 1500 bytes.

Important Topics

Pure Aloha

Ethernet Frame Format (IEEE 802.3)



Slotted Aloha

CSMA with Collision Detection (CSMA/CD)

Carrier Sense Multiple Access.

Summary

- In this session, we discuss that in some networks, if a single channel and many users use that channel, then, allocation strategy is required for the channel.
- Then, We discussed FDM and TDM allocation method.
- They are the simplest methods for allocation.
- They work efficiently for a small number of user. For a large number of users the ALOHA protocol is considered.
- There are two versions of ALOHA that is Pure ALOHA and Slotted ALOHA.
- In Pure ALOHA no slotting was done but the efficiency was poor.

Summary

- In Slotted ALOHA, slots have been made, so that every frame transmission starts at the beginning of the slot and throughput is increased by a factor of 2.
- For avoiding collision and to increase efficiency in sensing the channel, CSMA is used. Many versions of CSMA are persistent and non-persistent.
- In CSMA/CD collision detection process is added so that process can be aborted just after a collision is detected.
- Ethernet is a commonly used protocol for LAN.
- IEEE 802.3 Ethernet uses 1 persistent CSMA/CD access method.

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Data Communication and Computer Networks

Block-2 Unit-4

Wireless LAN and Data Link Layer Switching

Topics to be Covered

- Introduction
- Introduction to Wireless LAN
- Wireless LAN Architecture (IEEE 802.11)
- Hidden Station and Exposed Station Problems
- Wireless LAN Protocols: MACA and MACAW Flow Control
- IEEE 802.11 Protocol Stack
- The 802.11 Physical Layer
- The 802.11 MAC Sub-layer Protocol

Topics to be Covered

- Switching at Data Link Layer
- Operation of Bridges in Different LAN Environment
- Transparent Bridges
- Spanning Tree Bridges
- Source Routing Bridges
- Important Topics
- Summary

Introduction

- "Wireless Local Area Network" stands for A WLAN, or wireless LAN, is a network that allows devices to connect and communicate wirelessly.
- Unlike a traditional wired LAN, in which devices communicate over Ethernet cables, devices on a WLAN communicate via Wi-Fi.

marketbusinessnews.com/financial-glossary/wireless-lan-wlan-definition-meaning/

Introduction to Wireless LAN

- Now a days, Wireless LANs are taking a important position due to coverage of location difficult to wire, to satisfy the requirement of mobility and ad-hoc networking.
- In a few years from now, we can notice a broader range of wireless devices accessing the Internet, such as digital cameras, automobiles, security systems, and kitchen appliances.
- As KUROSE and ROSS write that some day wireless devices that communicate with the Internet may be present everywhere even on walls, in cars, in bedrooms, in our pockets, and in human bodies.
- Wireless LANs are commonly being used in academic institutions, companies, hospitals and homes to access the internet and the LAN server while on roaming.

Introduction to Wireless LAN

- As the number of mobile computing and communication devices grows, so does the demand to connect them to the outside world.
- Each cell (called basic service set or BSS, in the 802.11) is controlled by a base station (called access point or AP).
- Wireless LAN may be formed by a single cell, with a single access point (it can also work within an AP), most stations will be formed by several cells, where the APs are connected through some kind of backbone (called distribution system or DS).
- This backbone may be the Ethernet and in some cases, it can be the wireless system.

Wireless LAN Architecture (IEEE 802.11)

- The wireless LAN is based on a cellular architecture where the system is subdivided into cells as we can see in Figure.
- Each cell (called basic service set or BSS, in the 802.11) is controlled by a base station (called access point or AP).
- Wireless LAN may be formed by a single cell, with a single access point (it can also work within an AP).
- Most stations will be formed by several cells.

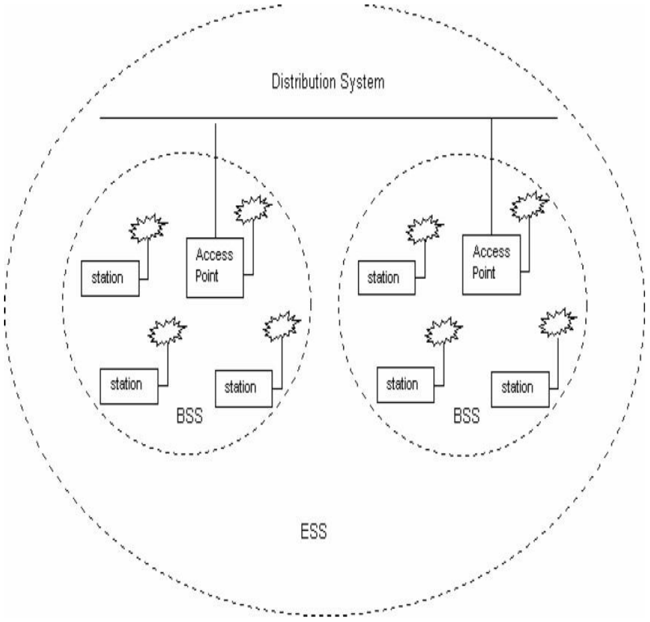
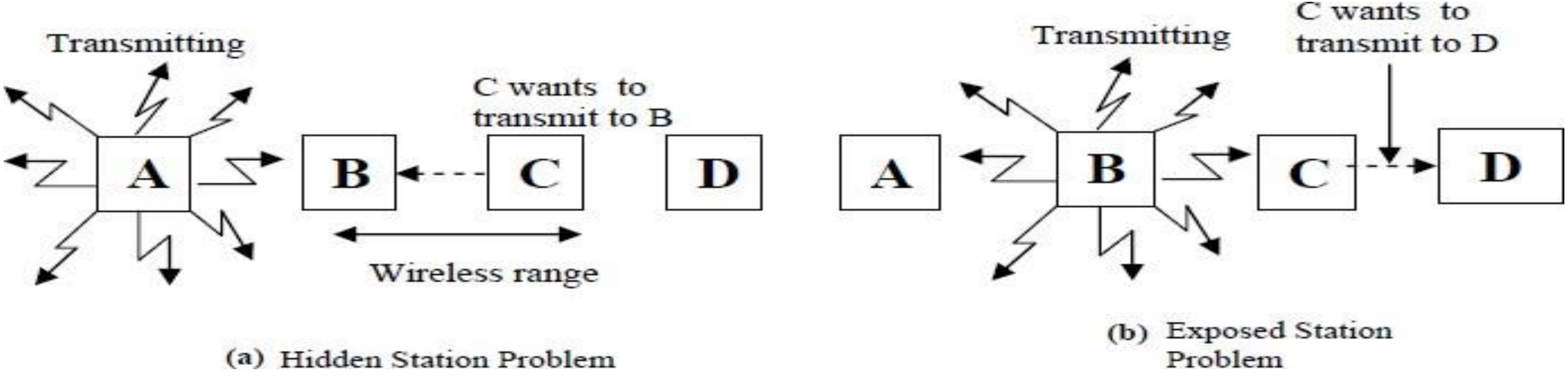


Figure 1: Wireless LAN architecture

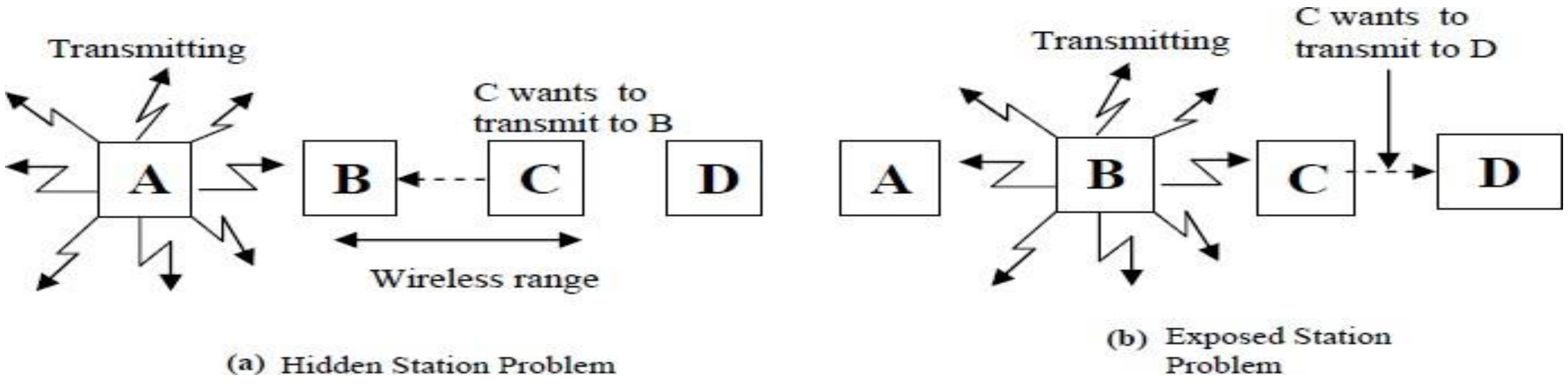
Hidden Station And Exposed Station Problems

- Assume that there are four nodes A, B, C and D. B and C are in the radio range of each other.
- Similarly A and B are in the radio range of each other. But C is not in the radio range of A.



Hidden Station And Exposed Station Problems

- In this case, B is transmitting to A. Both are within radio range of each other. Now C wants to transmit to D.



Wireless LAN Protocols: MACA AND MACAW

- MACA is the oldest protocol.
- MACA was proposed as an alternative to CSMA Protocol which has certain drawbacks.
- It senses the channel to see if the channel is free, it transmits a packet, otherwise it waits for a random amount of time.
- Hidden Station Problems leads to frequent collision.
- Generally, Exposed terminal problems leads to worse bandwidth utilization.

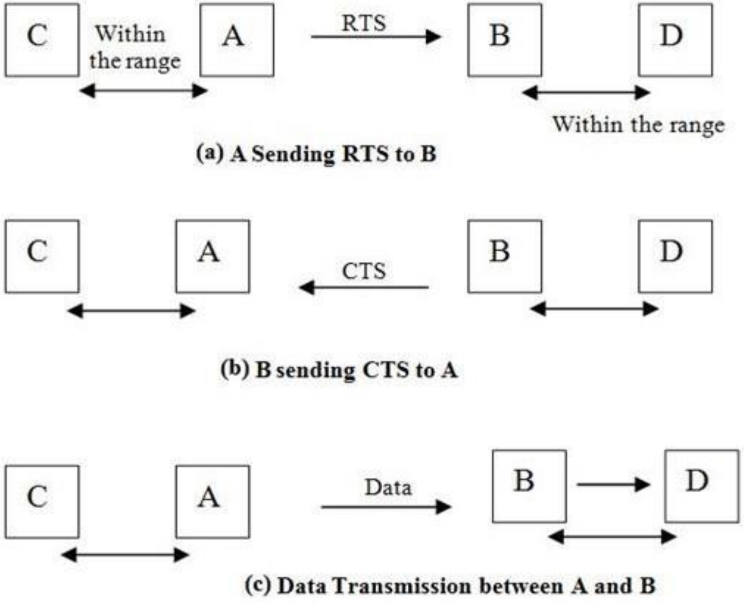
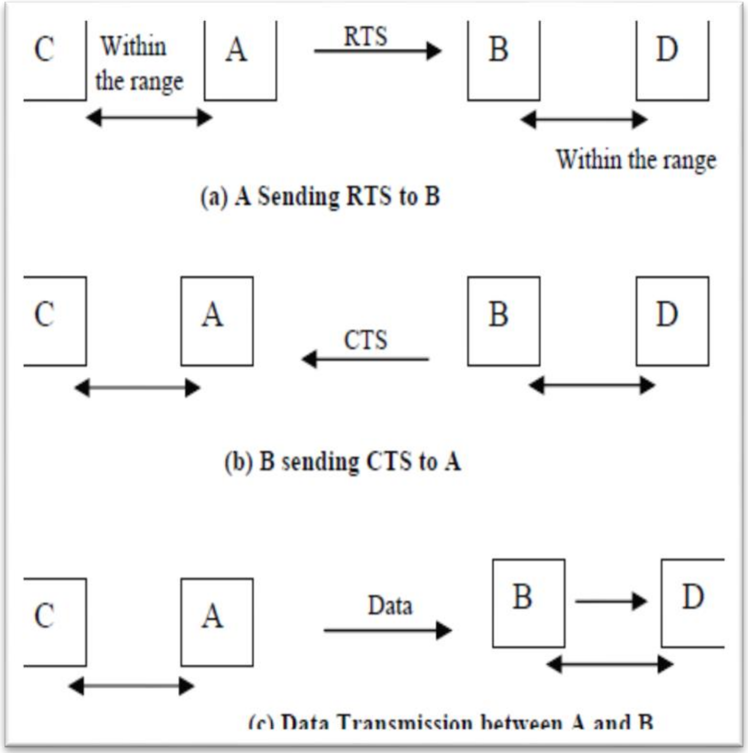


Figure 3: MACA Protocol

Maca Protocol

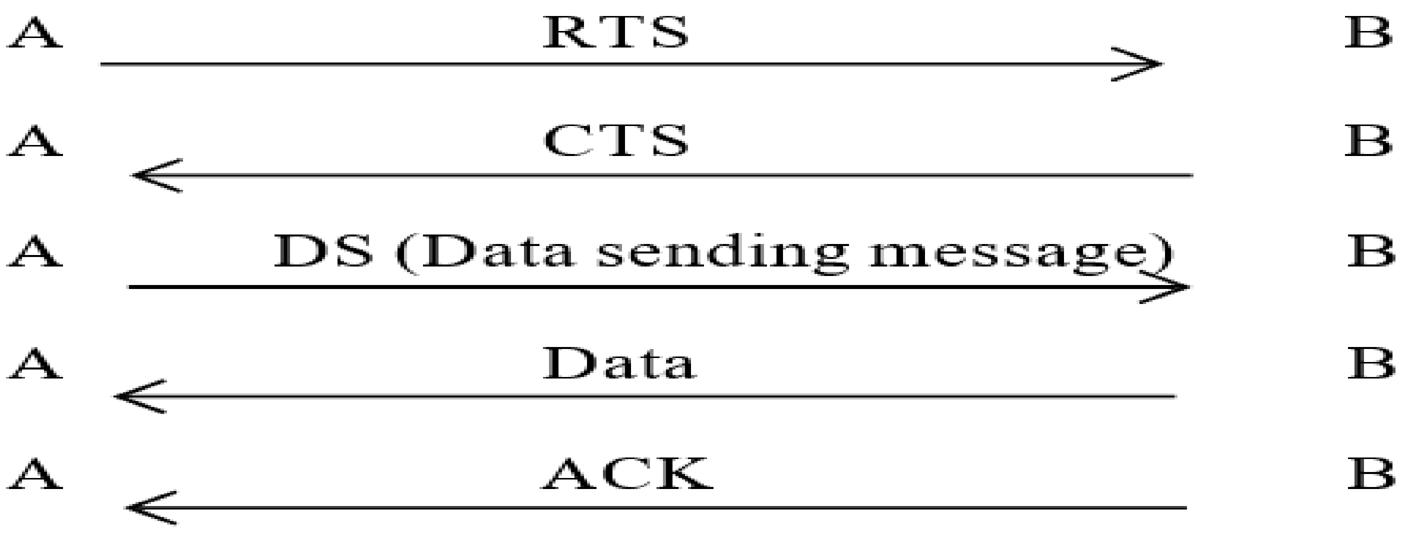
- It avoids some cases of hidden and exposed station problems.
- However, it does not always avoid these problems using RTS (Repeat to Send) and CTS (Clear to Send) handshake mechanism.
- All nodes near the sender/receiver after hearing RTS/CTS will defer the transmission
- If, the neighbor hears the RTS only, it is free to transmit if waiting interval is over.



Maca Protocol

- Despite these precautions, collisions can still occur.
- As an example, B and C could both send RTS frames to A at the same time.
- These frames will collide and will be lost.
- In the event of a collision, an unsuccessful transmitter (i.e. one that does not hear a CTS within the expected time interval) waits a random amount of time and tries again later.
- MACAW (MACA for wireless) extends MACA to improve its performance which will have the following handshaking mechanism.
- RTS-CTS-DS-Data ACK.
- It is also been explained through the diagram.

Maca Protocol



The MACAW protocol

Macaw Extends Maca

- **Data link layer acknowledgement:** Here, Data link layer acknowledgements is very important for the retransmission of lost frames retransition whenever transport layer noticed their absences.
- They solved this problem by introducing an ACK frame after each successful data frame.
- **Addition of carrier sensing:** It was observed that CSMA has some use, namely, to keep a station from transmitting a RTS while at the same time another nearby station is also doing so to the same destination, so carrier sensing was added.
- **An improved backoff mechanism:** It runs the backoff algorithm separately for each data stream in the form of source-destination pair).

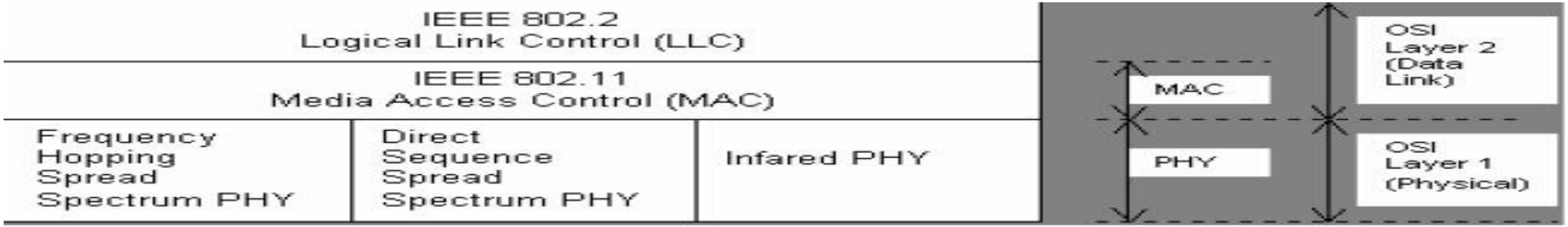
Macaw Extends Maca

- **DS (Data sending) message:** Say a neighbor of the sender overhears an RTS but not a CTS from a receiver.
- In this case it can tell if RTS-CTS was successful or not.
- When it overhears DS, it realizes that the RTS-CTS was successful and it defers its own transmission.

www.semanticscholar.org/paper/RTS-CTS-Framework-Paradigm-and-WLAN-QoS-Methods-Nj.-Sahib/0db39bfaa0eda2a502aa81aa9877f4bdc0938d70

IEEE 802.11 Protocol Stack

- IEEE 802.11 standard for wireless LAN is similar in most respects to the IEEE 802.3 Ethernet standard addresses.
- 802.11 corresponds to the OSI physical layer fairly well but, the data link layer in all the 802 protocols is split into two or more sub-layers.
- In 802.11, the MAC (Medium Access Control) sub-layer determines how the channel is allocated, that is.
- Above it is the LLC (Logical Link Control) sublayer, whose job is to hide the differences between the different 802 variants



The 802.11 Physical Layer

- The 802.11 physical layer (PHY) standard specifies three transmission techniques allowed in the physical layer.
- The infrared method used much the same technology as television remote controls do.
- The other two use short-range radio frequency (RF) using techniques known as FHSS (Frequency Hopped Spread Spectrum) and DSSS (Direct Sequence Spread Spectrum).
- Both these techniques, use a part of the spectrum that does not require licensing (the 2.4 –GHz ISM band).
- Radio-controlled garage door openers also use the same piece of the spectrum, so your notebook computer may find itself in competition with your garage door.

The 802.11 Physical Layer

- Cordless telephones and microwave ovens also use this band.
- All of these techniques operate at 1 or 2 Mbps and at low enough power that they do not conflict too much.
- RF is capable of being used for ‘not line of sight’ and longer distances.
- The other two short range radio frequency techniques are known as spread spectrum.
- It was initially developed for military and intelligence requirement.
- Both these techniques are used in wireless data network products as well as other communication application such as, a cordless telephone

The 802.11 Physical Layer

- There are two types of spread spectrum:
- Frequency Hopping (FH), and Direct Sequence (DS).
- DSSS(Direct Sequence Spread Spectrum) divides the 2.4 GHz band into 14 channels. Channels used at the same location should be separated 25 MHz from each other to avoid interference.
- FHSS(Frequency Hopped Spread Spectrum) and DHSS are fundamentally different signaling techniques and are not capable of interoperating with each other.
- Under this scheme, each bit in the original signal is represented by multiple bits in the transmitted signal, which is known as a chipping code.

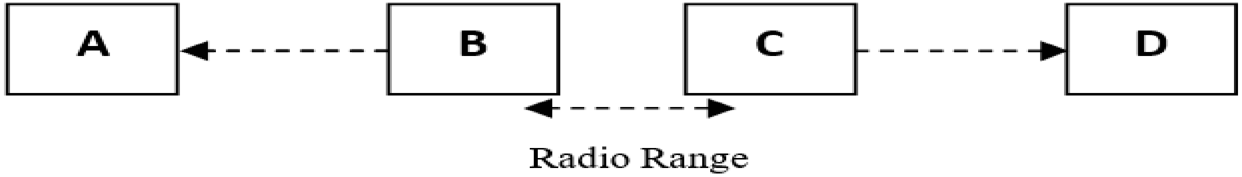
The 802.11 MAC Sub-layer Protocol

- MAC sub layer protocols which are quite different from that of the Ethernet due, to the inherent complexity of the wireless environment compared to that of a wired system.
- With Ethernet (IEEE 802.3) a node transmits, in case, it has sensed that the channel is free. If, it does not receive a noise burst back within the first 64 bytes, the frame has almost assuredly been delivered correctly. With wireless technology, this situation does not hold.

The hidden station problem

The Exposed Station Problem

- Most radios are half duplex, meaning that they cannot transmit and listen for noise bursts at the same time on a single frequency as Ethernet does. To deal with this problem, 802.11 supports two modes of operation.
  - DCF (Distributed Coordination Function)
  - PCF (Point Coordinated Function) (optional)



B Transmits to **A** which is heard by **C**
C unnecessarily avoids sending a message to **D** even though there would be no collision.

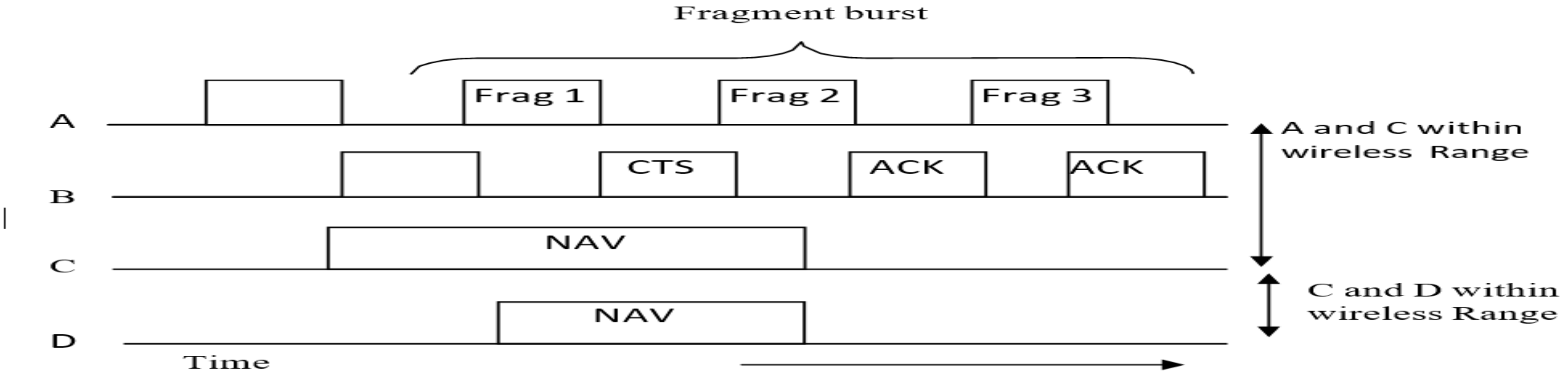
The exposed station problem

Virtual channel sensing using CSMA/CA

- When DCF is employed, 802.11 uses a protocol called CSMA/CA (CSMA with Collision Avoidance).
- In this protocol, both **physical channel sensing** and **virtual channel sensing** are used.
- Two methods of operation are supported by CSMA/CA. In the first method (Physical sensing), before the transmission, it senses the channel.
- If the channel is sensed idle, it just wants and then transmitting.
- But it does not sense the channel while transmitting but, emits its entire frame, which may well be destroyed at the receiver's end due to interference there.

Fragment Burst

- Once the channel has been acquired using RTS and CTS, multiple fragments can be sent in a row.
- A sequence of fragments is called a **fragment burst.**



A fragment burst

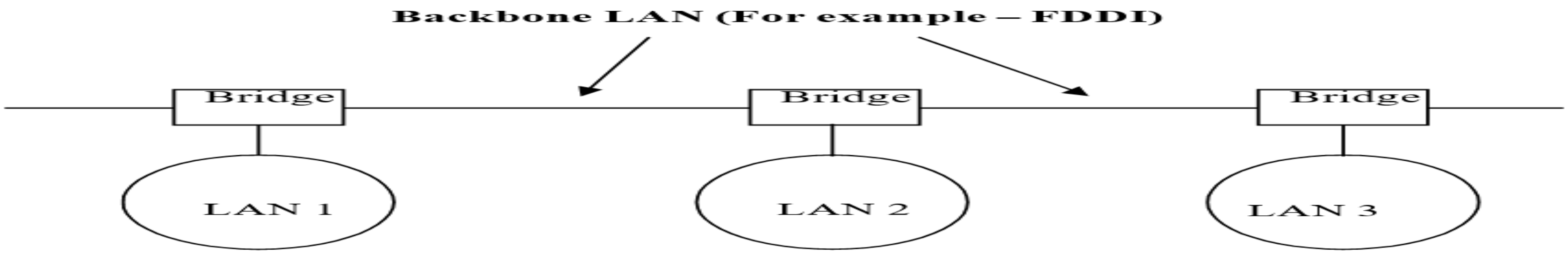
Switching at Data Link Layer

- Repeaters which are layer 1 devices, provide both physical and electrical connections.
- Their functions are to regenerating and propagate a signal in a channel.
- Repeaters are used to extend the length of the LAN which depends upon the type of medium.
- Many organizations have multiple LANs and wish to connect them.
- LANs can be connected by devices called **bridges,** which operate as the data link layer.
- Unlike repeaters, bridges connect networks that have different physical layers.
- Characteristics of Bridges are :
- Store and forward device and highly susceptible to broadcast storms.

Technology of Bridges

Reasons why a single organization may end up with multiple LANs.

- **Multiple LANs in organization:** Many university and corporate departments have their own LANs, primarily to connect their own personal computers, workstations, and servers.
- So, there is a need for interaction, so bridges are needed.



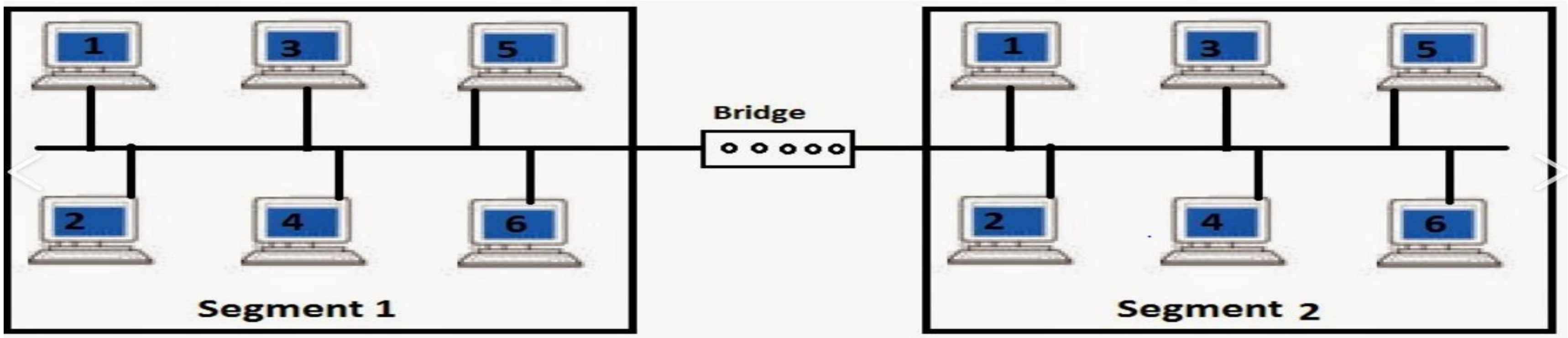
Multiple LANs connected by a backbone to handle a total load higher than the capacity of single LAN

Technology of Bridges

- **Geographical difference:** The organization may be geographically spread over several buildings separated by considerable distances.
- It may be cheaper to have separate LANs in each building and connect them with bridges and later link them to run a single cable over the entire site.
- **Load distribution**: It may be necessary to split what is logically a single LAN into separate LANs to accommodate the load.
- **Long Round Trip delay:** In some situations, a single LAN would be adequate in terms of the load, but the physical distance between the most distant machines is can be great (e.g. more than 2.5 km for Ethernet).
- The only solution is to partition the LAN and install bridges between the segments. Using bridges, the total physical distance covered can be increased.

Technology of Bridges

- **Reliability:** By inserting bridges at critical points reliability can be enforced in the network by isolating a defective node.
- Unlike a repeater, a bridge can be programmed to exercise some discretion regarding what forwards and what it does not forward.
- **Security** is a very important feature in bridges today, and can control the movement of sensitive traffic by isolating the different parts of the network.



Bridge with Characters

- In an ideal sense, a bridge should have the following characters:
  - **Fully transparent:** It means that it should allow the movement of a machine from one cable segment to another cable segment without change of hardware and software or configuration tools.
  - **Interpretability:** It should allow a machine on one LAN segment to talk to another machine on another LAN segment.

https://i.ytimg.com/vi/UX5NTWjLcwQ/maxresdefault.jpg

Operation of Bridges in Different LAN Environment

- LANs, Assume that, there are two machines A and B. Both of them are attached to different LANs. Machine A is on a wireless LAN (Ethernet IEEE 802.3).
- While B is on both LANs are connected through a bridge. A has a packet to be sent to B. The flow of information at Host A is :
- The packet at machine A arrives at the LLC sublayer from the application layer through the transport layer and network layer.
- LLC Sublayer header gets attached to the packet.
- Then it moves to the MAC Sublayer. The packet gets the MAC sublayer header for attachment.
- Since the node is part of a wireless LAN, the packet goes to the air using GRF.

Operation of Bridges in Different LAN Environment

- The packet is then picked up by the base station. It examines its destination address and figures that it should be forwarded to the fixed LAN (it is Ethernet in our case).
- When the packet arrives at the bridge which connects the wireless LAN and
- Ethernet LAN, it starts at the physical layer of the bridge and moves to its LLC layer. At the MAC sublayer its 802.11 header is removed.
- The packet arrives at the LLC of a bridge without any 802.11 header.
- Since the packet has to go to 802.3 LAN, the bridge prepares packets accordingly.

Note that a bridge connecting k different LANs will have K different MAC sublayers and k different physical layers. One for each type.

Difficulties Encounters while Building a Bridge

- **Different frame Format:** To start with, each of the LANs uses a different frame format.
- Unlike the differences between Ethernet, token bus, and token ring, which were due to history and big corporate egos, here the differences are to some extent legitimate.
- For example, the Duration field in 802.11 is there, due to the MACAW protocol and that makes no sense in Ethernet.
- As a result, any copying between different LANs requires reformatting, which takes CPU time, requires a new checksum calculation, and introduces the possibility of undetected errors due to bad bits in the bridge's memory.

Difficulties Encounters while Building a Bridge

- **Different data rates**: When forwarding a frame from a fast LAN to a slower one, the bridge will not be able to get rid of the frames as fast as they come in. Therefore, it has to be buffered.
- **Different frame lengths:** An obvious problem arises when a long frame must be forwarded onto a LAN that cannot accept it. This is the most serious problem.
- The problem comes when a long frames arrives. An obvious solution is that the frame must be split but, such a facility is not available at the data link layer.
- Therefore the solution is that such frames must be discarded.
- Basically, there is no solution to this problem.

Difficulties Encounters while Building a Bridge

- **Security:** Both 802.11 and 802.16 support encryption in the data link layer, but the Ethernet does not do so.
- This means that the various encryption services available to the wireless networks are lost when traffic passes over the Ethernet.
- **Quality of Service:** The Ethernet has no concept of quality of service, so traffic from other LANs will lose its quality of service when passing over an Ethernet.

www.learnabhi.com/bridge-features-advantages-disadvantages/

Transparent Bridges

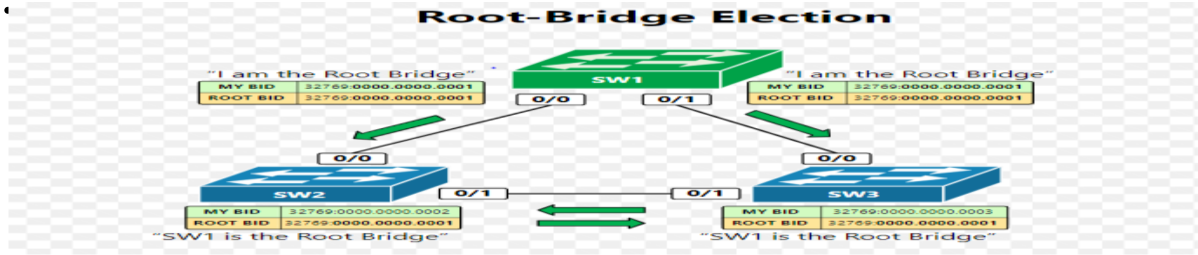
- In, the previous section, we dealt with the problems encount-ered in connecting two different IEEE 802 LANs via a single bridge.
- However, in large organizations with many LANs, just interconnecting them all raises a variety of issues, even if they are all just Ethernet.
- In this section, we introduce a type of bridge called Transparent Bridge, which is a plug and play unit which you connect to your network and switch it on.
- There is no requirement of hardware and software changes, no setting of address switches , no downloading of routing tables, just plug and play.

Configuration with four LANs and Two Bridges

- For reliability, some networks contain more than one bridge, which increases the likelihood of networking loops.
- A networking loop occurs when frames are passed from bridge to bridge in a circular manner, never reaching its destination.
- To prevent networking loops when multiple bridges are used, the bridges communicate with each other and establish a map of the network to derive what is called a spanning tree for all the networks.
- A spanning tree consists of a single path between source and destination nodes that does not include any loops.
- Thus, a spanning tree can be considered to be a loop-free subset of a network's topology. The spanning tree algorithm, specified in IEEE 802.1d, describes how bridges (and switches) can communicate to avoid network loops.

Spanning Tree Bridges

- For reliability, some networks contain more than one bridge, which increases the likelihood of networking loops.
- A networking loop occurs when frames are passed from bridge to bridge in a circular manner, never reaching its destination.
- To prevent networking loops when multiple bridges are used, the bridges communicate with each other.
- It establish a map of the network to derive what is called a spanning tree for all the networks.



www.networkacademy.io/ccna/spanning-tree/how-stp-works

Source Routing Bridges

- IBM introduced source routing bridges for use in Token Ring networks.
- With source routing, the sending machine is responsible for determining whether, a frame is destined for a node on the same network or on a different network.
- If, the frame is destined for a different network, then, the source machine designates this by setting the high-order bit of the group address bit of the source address to 1.
- It also includes in the frame's header the path the frame is to follow from source to destination.

https://image3.slideserve.com/5652876/source-route-bridging6-l.jpg

Important Topics

- Wireless LAN Architecture
- Wireless LAN Protocols MACA and MACAW
- IEEE 802.11 Protocol Stack
- Operation of Bridges in different LAN Environment

Summary

- In this Session, we discussed two major topics wireless LANs and switching mechanism at the data link layer with IEEE at the data link layer.
- With IEEE 802.11 standardization, wireless LANs are becoming common in most of the organizations but, they have their own problems and solutions CSMA/CD does not work due to hidden station problem.
- To make CSMA work better two new protocols, MACA and MACAW were discussed.
- The physical layer of wireless LAN standard i.e. IEEE 802.11 allows five different transmission modes, including infrared, various spread spectrum schemes etc.

Thank you

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Data Communication and Computer Networks

Block-3 Unit-1_Part2

Introduction to Layer Functionality and Design Issues

Topics to be Covered

- Introduction
- Services of the Network Layer
- Packet Switching
- Virtual Circuit Approach (Connection Oriented Service)
- Datagram Approach (Connectionless Service)
- Comparison of Virtual Circuit and Datagram Approach
- View of some network service models
- Network Addressing
- IP Addressing

Topics to be Covered

- Hierarchy in addressing
- Getting an IP Address
- Congestion
- Routing
- Classification of Routing Algorithm
- Delay in Packet Switched Network
- Types of delay
- Numerical
- Important Topics
- Summary

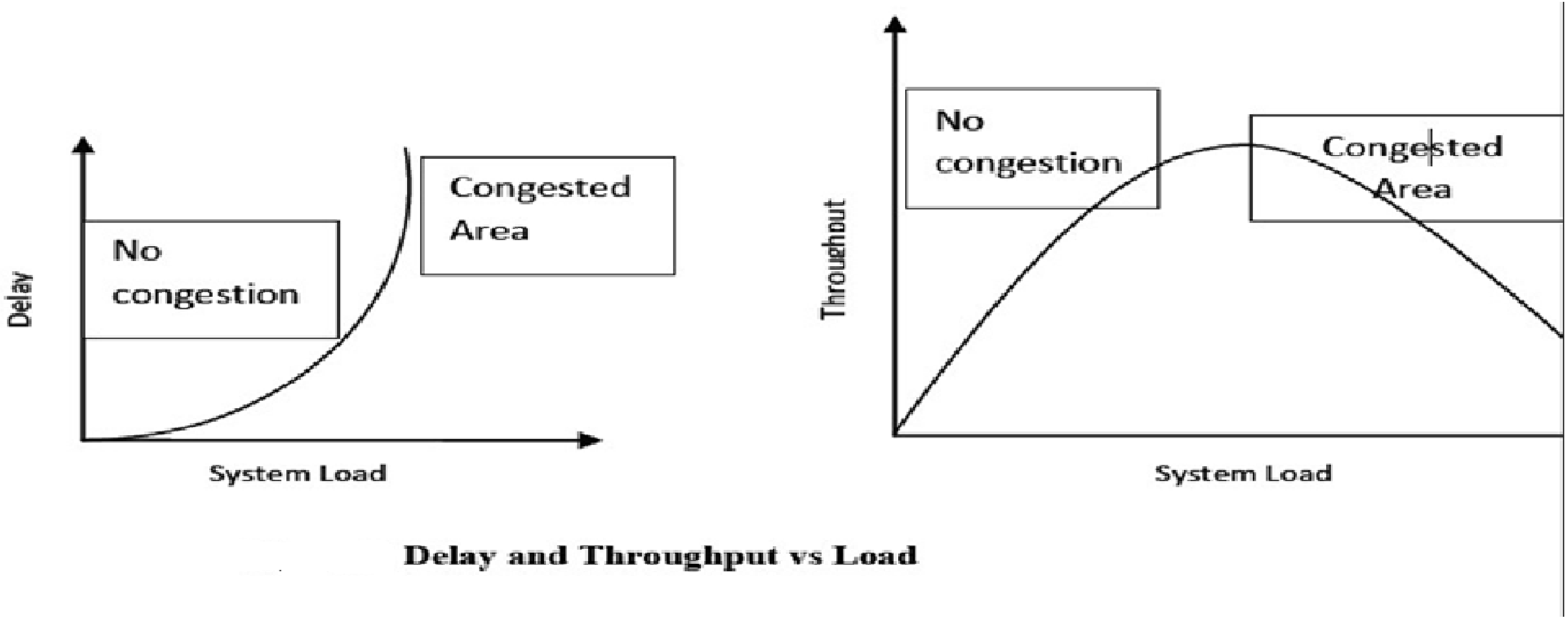
Introduction

- As we are aware about the network layer that the network connections are make possible at this layer
- It is part of the Internet communication process where connections occur and packets are sent between network to network.
- Sometimes, congestion is the temporary issue with network.
- As because of this reason, sometimes network is continuously congested and may cause problems.
- For the solution of this problem, Routing Algorithms are made and are implemented.

Congestion

- If the packets are coming at a faster rate than the handling capacity of the network, this leads to a situation known as congestion.
- Initially, when the packet arrival rate starts getting higher than the packet processing rate, queue starts filling up.
- If the same situation continues, queue becomes full and packet drop starts.
- In this situation, source does not receive any acknowledgement and for a large number of the packets, the timer is up; which leads to unnecessary retransmissions.
- Sometimes the situation becomes worse and reaches to a deadlock point and whole system gets collapsed.

Delay And Throughput Vs. Load



Delay and Throughput vs Load

Open Loop Congestion Control Policies

- **Admission policy:** A router can visualize the possibility of congestion and if there are chances of congestion then the new virtual connection request can be rejected.
- **Retransmission policy:** When the sender does not receive any acknowledgement of the sent packet, it does the retransmission.
- For how long the sender has to wait or after how many lost packets, the packet needs to be retransmitted.
- All these kinds of retransmission policies should be designed in such a way that it will not add more congestion in the network.

Open Loop Congestion Control Policies

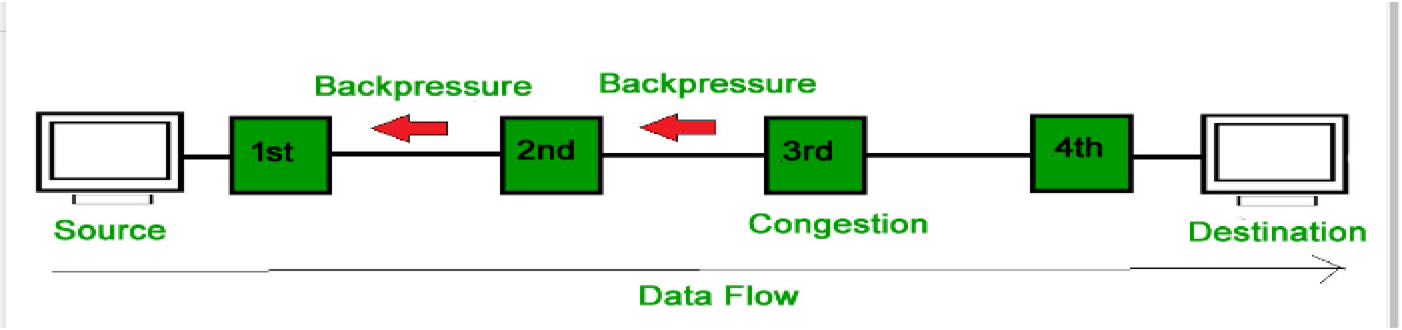
- **Acknowledgement policy:** Receiver's acknowledgement policy can also control the congestion at some level.
- As an example, if receiver sends the acknowledgement packet after receiving some packets, it will slow the sender as well as not add a burden of sending acknowledgement packets.
- **Discard Policy:** If a router implements the good discarding policy then it can also prevent congestion at some level.
- The good discarding policy which does not impact the overall quality of transmission.

Closed Loop Congestion Control Policies

- **Sending of Choke packet:** A choke packet is a control packet sent by the router to the source node. This method is implemented by protocols like ICMP, etc.
- **Signaling:** In this method, a signal would be sent by the congested node to inform the sender about congestion.
- **Implicit signaling**: In this method, no specific information is sent to the source rather sender itself guesses about congestion.
- As an example, if the sender does not receive any acknowledgements is a signal for the sender that there is congestion in the network.

Closed Loop Congestion Control Policies

- **Implicit signaling**: In this method, no specific information is sent to the source rather sender itself guesses about congestion.
- For example, if the sender does not receive any acknowledgements of several packets with in timeout period is a signal for the sender that there is congestion in the network.



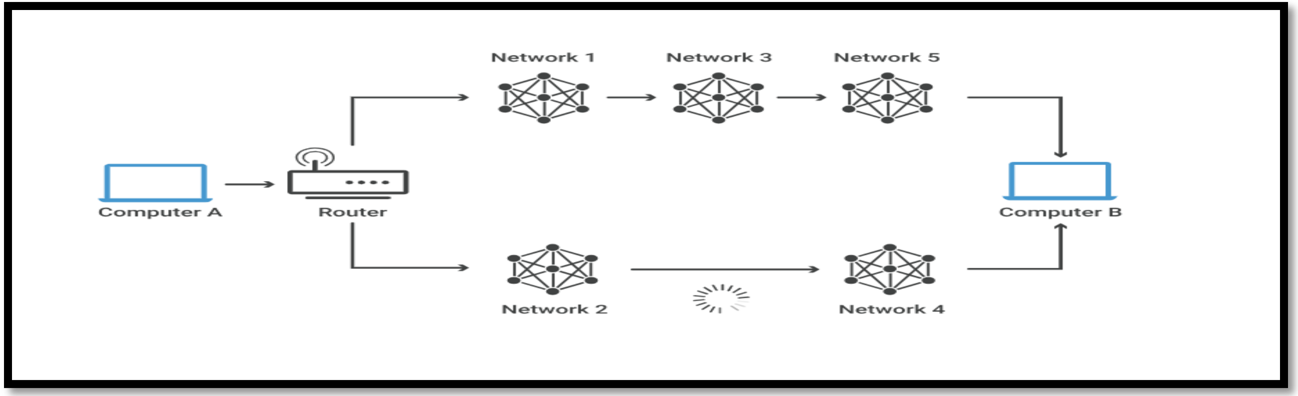
Routing

- At Network layer data may travel through different paths or through multiple hops to reach to the destination.
- The process of deciding about the path to reach to the destination is known as routing.
- There are routing protocols or which helps in constructing the forwarding table.
- The forwarding table or routing table is stored at every end system and router. Whenever router receives a packet, it consults its routing table to decide the output interface. this process is known as forwarding.

Routing

- Filling up of routing tables, their maintenance and regular updation is done at continuous intervals by routing protocols or routing algorithms.

- Routing algorithms is a part of network layer software.



www.cloudflare.com/learning/network-layer/what-is-routing/

Properties of Routing Algorithm

- Routing protocols decide the best route from a source S to destination D.
- This best route can be best in terms of any of the metrics like delay, throughput, packet loss, etc.
- So, here in communication networks, the path is decided by looking into various metrics.
- Other than the metrics, paths should be decided or updated in between by looking into the conditions of congestion into the network.
- Routing is successful only when all the nodes are cooperative with each other. So, cooperation among the nodes is must.

Properties of Routing Algorithm

- Suppose, a link gets down or a router become fail, then all the routing tables to be quickly updated. This knowledge should be reflected in all the routing tables, so that packet loss will be minimal. Quick convergence and stability is very important in a routing algorithm.
- To solve the routing problems, network is represented in terms of graphs.
- While formulations as a graph, routers are represented as nodes and the links connecting the routers are represented as edges.
- A graph G is represented as (V, E). Where, V is a set of nodes and E is a set of edges connecting those nodes. Each edge is labeled with a value representing its cost.

Properties of Routing Algorithm

- This cost could be directly or indirectly related to the link type or metric value. For example, cost could be directly proportional to the congestion or inversely proportional to the bandwidth.
- Higher bandwidth link or less congested link gives a low cost value of that link.
- If the nodes are connected by an edge then the cost is associated with that edge else it is infinity.
- Undirected graphs are considered for formulation.
- Thus, the cost associated with edge (a, b) is same as edge (b, a).
- A path is a sequence of edges traversed from chosen starting node to chosen destination node. The cost of a path is sum of all the edges cost of that path.

Properties of Routing Algorithm

- It can be visualized from the figure 7, set V consist of routers {A, B, C, D, E} and set E consist of edges such as (A,B), (A,C), (D, E), etc.
- Each edge is labeled with a value known as cost.
- Let us calculate the path from node A to node E.
- It can be visualized from the figure there are multiple paths from A to E.

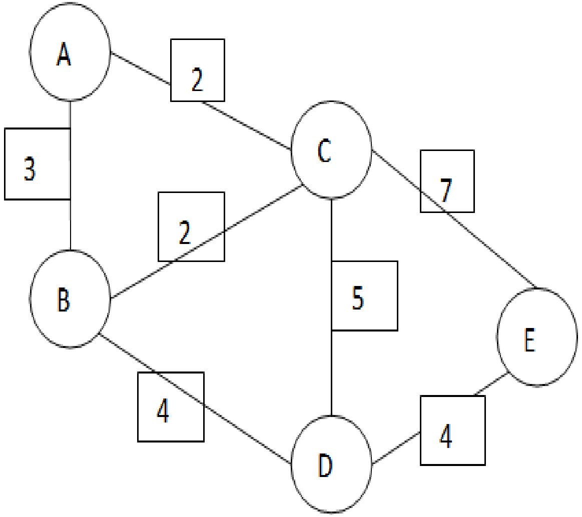


Figure 7: Graph representation of a network

Properties of Routing Algorithm

Some of the paths and associated costs are as follows

- The cost of a path is sum of the costs of all traversed edges from A to E.
- On the basis of low cost, routing algorithm chose the path A-C-E. Suppose, all the edges have same unit cost, then the path with less number of hops is chosen.
- For this scenario, again path A-C-E would be selected because it has less number of hops in comparison to other paths.

Classification of Routing Algorithms

- Routing algorithms for the computation of routes would be can be classified into different types like:
  - Centralized or Decentralized based on.
- Another categorization based on strategy of updation of routes is:
  - Adaptive routing algorithms or Non-adaptive routing algorithms.
- Further these are as:
- Global Routing Algorithm (centralized routing algorithm)
- Decentralized Routing Algorithm(Distance Vector Routing)
- Adaptive Routing Algorithm (Dynamic routing algorithms)
- Non-adaptive Routing Algorithm(Static Algorithm)

Delay in Packet Switched Networks

- **Transmission delay:** This is the time it takes to transmit a packet over a link. It is affected by the size of the packet and the bandwidth of the link.
- **Propagation delay:** This is the time it takes for a packet to travel from the source to the destination. It is affected by the distance between the two nodes and the speed of light.
- **Processing delay:** This is the time it takes for a packet to be processed by a node, such as a router or switch. It is affected by the processing capabilities of the node and the complexity of the routing algorithm.
- **Queuing delay:** This is the time a packet spends waiting in a queue before it can be transmitted. It is affected by the number of packets in the queue and the priority of the packets.

Important Topics

Packet Switching

Network Addressing

Congestion

Routing

Delay in Packet Switched Network

Summary

- In this Session, we understood the concepts of packet switching. Network layer follows the concept of packet switching as packet is the basic data unit used at this layer.
- There are two types of packet switching techniques, virtual circuit and datagram approach. In virtual circuit approach, before sending any data between a source destination pair, end to end logical connection needs to be established between them.
- Datagram approach is a connection less service.
- There is no handshaking between source and destination and each packet follows its own route.

Summary

- Each machine is identified by a logical address, known as IP address.
- It is a 32 bit address which is unique for an individual. Due to overload of the network, network layer faces a problem which is known as congestion.
- Open loop congestion control policies for avoiding the congestion and Closed loop congestion control policies for congestion removal phase has been studied.
- A very important job of network layer is route the packets, thus concepts of routing has been discussed.
- It is followed by the computation of delay metric which is a very important in network layer as packet travels through a number of paths and routers.

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Thank you

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Data Communication and Computer Networks

Block-3 Unit-4_Part1

Emerging Network Technology

Topics to be Covered

- Introduction
- Mobile Ad hoc Networks (MANETs)
- MANET Characteristics
- MANET Limitations
- Advantages of MANET
- Applications of MANET
- MANET Architecture
- MANET Operating Principle

Topics to be Covered

- Challenges facing Ad hoc Networks
- Wireless Sensor Networks(WSN)
- Applications of Wireless Sensor Networks
- Structure of WSN
- Classification of WSNs
- WSN Topologies
- IoT Introduction
- Characteristics of Applications of IoT
- IoT in home Automation Intrusion Detection Use Case

Topics to be Covered

- IoT in Smart Cities: Smart Lightening Use Case
- IoT in Environment: Air Pollution Monitoring Use Case
- IoT in Health Wearable Devices Use Case
- IoT System Architecture
- Important Topics
- Summary

Introduction

- MANET stands for Mobile Adhoc Network also called a wireless Adhoc network or Adhoc wireless network.
- It usually has a routable networking environment on top of a Link Layer ad hoc network.
- They consist of a set of mobile nodes connected wirelessly in a self-configured, self-healing network without having a fixed infrastructure.
- MANET nodes are free to move randomly as the network topology changes frequently.
- Each node behaves as a router as they forward traffic to other specified nodes in the network.

Mobile Ad Hoc Networks (MANETS)

- Now a days, we live in a digital age and are constantly surrounded by digital devices, such as laptops, mobiles, cameras, music players, etc.
- These devices are connected through various communication networks like WiFi, Bluetooth, Cellular Networks etc. In the earlier times, communication infrastructure had a fixed wired backbone with stationary cell towers and access points.
- Such infrastructure networks, for example, cellular networks, were suitable for locations where access points were easy to install. However, today, the nodes (laptops, mobiles, etc.) are not always fixed at one point but can move in the system.

Mobile Ad Hoc Networks (MANETS)

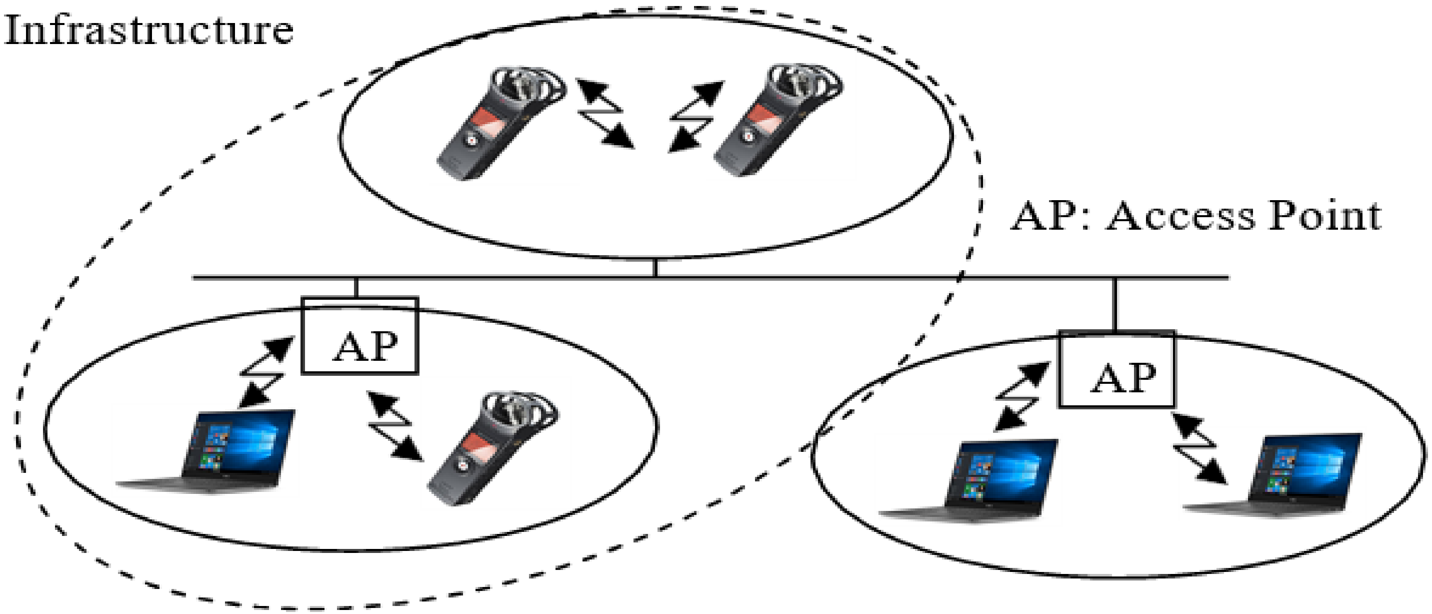


Figure 1: Typical communication between Infra. and APs.

Mobile Ad Hoc Networks (MANETS)

- As today, the nodes (laptops, mobiles, etc.) are not always fixed at one point but can move in the system.
- They connect to the fixed communication infrastructure of cell towers and communication access points (routers) further connected to the Internet backbone.
- However, networks can be configured without APs too.
- A collection of at least two or more electronic devices equipped with wireless communication and networking capability forming infrastructure-less networks without base stations is called a mobile ad-hoc network.

Mobile Ad Hoc Networks (MANETS)

- The term "Ad Hoc" network means a network that is temporary or implemented immediately when needed for a particular purpose.
- With the technological advancements in recent times, it is possible to use mobile nodes and mobile routers (a node can act as a router) to create a communication network where everything is mobile and nothing is fixed.
- In such a network, the topology of the network changes continuously as the neighbor's of any given node are likely to change frequently.
- A few nodes will join the network at any point in time, while a few will disengage from the network, making the environment ad-hoc.

Mobile Ad Hoc Networks (MANETS)

- With minimal or no dependence on infrastructure, a MANET is easy to deploy to support communication and computing anywhere, anytime.
- In heterogeneous networks, the devices have varying capabilities, while all the nodes have identical capabilities and responsibilities in homogenous networks.

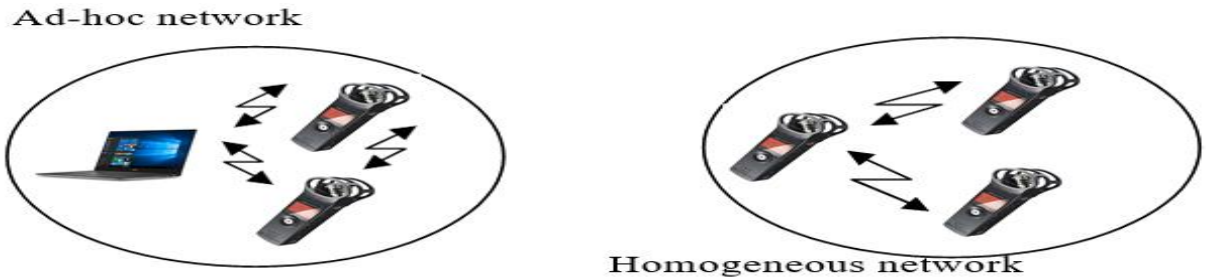


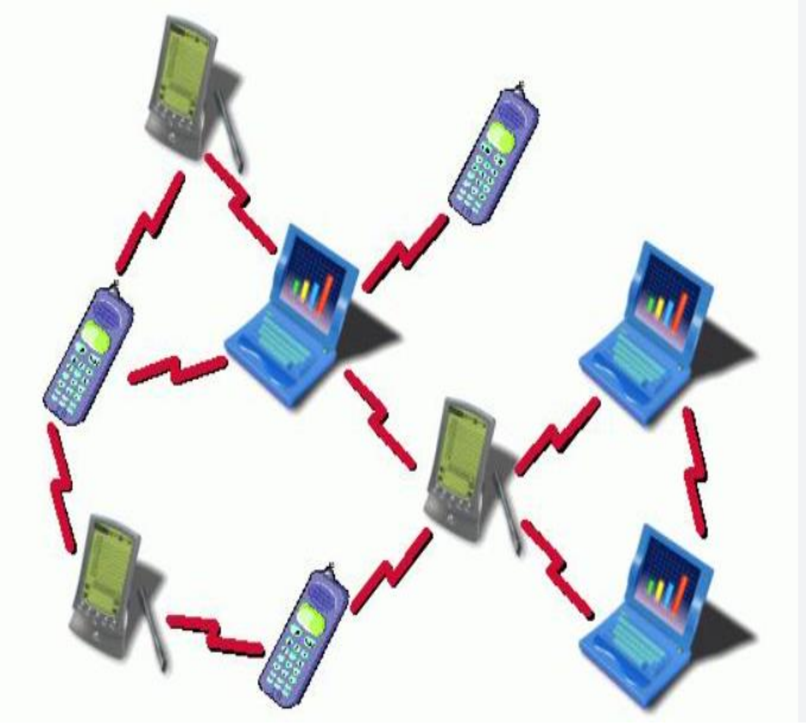
Figure 2: Ad-hoc and Homogenous network

MANET Characteristics

- A MANET does not have a centralized infrastructure, unlike the traditional mobile wireless networks that need base stations, access points and servers to make it operational.
- Its decentralized ad hoc network with all interconnected mobile nodes receiving data while functioning as routers simultaneously.
- One crucial characteristic of MANET is that it is a self-configuring network in which the network activities such as discovering the dynamic topology, delivery of messages and receiving data packets are executed by the nodes themselves.
- MANET has a dynamic topology in which node scan freely move in and out of the network arbitrarily. It causes the network topology to change rapidly and unpredictably over time.

Manet Limitations

- The present and future need for dynamic ad-hoc networking technology is tremendous.
- This highly adaptive networking technology, however, still faces various limitations.
- Throughput Drops with more Hops
- Throughput Drops with Increasing Mobility
- Delay Time



https://www.eexploria.com/manet-mobile-ad-hoc-network-characteristics-and-features/

History of Ad-hoc Networks

- The first generation of ad hoc network, called Packet Radio Network (PRNet), can be traced back to the 1970s when the Defence Advanced Research Project Agency (DARPA) launched packet-switched radio communication to provide reliable communication.
- PRNet provided an efficient means of sharing broadcast radio channels among many radios.
- PRNet, with low throughput (2 kbps per subscriber approximately), was not entirely infrastructure-less as it required a static station for routing.

History of AD-HOC Networks

- The PRNET then evolved into the smaller, cheaper, energy-efficient and cyber-resilient Survivable Adaptive Radio Network (SURAN) in the early 1980s.
- The United States Department of Defence (DOD) developed Globe Mobile Information System (GloMo) and Near Term Digital Radio (NTDR), providing a self-organizing and self-healing network.
- The NTDR, which uses clustering and link-state routing, was a widely used ad hoc network.
- The Internet Engineering Task Force (IETF) adopted IEEE 802.11 as the standard routing protocols for MANET for using mobile devices like PDA's, palmtops, notebooks, etc.

Advantages of Manet

- If we compared it to traditional networks, MANETs are cost-effective solutions and easy to set up, especially.
- This solution is in those geographical locations where fixed infrastructure is not feasible, as the nodes are not tethered.
- The device-to-device communication is used to create, implement and manage the MANET without any need for fixed hardware infrastructure.
- The ability to connect to the internet without any wireless router is the main advantages of MANET, making it more affordable than a traditional network.

Advantages of Manet

- Their ability to handle connection failures is also created as routing and transmission protocols are designed to manage these situations.
- MANETs are extensively chosen due to their other distinct advantages, such as the ability to auto-configure, adaptive, self-diagnosing capability and cost-effectiveness

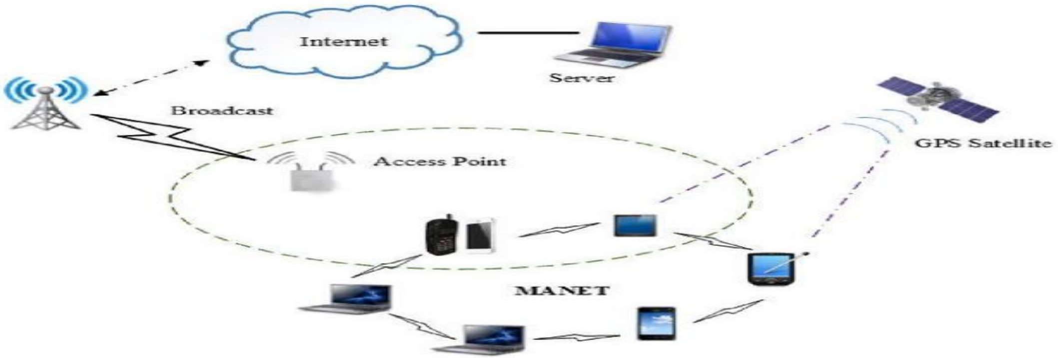


Fig.1 Example of mobile ad-hoc network

https://www.ripublication.com/acst17/acstv10n5_19.pdf

Applications of Manet

- The applications of MANET are not limited to a single area.
- There are many applications of MANET in industry, defence, medical and environmental sciences. A few of the applications include:

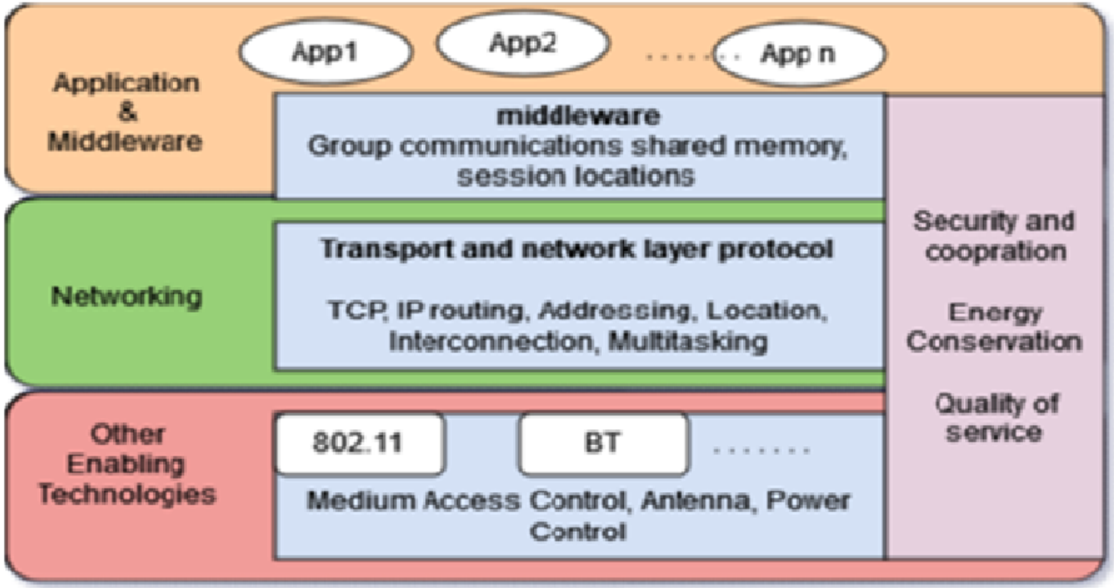
Tactical networks	Military Communication automated Battlefields
Sensor Network	Remote weathers for sensors, earth activities
Emergency Services	Disaster recovery, earthquakes, crowd control and commando operations
Educational Applications	Setup virtual class & conference rooms
Entertainment	Multi-user games, robotics pets
Location-Aware Services	Automatic Call forwarding, advertise location-Specific services, Location–dependent travel guide.

Manet Architecture

- Depending on the application and the area to be covered, the MANET architecture may use body area networks (BAN), personal area networks (PAN), local area networks (LAN), metropolitan area networks (MAN) and wide (WAN) area networks as enabling technologies.
- BAN coverage extends to about 2m and is usually deployed using wearable devices. When the communication requirement is of the order of 10-15m, PAN is used.
- MANETs using LAN easily cover an area ranging up to 500m and are widely deployed for communicating within residential colonies, markets, or a cluster of buildings.
- WAN can also be used for even larger areas.

Manet Architecture Layers

- The MANET architecture layers,
- Link Layer
- Network Layer
- Transport Layer
- Application Layer



Reference MANET layer architecture

Link Layer

- The joint design of the link and physical layer can significantly enhance a MANET's efficiency and reliability.
- The link layer identifies priority packets and schedules packet delivery according to priority levels.
- It is accomplished through a Media access control (MAC) layer with a MAC discriminator and priority classifier.
- MAC is a protocol that enforces a methodology to allow multiple devices access to a shared media network.
- The queue management section schedules the packets according to the priority levels.
- Data for real-time applications like live movies, video conferencing, etc.

Link Layer

- When the network is congested, some packets may be dropped.
- Assigning higher priority to real-time packets ensures that they are not dropped in case of network congestion.
- Queue management thus helps in the timely delivery of data packets in real-time applications and improves the packet delivery ratio.
- The MAC discriminator in the layer differentiates data packets that arrive from the wireless channel and sends them to the network layer.
- The address resolution protocol (ARP) packets go to the queue directly. The MAC packets used in IEEE 802.11 stay in the MAC layer.
- The bandwidth estimation control packets are sent to the bandwidth estimation module for use in the routing layer's adaptive scheme.

Network Layer

- As described earlier, MANETs are self-configuring and have dynamic topologies with several addressing and routing issues.
- In large networks, the communication between nodes becomes unstable and the network itself becomes prone to cyber-attacks.
- For reliable, error-free data transfers, various protocols connect to enabling technologies to construct an end-to-end delivery mechanism between a sender and one or more receivers.
- The routing protocol should have provisions such as call admission or adaptive feedback to provide quality service.
- The network layer provides IP addresses to hosts and establishes routes between senders (sources) and receivers (destinations).

Network Layer

- For this, the network must map the dynamic topology and overcome the challenging addressing and routing issues.
- The network layer obtains the network resource information from lower layers and sends the network status to the applications layer.
- The main objectives of the network layer are to enhance the efficiency of the network by minimizing overheads in routing, stability and reliability in computing routes with a high convergence rate, Congestion control, a better quality of service (QoS) and scalability by providing quality service in a large network with a large number of devices.
- The important parameters are delay and throughput.

Transport Layer

- Transmission control protocol (TCP) and user datagram protocol (UDP) are two transport layer protocols widely used in wired networks.
- In TCP, a connection must be established between communicating devices before the transmission of data.
- After transmitting the data, the connection must close. On the other hand, in UDP, there are no overheads for opening, maintaining and terminating a connection.
- In TCP, the delivery of data at the destination is guaranteed and it has extensive error checking mechanisms.
- UDP has a primary error control mechanism and the delivery of data to the destination is not guaranteed.

Transport Layer

- UDP is faster, simpler and preferred for broadcast and has no congestion control mechanism to react to network congestion.
- Due to this, the applications using UDP as the transport protocol to transmit packets in MANETs can easily overwhelm the network with data.
- Such a scenario may result in considerable power wastage and limited use of available bandwidth in transmitting packets.
- TCP is comparatively slower and doesn't support broadcasting but has an inherent congestion control mechanism (not clogging the network and overloading the capacity in the routers).
- The standard TCP assumes that the end-to-end congestion is due to increased flow in the transport pipe and/or reduced available bandwidth.

Applications Layer

- The flexibility, ease of deployment and cost-effectiveness of MANETs make them a highly attractive option for several application scenarios.
- Based on the sensitivity to transmission delay, the applications are classified into real- time and non-real-time.
- Online live movies, live sports, and video conferencing are a few examples of real-time applications.
- Applications such as Email and FTP are non-real-time applications.
- Applications need to be designed to handle frequent disconnection and reconnection, time-varying delays and data packet losses.
- Initially it was deployed for the military operation,

Manet Operating Principle

- The nodes in an ad hoc network are initially unaware of the network's topology.
- They eventually have to discover the topology. During the discovery process, every node will learn about its neighbor nodes and the distance between them.
- This way, it also lets the other nodes know about its existence in the network.
- For efficient routing, routing tables are initialized and maintained by the routers using routing protocols and these tables are stored in their memory.
- The routers use the routing protocols to decide the path of the packet from the source node to the destination.
- Finding stable routes decrease the route-related overhead and finding the shortest routes are the main goals of the routing protocols.

Manet Operating Principle

- As in given figure, It depicts a peer-to-peer multi-hop ad hoc network.
- Mobile node A communicates directly with B (single hop) when a channel is available
- If a channel is not available, then multi-hop communication is necessary,
- For multi-hop communication to work, the intermediate nodes should route the packet i.e., they should act as a router
- For communication between A-C, B, or D & E, should serve as routers.

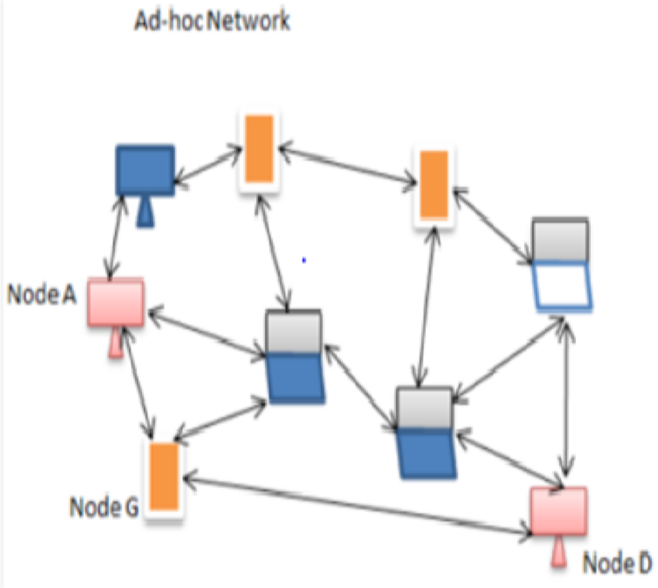
Example of an Ad Hoc Network

Challenges facing Ad Hoc Networks

- Regardless of the variety of mobile ad hoc network applications, there are still some issues and design challenges that we have to overcome.
- MANET being a wireless network, it inherits the traditional problem of wireless networking:
  - The channel is unprotected from the outside interfering signal.
  - The wireless media is unreliable as compared to the wired media.
  - The channel has time-varying and asymmetric propagation properties.

Other Challenges and Complexities

- With these problems, there are some other challenges and complexities:
  - Bandwidth Constraints
  - Energy constraints
  - High Latency
  - Transmission Errors
  - Scalability Concerns
  - Fault Detection and Management



https://study.com/academy/lesson/security-issues-with-mobile-ad-hoc-networks.html

Manet threats

- Limited physical security: The physical sizes of the devices involved in these networks are so small that they can easily be victims of theft.
- Cyber security: They are vulnerable to cyber-attacks and can be compromised easily.
- Each node acts as a router and a receiver in the network, so each node plays two roles becoming significantly more vulnerable.

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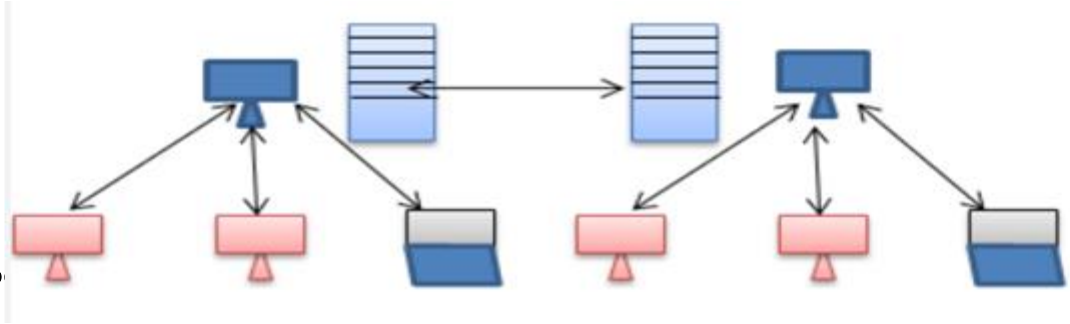
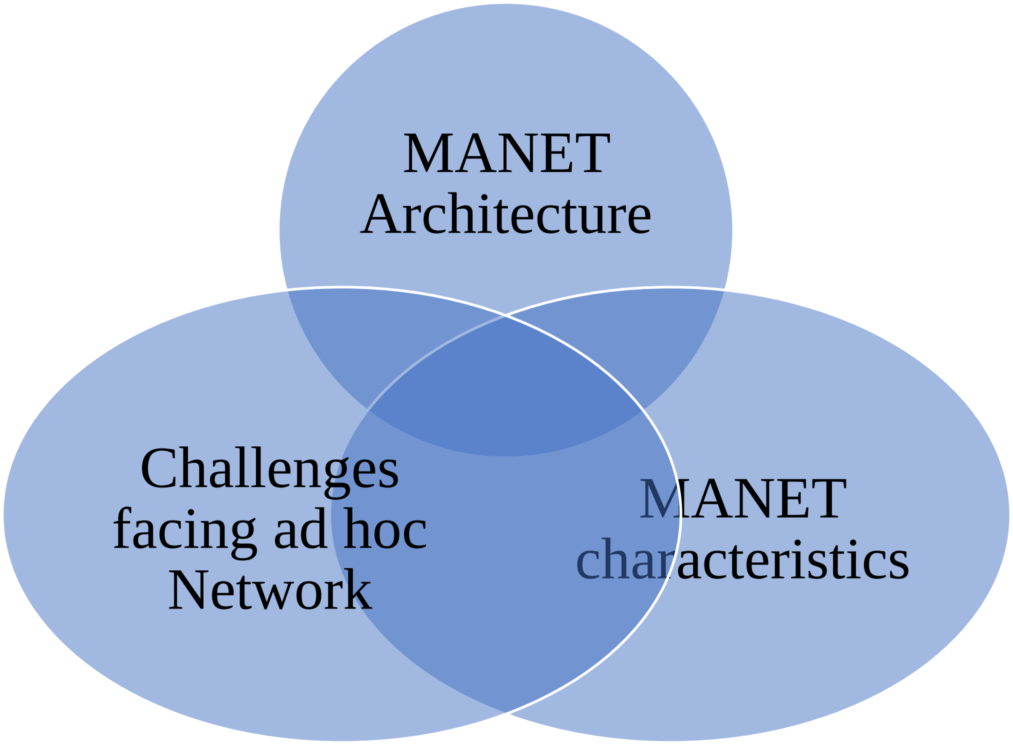


Important Topics



Summary

- In this session, we have learnt about different emerging technologies in networking namely MANET, WSN and IOT. Details about the architectures, usage, advantages and disadvantages are given.
- We have also learned about the various scenarios of using these technologies in day to day life.
- In MANET, characteristics and operating principles are described in detail. The structure of wireless sensor networks with its types and classification is also mentioned.
- WSN topologies like star and mesh are discussed in detail.

Thank you

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Data Communication and Computer Networks

Block-4 Unit-1

Transport Services and Mechanism

Topics to be Covered

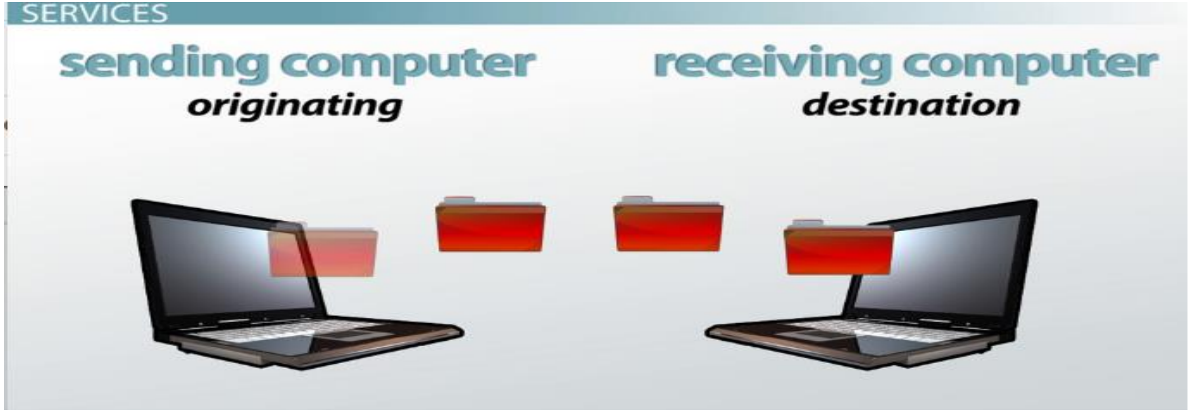
- Introduction
- Transport Services
    - Types of Services
    - Quality of Service
    - Data Transfer
    - Connection Management
    - Expedited Delivery
- Elements of Transport Layer Protocols
    - Addressing

Topics to be Covered

- Multiplexing
- Flow Control and Buffering
- Connection Establishment
- Crash Recovery
- Important Topics
- Summary

Introduction

- The transport layer is the core of the OSI model. It is not just another layer but the heart of the whole protocol hierarchy.
- Protocols at this layer oversee the delivery of data from an application program on one device to an application program on another device.
- It is the first end-to-end layer in the OSI model.



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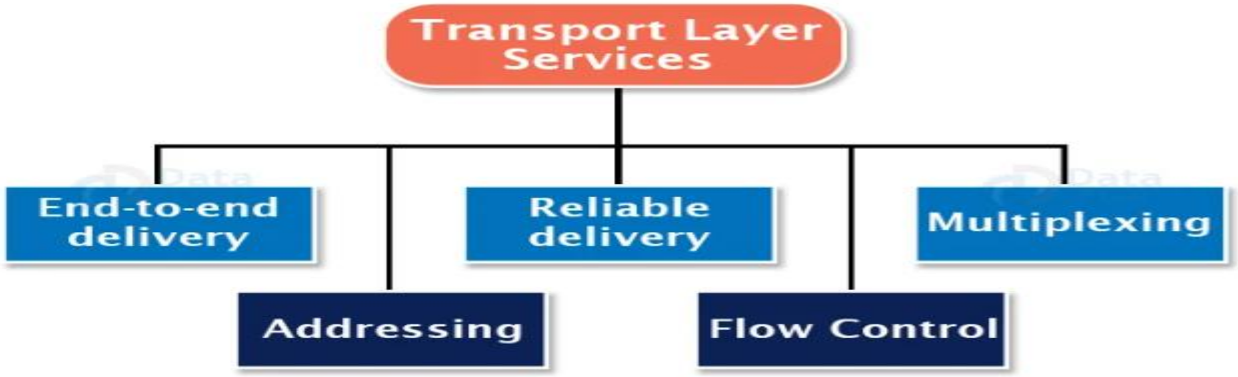
Transport Services

- The kinds of services that a transport protocol can or should provide to upper layer protocols.
- There is a transport entity that provides services to the upper layer users, which might be an application process such as http, telnet etc.



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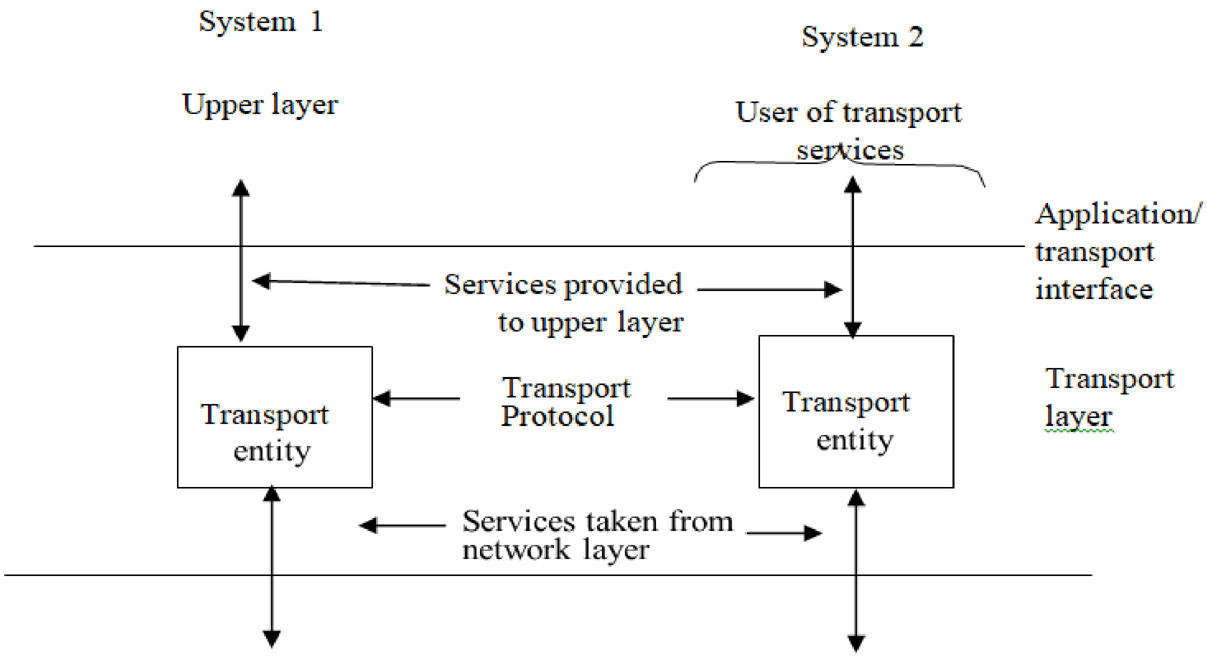
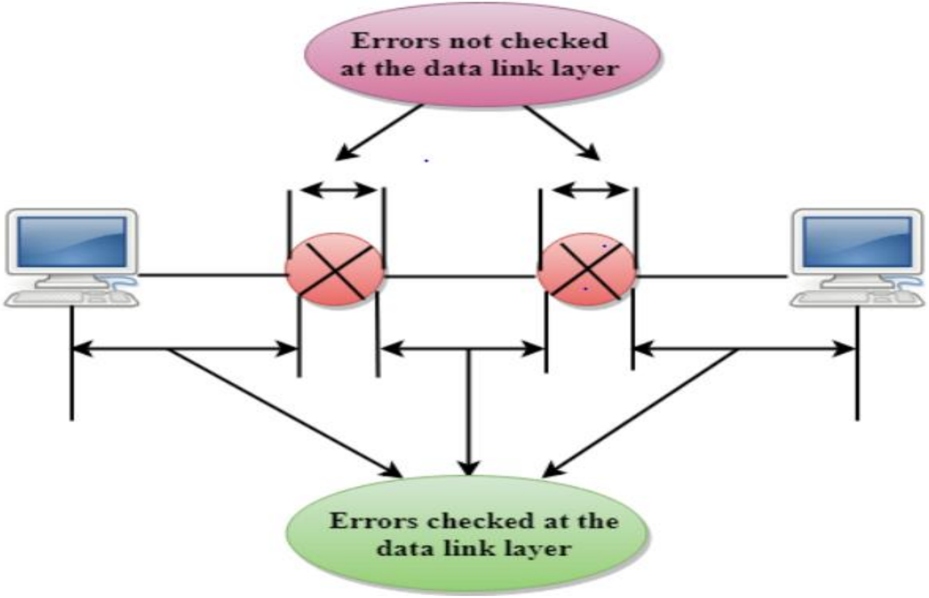
Transport Services



Transport entity environment

Categories of services by the transport layers

- Types of services
- Quality of service
- Data transfer
- Connection management
- Expedited delivery

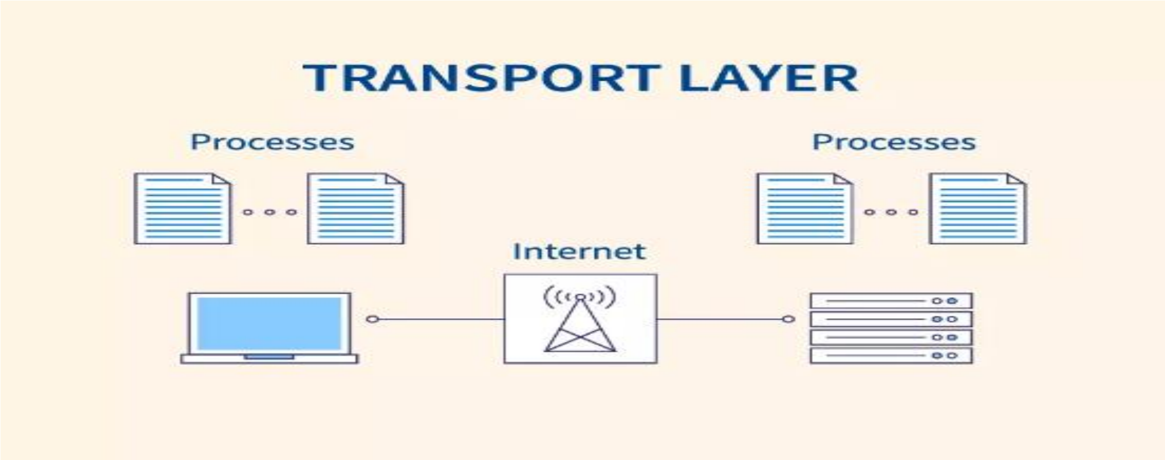


https://www.javatpoint.com/computer-network-transport-layer

Types of Services

There are generally two basic types of services:

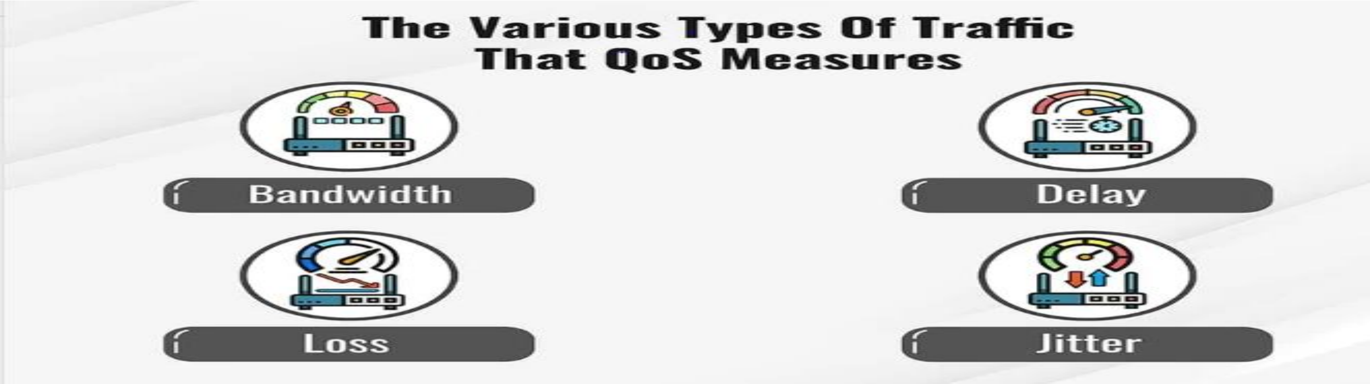
- Connection-oriented
- Connectionless



https://www.scaler.com/topics/computer-network/transport-layer-protocols/

Quality of Service

- The transport protocol entity should allow the upper layer protocol users to specify the types of quality of transmission service to be provided.
- Based on these specifications the transport entity attempts to optimise the use of the underlying link, network, and other resources to the best of its ability.



https://www.fortinet.com/resources/cyberglossary/qos-quality-of-service

Quality of Service

- Examples of services that might be requested are:
- Expedited delivery of packet
- Acceptable error and loss levels of packet
- Desired average and maximum delay in transmission of a packet
- Desired average and minimum throughout
- Priority levels

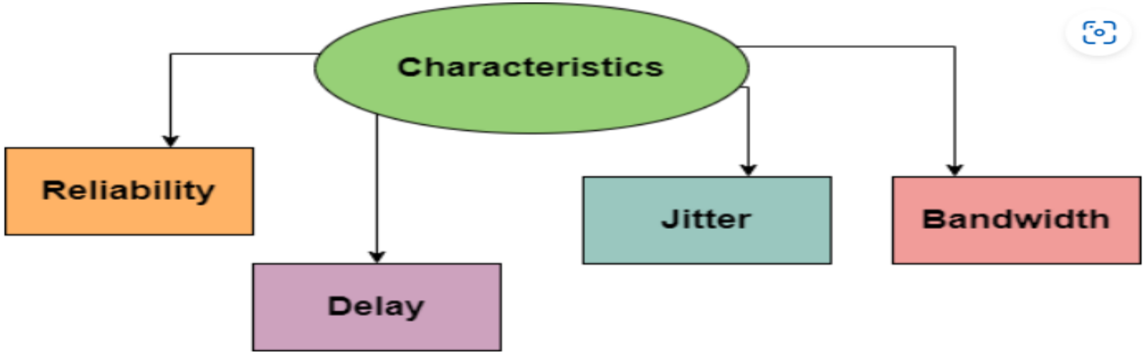
https://www.section.io/engineering-education/understanding-quality-of-service/

An application Layer Protocols

An application layer protocols need from the Transport layer

These are:

- Reliable data transfer
- Bandwidth
- Timing/delay



https://www.studytonight.com/computer-networks/quality-of-serviceqos

Data Transfer

- The whole purpose, of course, of a transport protocol is to transfer data between two transport entities.
- Both user data and control data must be transferred from one end to the other end either on the same channel or separate channels. Full-duplex service must be provided.
- Half-duplex and simplex modes may also be offered to support peculiarities of particular transport users.
- The network layer overseas the hop by hop delivery of the individual packet but does not see any relationship between those packets even those belonging to a single message. Each packet is treated as an independent entity. The transport layer on the other hand makes sure that not just a single packet but the entire sequence of packets.

Connection Management

- When connection-oriented service is provided, the transport entity is responsible for establishing and terminating connections.
- Both symmetric and asymmetric procedure for connecting establishment may be provided.
- Connection termination can be either abrupt or graceful. With an abrupt termination, data in transmit may be lost. A graceful termination prevents either side from shutting down until all data has been delivered.

https://www.istockphoto.com/search/2/image?phrase=computer+network+management

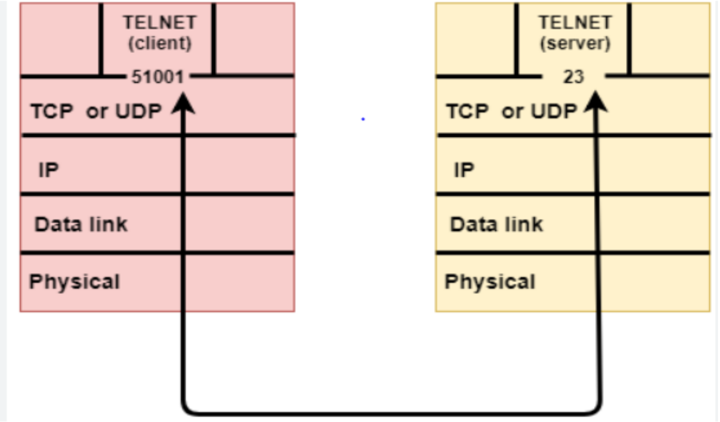
Expedited Delivery

- A service similar to that provided by priority classes is the expedited delivery of data.
- Some data submitted to the transport service may supersede data submitted previously.
- The transport entity will endeavour to have the transmission facility transfer th data as rapidly as possible

https://onlineclassnotes.com/define-delivery-explain-various-delivery-concepts-or-methods/

Elements of Transport Layer Protocols

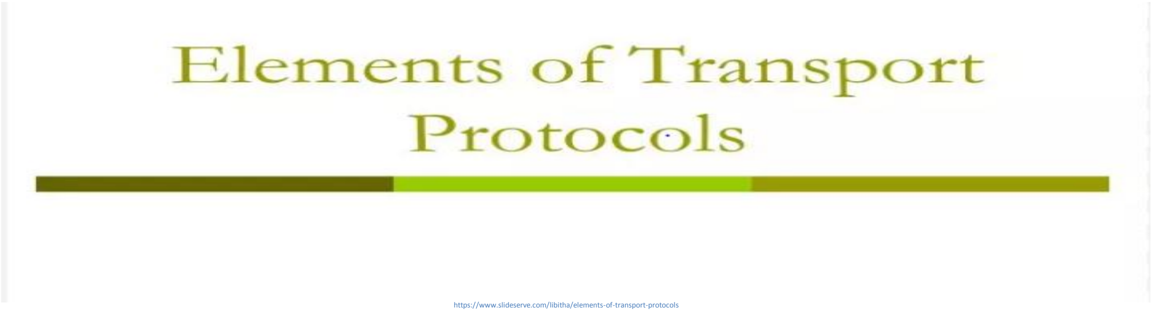
- The services provided by the transport layer are implemented by the transport layer protocols such as TCP and UDP.
- In this section, we will take up important components of the transport layer for discussion. A transport layer deals with hop-by-hop processes.
- **Addressing**
- **Multiplexing**
- **Flow Control and Buffering**
- **Connection Establishment**



https://www.javatpoint.com/computer-network-transport-layer-protocols

Elements of Transport Layer Protocols

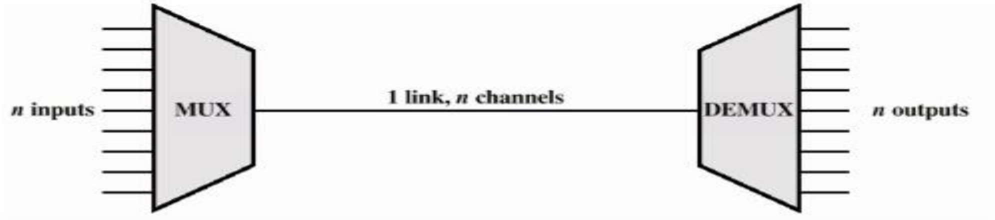
- **Addressing**
- how a user process will set up a connection to a remote user process? How the address of a remote process should be specified?
- The method generally used is to define transport addressto which the process can listen for connection requests. These are called TSAP (Transport Services Access Points)
- These end points at the network layer are then called NSAP (Network Services Access Points).



https://www.slideserve.com/libitha/elements-of-transport-protocols

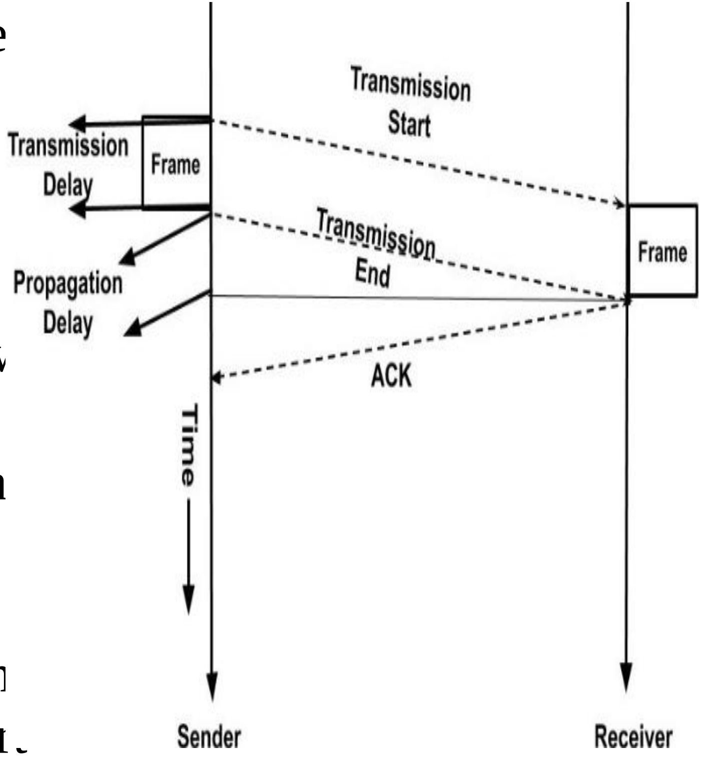
Multiplexing

- The transport entity may also perform a multiplexing function
- with respect to the network services that it uses.
- There are two types of multiplexing techniques that define
  - Upward multiplexing as the multiplexing of multiple transport connections on a single network connection
  - Downward multiplexing as the splitting of a single transport connection among multiple lower-level connections.



Flow Control and Buffering

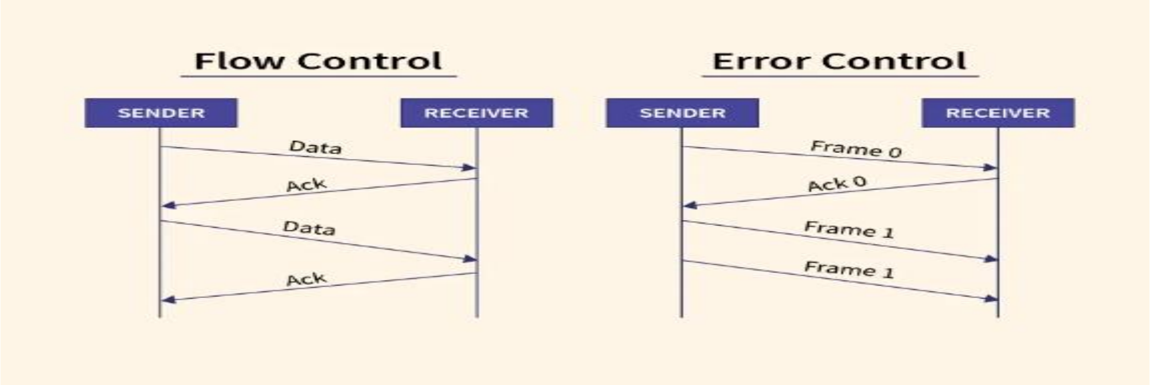
- There are similarities as well as difference between:
- Data link layer and Transport layer in order to support flow control mechanism.
- The similarity is in using the sliding window protocol.
- The main difference is that a router usually has relatively few lines,
- whereas a host may have numerous connections.
- Due to this reason it is impractical to implement the data link buffering strategy in the transport layer.



afteracademy.com/blog/what-is-flow-control-in-networking/

Flow control at Data link Layer vs Transport Layer

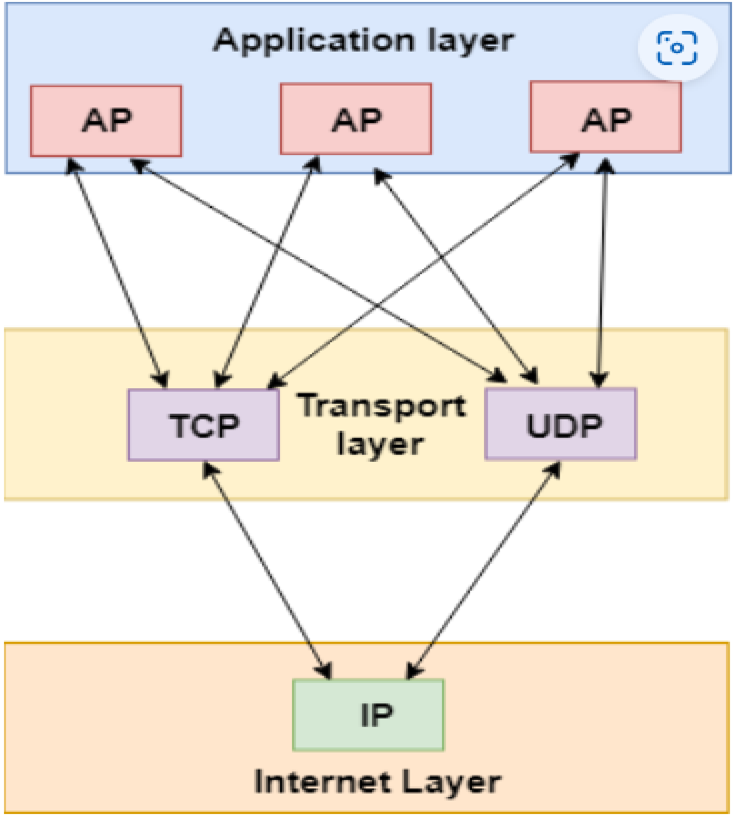
- Flow control is a relatively simple mechanism at the data link layer, it is a rather complex mechanism at the transport layer, for two main reasons.
- Flow control at the transport level involves the interaction of three components:
- TS users, transport entities, and the network service. The transmission delay between transport entities is variable and generally long compared to actual transmission time.



https://www.scaler.com/topics/computer-network/flow-control-and-error-control/

Connection Establishment

- End-to-end delivery can be accomplished in either of two modes:
  - Connection-oriented
  - Connectionless
- Both Of these two, the connection-oriented mode is the more Commonly used.
- A connection-oriented protocol establishes a virtual circuit or pathway through the Internet between the sender and receiver.



Connection Establishment

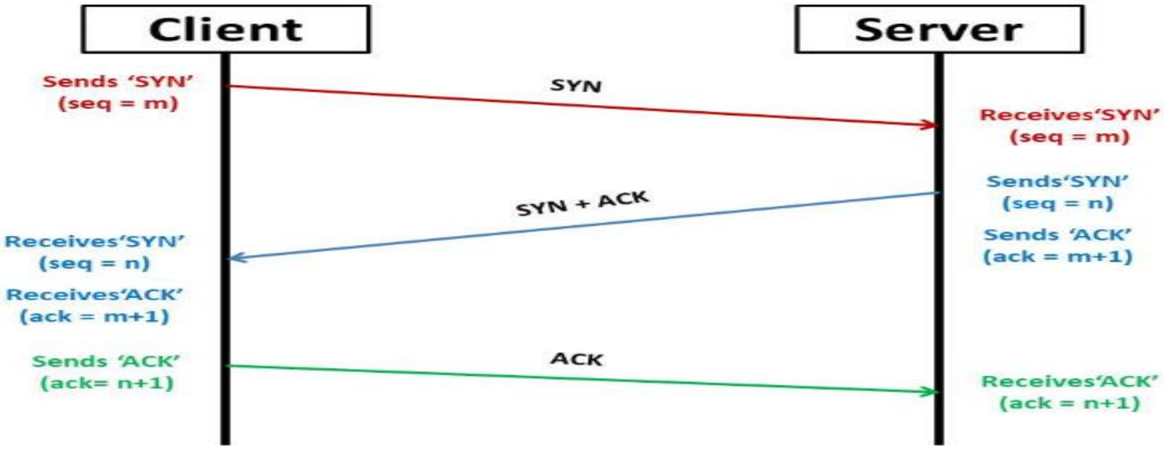
- Connection establishment serves three main purposes:
- It allows each end to assure that the other exists.
- It allows negotiation of optional parameters (e.g., maximum segment size, maximum window size, quality of service).
- It triggers allocation of transport entity resources (e.g., buffer space, entry in connection table).

https://data-flair.training/blogs/connection-oriented-vs-connectionless-service/

Connection-Oriented TransmissionStages

3-way handshake procedure:

- Connection initiation / requirement
- Connection confirmation
- Acknowledgement of confirmation and data transfer



https://afteracademy.com/blog/what-is-a-tcp-3-way-handshake-process/

Crash Recovery

- In case of the crash delivery, a new virtual circuit may be established then probing the transport entity to ask which TPDU(Transaction Protocol Data Unit) it has received and which one it has not received so that the latter can be retransmitted.
- In case of datagram subnet, the frequency of the lost TPDU is high but the network layer knows how to handle that. The real problem is the recovery from the host crash.

Important Topics

- Transport Services.
- Transport Layer Protocols

Summary

- In this session, we have discussed two important components with respect to transport layer namely, transport services and the elements of transport layer protocols.
- As a part of transport services we discussed type of services, quality of services, connection mechanism.
- We also outlined three types of services provided by the transport layer through the applications layer. We listed the various types of protocols (applications layer) and the kind of services.
- We also differentiated between the data link layer and transport layer with respect to flow control mechanism. Similarly, we also differentiated between two types of multiplexing mechanism at the transport layer.

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Data Communication and Computer Networks

Block-4 Unit-2

TCP UDP

Topics to be Covered

- Introduction
- Services Provided by Internet Transport Protocols
- TCP Services
- UDP Services
- Introduction to UDP
- Introduction to TCP
- TCP Segment Header

Topics to be Covered

- TCP Connection Establishment
- TCP Connection Termination
- TCP Flow Control
- TCP Congestion Control
- Remote Procedure Call
- Important Topics
- Summary

Introduction

- TCP is the more sophisticated protocol of the two and is used for applications that need connection establishment before actual data transmission.
- Applications such as electronic mail, remote terminal access, web surfing and file transfer are based on TCP.
- UDP is a much simpler protocol than TCP because, it does not establish any connection between the two nodes.
- Unlike TCP, UDP does not guarantee the delivery of data to the destination. It is the responsibility of application layer protocols to make UDP as reliable as possible.

Services Provided by Internet Transport Protocols

- **TCP Services**
- **Connection oriented service** is comprised of the handshake procedure which is a full duplex connection in that, two processes can send messages to each other over the connection at the same time.
- **Reliable transport service** Communicating processes can rely on TCP to deliver all the data that is sent, without error and in the proper order.
- **Congestion-control mechanism** TCP also includes a congestion-control mechanism for the smooth functioning of Internet processes.

Services Provided by Internet Transport Protocols

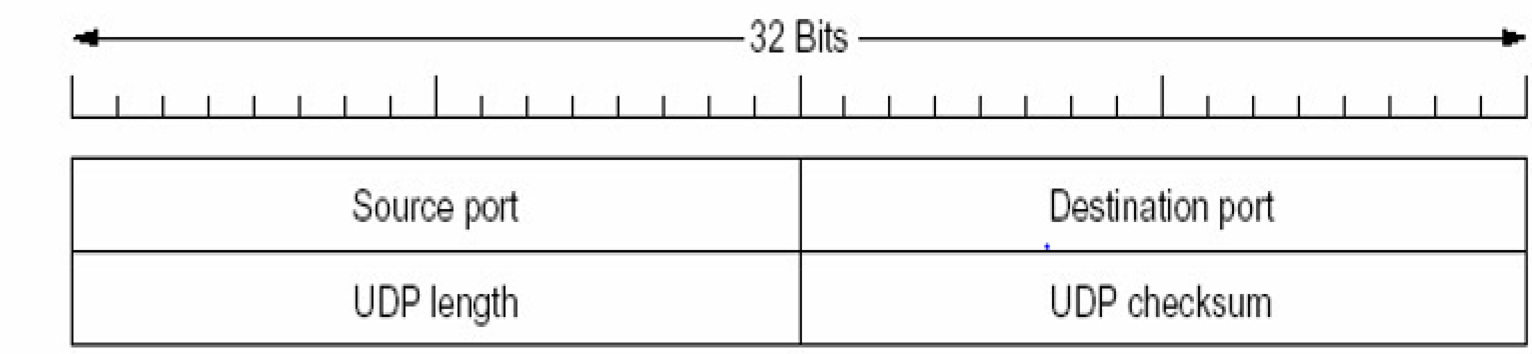
- **UDP Services**
- UDP is connectionless
- It provides unreliable data transfer service. there is no guarantee that the message will reach. Due to this, the message may arrive as the receiving process at a random time.
- UDP does not provide a congestion-control service.
- Therefore, the sending process can pump data into a UDP socket at any rate it pleases. SMTP (E-mail), Telnet, HTTP, FTP exclusively uses TCP whereas NFS and Streaming Multimedia may use TCP or UDP.
- But Internet Telephony uses UDP exclusively.

Introduction to UDP

- UDP is unreliable transport protocol. Apart from multiplexing /DE multiplexing and some error correction UDP adds little to the IP protocol.
- In case, we select UDP then the application is almost directly talking to the IP layer. UDP takes messages from the application process, attaches the source and destination port number fields for the multiplexing or DE multiplexing service, adds two other small fields, and passes the resulting segment to the network layer.
- Without establishing handshaking between sending and receiving transport-layer entities, the network layer encapsulates the segment into an IP datagram and then makes a **best-effort** attempt to deliver the segment to the receiving host.

UDP Segment Structure

- UDP is an end to end transport level protocol that adds only port addresses, checksum error control and length information to the data from the upper layer.

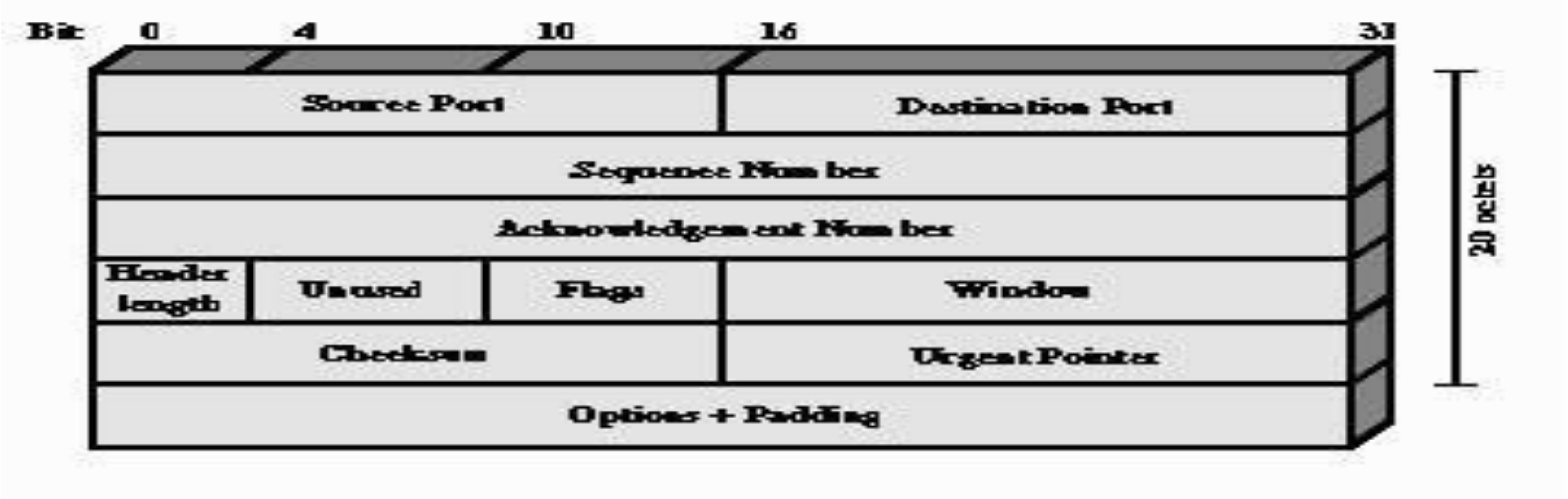


Introduction to TCP

- TCP is designed to provide reliable communication between pairs of processes (TCP users) across a variety of reliable and unreliable networks and Internets. TCP is stream oriented (Connection Oriented).
- Stream means that every connection is treated as stream of bytes.
- The user application does not need to package data in individual datagram as it is done in UDP.
- In TCP a connection must be established between the sender and the receiver. By creating this connection, TCP requires a VC at IP layers that will be active for the entire duration of a transmission.
- The data is placed in allocated buffers and transmitted by TCP in segments.

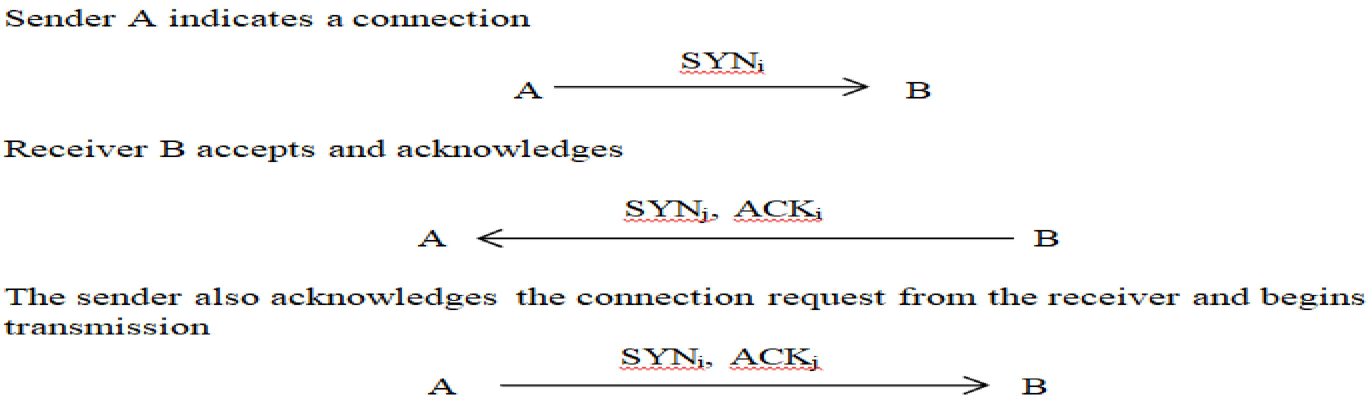
TCP Segment Header

- TCP uses only a single type of protocol data unit, called a TCP segment. The header is:



TCP Connection Establishment

- Before data transmission a connection must be established. Connection establishment must take into account the unreliability of a network service, which leads to a loss of ACK, data as well as SYN. Therefore, the retransmission of these packets have to be done.
- Recall that a connection establishment calls for the exchange of SYNs.



TCP Connection Establishment

Obsolete SYN arrives from the previous connection at B

$$A \xrightarrow{SYN_i} B$$

B accepts and acknowledges

$$A \xleftarrow{SYN_j,\ ACK_i} B$$

Sender A rejects Receiver B's connection

$$A \xrightarrow{RST,\ ACK_j} B$$

(a) Delayed arrived of SYN packet

TCP Connection Establishment

A initiates a connection

$$A \xrightarrow{SYN_i} B$$

old SYN arrives at A

$$A \xleftarrow{SYNK,\ ACK = X} B$$

A rejects it

$$A \xrightarrow{RST,\ ACK = K} B$$

TCP Connection Establishment

B accepts and acknowledge it

$$A \xleftarrow{SYN_j,\ ACK\ =\ i} B$$

A acknowledges and begins transmission

$$A \xrightarrow{SN_i,\ ACK\ =\ j} B$$

(a) Delayed SYN and ACK

Figure 3: Three-way handshake mechanism

(Ctrl)

TCP Connection Establishment

- TCP adopts a similar approach to that used for connection establishment. TCP provides for a graceful close that involves the independent termination of each direction of the connection.
- A termination is initiated when an application tells TCP that it has no more data to send. Each side must explicitly acknowledge the FIN parcel of the other, to be acknowledged.
- The following steps are required for a graceful close.
- The sender must send FIN j and receive an ACK j
- It must receive a FIN j and send an ACK j
- It must wait for an interval of time, to twice the maximum expected segment lifetime

Acknowledgement Policy

- When a data segment arrives in sequence, the receiving TCP entity has two options concerning the timing of acknowledgment:
- **Immediate:** When data are accepted, immediately transmit an empty (no data) segment containing the appropriate acknowledgment number.
- **Cumulative:** When data are accepted, record the need for acknowledgment, but wait for an outbound segment with data on which to piggyback the acknowledgement. To avoid a long delay, set a window timer. If the timer expires before an acknowledgement is sent, transmit an empty segment containing the appropriate acknowledgement number.

TCP Flow Control

- Flow Control is the process of regulating the traffic between
  - two end points and is used to prevent the sender from flooding the
  - receiver with too much data.
- TCP provides a flow control service to its application to eliminate the possibility of the sender overflowing.
- At the receivers buffer TCP uses sliding window with credit scheme to handle flow control.
- The scheme provides the receiver with a greater degree of control over data flow. In a credit scheme a segment may be acknowledged without the guarantee of a new credit and vice-versa.
- Whereas in a fixed sliding window control (used at the data link layer),
  - the two are interlinked (tied).

TCPCongestion Control

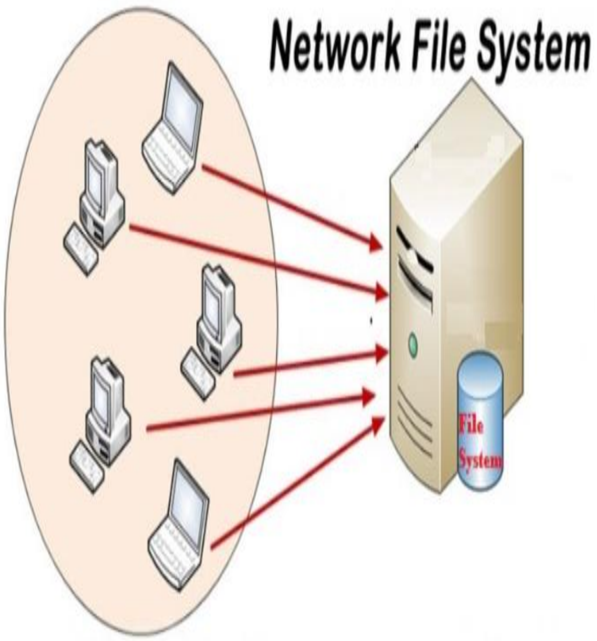
- TCP provides many services to the application process.
- Congestion Control is one such service.
- TCP uses end-to-end Congestion Control rather than the network supported Congestion Control, since the IP Protocol does not provide Congestion released support to the end system.
- The basic idea of TCP congestion control is to have each sender transmit just the right amount of data to keep the network resources utilised but not overloaded.
- TCP flow control mechanism uses a sliding-window protocol for end-to-end flow control.

Remote Procedure Call

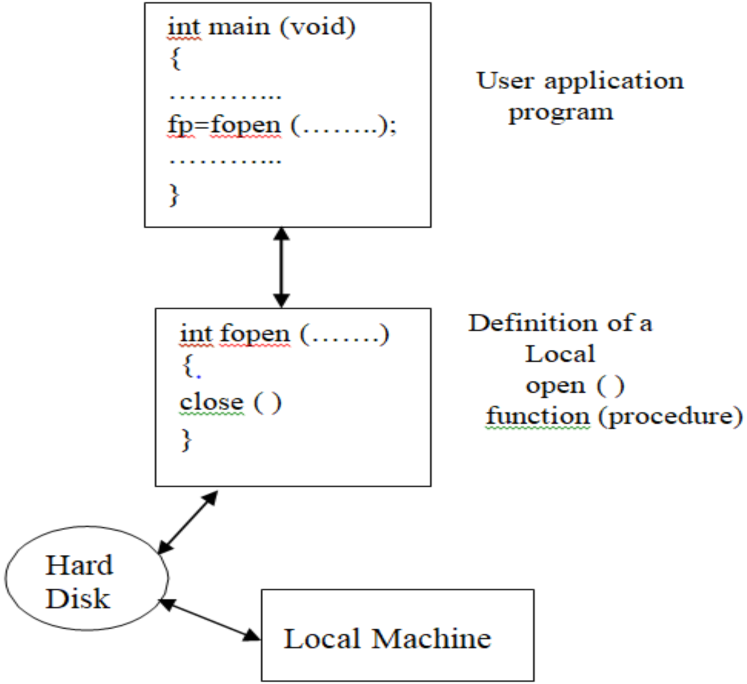
- Remote Procedure Call is a software communication protocol that one program can use to request a service from a program located in another computer on a network without having to understand the network's details.
- Files access mechanisms such as **Network File System** and **Remote Procedure Call** used for accessing files from a server at the client machine.
- These two mechanism allow programs to call procedures located on remote machines.

Network File System

- The network file system (NFS) is a file access protocol.
- FTP and TFTP transfer entire files from a server to the client host.
- A file access service, on the other hand, makes file systems on a remote machine visible, though they were on your own machine but without actually transferring the files.
- NFS provides a number of features.



Remote Procedure Call (RPC)



- NFS works by invoking the services of a second protocol called remote procedure call Figure shows a local procedure call (in this case, a C program calls the open function to access a file stored on a disk).

Remote Procedure Call

- RPC transfers the procedure call to another machine.
- Using RPC, local procedure calls are mapped onto appropriate RPC function calls.
- Figure illustrates the process: a program issues a call to the NFS client process.

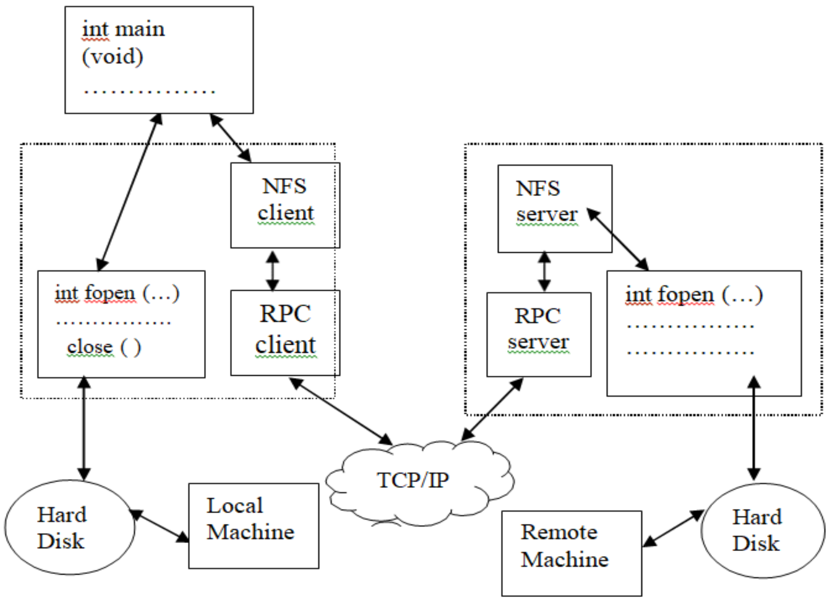
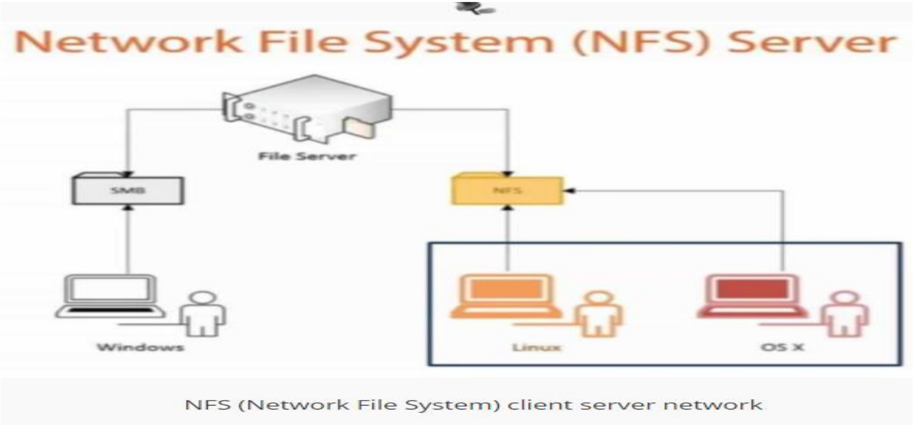


Figure 6: Concept of remote procedure call

Components In NFS Protocol

There are three independent components in NFS protocol:

- NFS itself
- General purpose RPC
- Data representing format (DRF).



http://storagegaga.com/a-paean-to-nfs/

Important Topics

- Services provided by internet Transport Protocols
- Introduction to TCP and
- Introduction to UDP)
- TCP Segment Header
- TCP Connection Establishment
- TCP Connection Termination
- TCP Control Flow
- TCP Congestion Control
- Remote Procedure Call

Summary

- In this session, we discussed the two transport layer protocols (TCP & UDP) in detail. The transport port can be light weight (very simple) which provide very little facility.
- In such cases, the application directly talks to the IP. UDP is an example of a transport protocol, which provides minimum facility, with no control, no connection establishment, no acknowledgement and no congestion handling feature.
- Therefore, if the application is to be designed around the UDP, then such features have to be supported in the application itself.

Summary

- On the other hand, there is another protocol-TCP which provides several feature such as reliability, flow control, connection establishment to the various application services.
- Nevertheless, the services that the transport layer can provide often constrain the network layer protocol.
- We had a close look at the TCP connection establishment, TCP flow control, TCP congestion handling mechanism and finally UDP and TCP header formats.
- With respect to congestion control, we learned that congestion control is essential for the well-being of the network. Without congestion control the network can be blocked.

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Data Communication and Computer Networks

Block-4 Unit-3

Network Security-1

Topics To be Covered

- Introduction
- Cryptography
- Symmetric Key Cryptography
- Public Key Cryptography
- RSA Public Key Algorithm
  - Diffie-Hellman
  - Elliptic Curve Public Key Cryptosystems
  - DSA

Topics To be Covered

Mathematical Background

- Exclusive OR
- The Modulo Function

Important Topics

Summary

Introduction

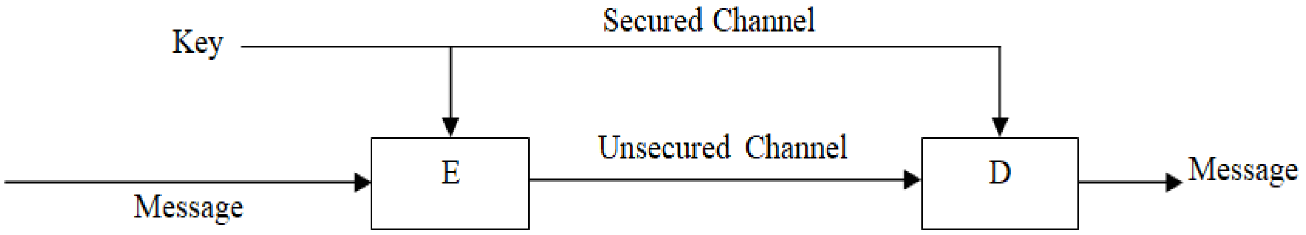
- Cryptography plays an important role in network security. The purpose of cryptography is to protect transmitted information from being read and understood by anyone except the intended recipient.
- In the ideal sense unauthorized individuals can never read an enciphered message.
- Secret writing can be traced back to 3,000 B.C.
- During 50 C.Julius Caesar, the emperor of Rome, used a substitution cipher to transmit messages to Marcus Tullius Cicero. In this, letters of t e alphabets are substituted for other letters of the same alphabet. Since, only one alphabet was used, this is also called monoalphabetic substitution.

Cryptography

- Cryptography is the science of writing in secret code and is an ancient art; the first documented use of cryptography in writing dates back to circa 1900 B.C. when an Egyptian scribe used non-standard hieroglyphics in an inscription.
- Cryptography is a method of protecting information and communications through the use of codes, so that only those for whom the information is intended can read and process it.
- Cryptographic systems are generally classified among independent dimensions.
- **Types of Operation**
- **Key Used**

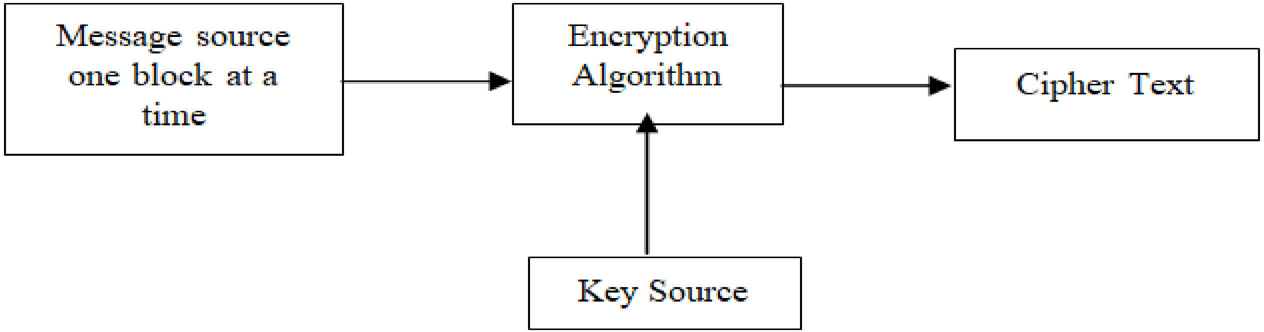
Symmetric Key Cryptography

- Symmetric Key Cryptography is also known as Symmetric Encryption is when a secret key is leveraged for both encryption and decryption functions.
- This method is the opposite of Asymmetric Encryption where one key is used to encrypt and another is used to decrypt.
- The transmission of a message is done using an encryption function E that converts the message or plaintext using the key into a cipher text. The receiver does the reverse of this operation, using the decryption function D and a decryption key to recover the plaintext from the cipher text. The key is distributed in advance over a secure channel, for example by courier.



Block Ciphers

- A block of cipher transforms n-bit plaintext blocks to n-bit cipher text blocks on the application of a cipher key k.
- The key is generated at random using various mathematical functions.



Stream Ciphers

- A stream cipher consists of a state machine that outputs at each state transition one bit of information. This stream of output bits is called the running key.
- Just XOR ring the running key to the plaintext message can implement the encryption. The state machine is a pseudo-random number generator. For example, we can build one from a block cipher by encr

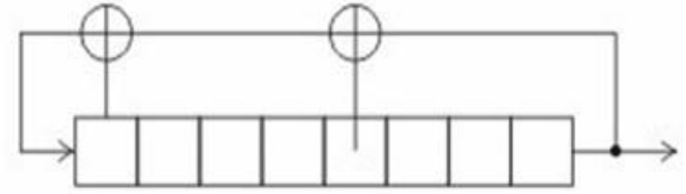


Figure 3: Stream cipher linear feedback shift register (LFSR)

SSSC (Self-Synchronising Stream Ciphers)

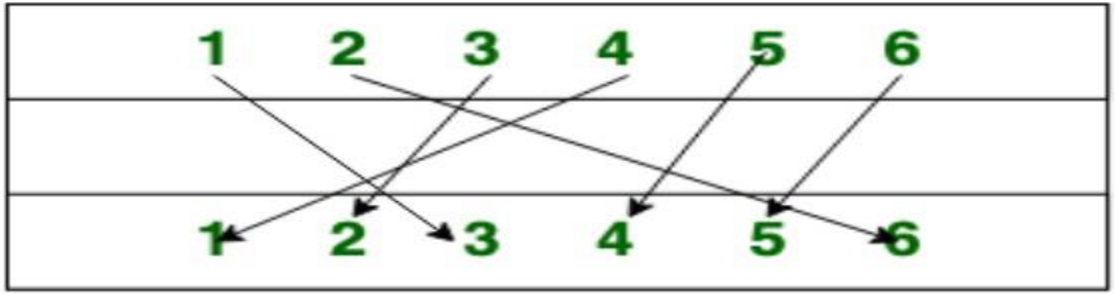
- The class of self-synchronizing stream ciphers has property that it corrects the output stream after the bit flips or even drops bits after a short time.
- In other words, A **stream cipher** is a symmetric key cipher where

plaintext digits are combined with a pseudorandom cipher.

- (A **pseudorandom** sequence of numbers is one that appears to be statistically random), digit stream keystream.

Substitution and Transposition

- A substitution means replacing symbols or group of symbols by other symbols or groups of symbols.
- Transposition, on the other hand, means permuting the symbols in a block.



Transportation Cipher

Feistel Networks

Feistel Cipher model is a structure or a design used to develop many block ciphers such as DES.

Feistel cipher may have invertible, non- invertible and self invertible components in its design.

Expansion, Permutation

- They are linear operations, and thus, not sufficient to guarantee security.
- They are common tools in mixing bits in a round function and when used with good non-linear S-boxes (as in DES) they are vital for the security because they propagate the nonlinearity uniformly over all bits.

Modes of Operation

- Block ciphers are the most commonly used cipher.
- Block ciphers transform a fixed size block of data (usually 64 bits) into another fixed-size block (possibly 64 bits wide again) using a function selected by the key.
- If the key, input block and output block have all n bits, a block cipher basically defines a one-to-one mapping from n-bit integers to permutations of n-bit integers.
- If the same block is encrypted twice with the same key, the resulting cipher text blocks are also the same and is called Electronic Code Book.

CBC (Cipher Block Chaining)

- First XORing the plaintext block with the previous cipher text block obtains a cipher text block, and then the resulting value needs to be encrypted.
- In this, leading blocks influence all trailing blocks, which increases the number of plaintext bits one cipher text bit depends on, but this may also leads to synchronization problems if one block is lost.

CFB (Cipher Feedback)

- In cryptography, cipher text feedback (CFB), also known as cipher feedback, is a mode of operation for a block cipher.
- Cipher text refers to encrypted text transferred from plaintext using an encryption algorithm, or cipher.
- A block cipher is a method of encrypting data in blocksto produce cipher text using a cryptographic key and algorithm

Design Patterns

OFB (Output Feedback Block):

It is used as a synchronous key-stream

generator, whose output is XORed with the plaintext to obtain cipher text, block by block.

The One-Time Pad

- **The One-Time Pad** (OTP) is the only cipher that has been proven to be unconditionally secure, i.e., unbreakable in practice.
- Further, it has been proven that any unbreakable, unconditionally secure cipher must be a one-time pad.

Data Encryption Standard (DES)

- DES is a block cipher with a 64-bit block size.
- It uses 56-bit keys.
- This makes it susceptible to exhaustive key search with modern computing powers and special purpose hardware.
- DES is still strong enough to keep most random hackers.
- A variant of DES, Triple-DES (also 3DES) is based on using DES three times (normally in an encrypt-decrypt-encrypt sequence with three different, unrelated keys).

Breaking DES

- DES's 56-bit key was too short to withstand a brute-force attack from computing power of modern computers.
- DES is even more vulnerable to a brute-force attack as it is commonly used to encrypt words, meaning that the entropy of the 64-bit block is greatly reduced.

Breaking DES Cont

- **DES Challenge I:** was launched in March 1997.
  - R. Verser completed it in 84 days
- **DES II challenge:** lasted just less than 3 days. On July 17, 1998, the Electronic Frontier Foundation (EFF)
  - announced the construction of a hardware that could
  - brute-force a DES key in an average of 4.5 days.
- **DES III challenge:** launched in January 1999, was broken in less than a day by the coordinated efforts of Deep –
  - Crack and distributed.net.

DES Variants

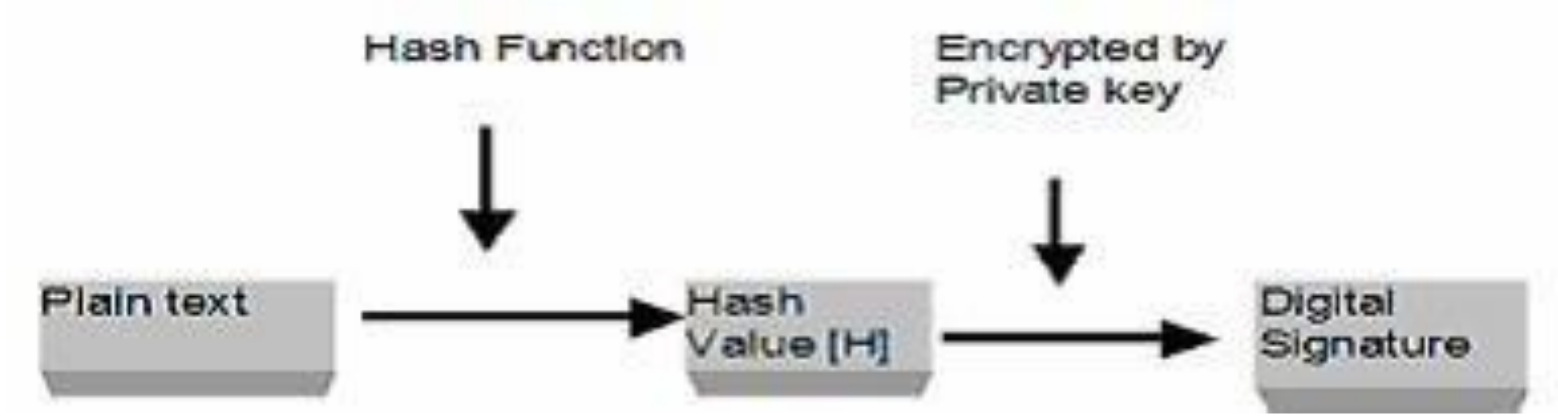
- After DES algorithm was “officially” broken, several variants appeared. In the early 1990s, there was a proposal to increase the security of DES by effectively increasing the key length by using multiple keys with multiple passes etc.
- But for this scheme to work, it had to first be shown that, the DES
- function is not a group.
- **Triple-DES (3DES)** is based upon the Triple Data Encryption Algorithm (TDEA)
- **AES :** In response to cryptographic attacks on DES, search for a replacement to DES started in January 1997.

Other Symmetric Ciphers

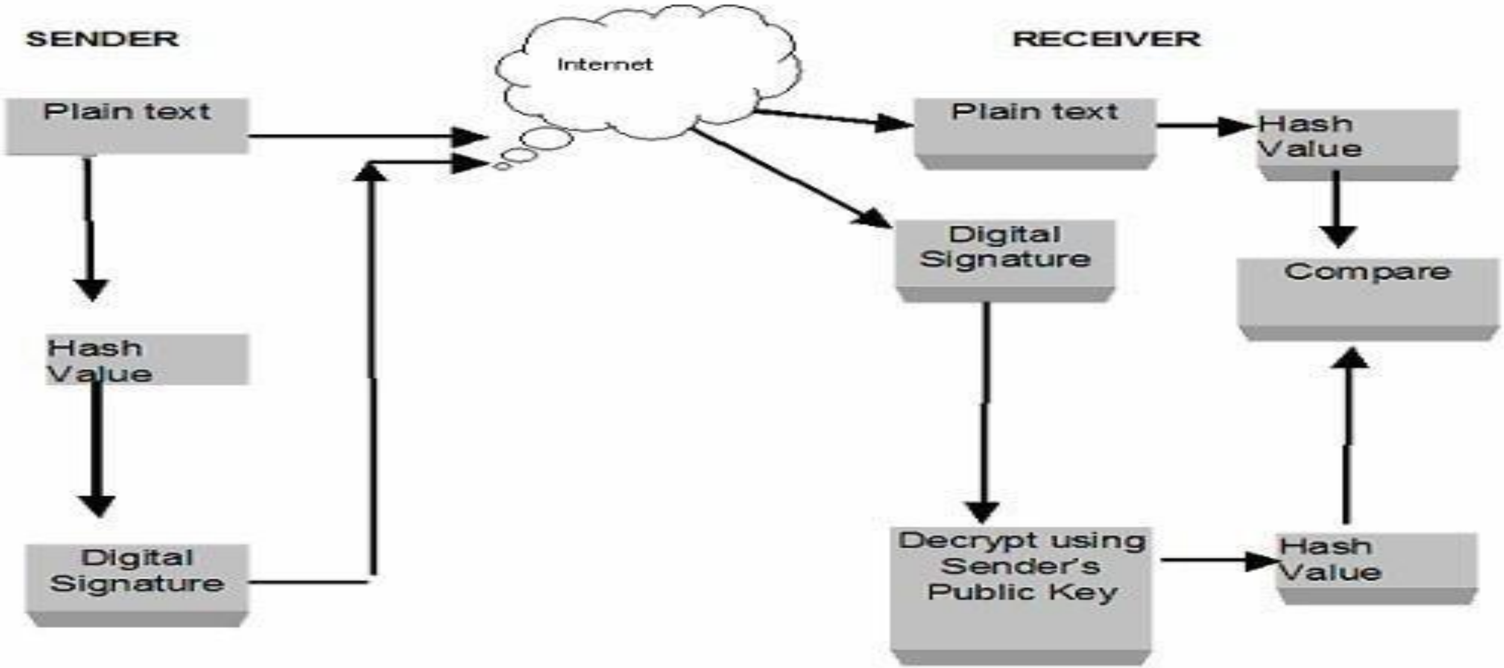
- **Blowfish:** Bruce Schneier designed blowfish and it is a block cipher with a 64-bit block size.
- **CAST-128:** CAST-128, Substitution-Permutation Network (SPN) cryptosystem, is similar to DES, which have good resistance to differential, linear and related-key crypt analysis.
- **IDEA :** IDEA (International Data Encryption Algorithm), was developed at
- ETH Zurich in Switzerland by Xuejia Lai uses a 128-bit key.
- **Rabbit:** Rabbit is a stream cipher, based on iterating a set of coupled nonlinear function
- **RC4:** RC4 is a stream cipher by Ron Rivest at RSA Data Security, Inc. It accepts keys of arbitrary length.

Public Key Cryptography

- Modern cryptography deals with communication between many different parties, without using a secret key for every pair of parties.
- In 1976, Whitfield Diffie and Martin Hellman published an invited paper in the ÏEEE Transactions on Information Theory titled “New Directions in Cryptography”.



Digital Signature Verification Process

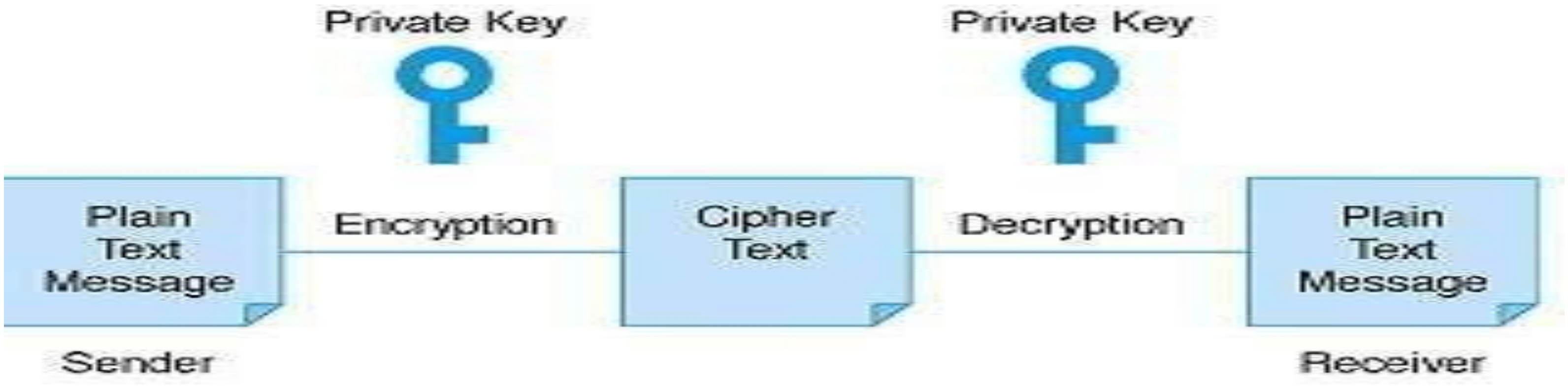


Public Key Infrastructure (PKI)

- The Act provides the Controller of Certifying Authorities to license and
- regulate the working of CAs and also to ensure that CAs does not violate any of the provisions of the Act.
- CCA has created the framework for Public Key Infrastructure in India.
- It has prescribed technical standards for cryptography and physical security based on international standards/best practices of ITU, IETF, IEEE etc.

Public Key Cryptosystem

- Asymmetric key algorithms or Public key algorithm use a different key for encryption and decryption, and the decryption key cannot (practically) be derived from the encryption key.
- Public key methods are important because they can be used to distribute encryption keys or other data securely even when the parties have no opportunity to agree on a secret key in private.



RSA Public Key(Rivest-Shamir-Adelman Algorithm)

- RSA Public Key Algorithm is the most commonly used public key algorithm and this algorithm can be used both for encryption and for signing.
- It is generally considered to be secure when sufficiently long keys are used (512 bits is insecure, 768 bits is moderately secure, and 1024 bits is good).
- The security of RSA relies on the difficulty of factoring large integers.

Diffie-Hellman

- **Diffie-Hellman** is a commonly used public-key algorithm for key exchange.
- It is generally considered to be secure when sufficiently long keys and proper generators are used.
- The security of Diffie-Hellman relies on the difficulty of the discrete logarithm problem (which is believed to be computationally equivalent to factoring large integers).

Elliptic Curve Public Key Cryptosystems

- Elliptic Curve Public Key Cryptosystems is an emerging field.
- They have been slow to execute, but have become feasible with modern computers.
- They are considered to be fairly secure, but haven’t yet undergone the same scrutiny as done on RSA algorithm.

Mathematical Background

- **Exclusive OR (XOR):** Boole defined a bunch of primitive logical operations where there are one or two inputs and a single output Depending upon the operation; the input and output are either TRUE or FALSE. The most elemental Boolean operations are:
- **NOT: T**he output is TRUE if the input is false, FALSE if the input is true. **AND:** “The sky is blue AND the world is flat” is FALSE while “the sky is blue AND security is a process” is TRUE.
- **OR:** “The sky is blue OR the world is flat” is TRUE and “the sky is blue OR security is a process” is TRUE).
- **XOR (Exclusive OR):** the sky is blue XOR the world is flat” is
- TRUE while “the sky is blue XOR security is a process” is FALSE

Mathematical Background

- **The Modulo Function:** The modulo function is, simply, the remainder function.
- To calculate X modulo Y (usually written X mod Y), you merely determine the remainder after removing all multiples of Y from X. Clearly, the value X mod Y will be in the range from 0 to Y-1.
- **Some examples should clear up any remaining confusion**
- 15 mod 7 = 1
- 25 mod 5 = 0
- 33 mod 12 = 9
- 203 mod 256 = 203

Important Topics

- Cryptography
- Symmetric Key Cryptography
- Public Key Cryptography
- RSA Public Key Algorithm
- Mathematical Background

Summary

- Information security can be defined as technological and managerial procedures applied to computer systems to ensure the availability, integrity, and confidentiality of the information managed by computer.
- Cryptography is a particularly interesting field because of the amount of work that is, by necessity, done in secret. The irony is that today, secrecy is not the key to the goodness of a cryptographic algorithm.
- Regardless of the mathematical theory behind an algorithm, the best algorithms are those that are well known and well documented because they are also well tested and well studied! In fact, time is the only true test of good cryptography; any cryptographic scheme that stays in use year after year is most likely to be a good one. The strength of cryptography lies in the choice and management of the keys; longer keys will resist attack better than shorter keys.

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Data Communication and Computer Networks

Block-4 Unit-4

Network Security -II

Topics to be Covered

- Introduction
- Cyber Threats, Attacks and Counter Measures
    - Cyber-Threats
    - Cyber Attacks
    - Counter Measures
- Taxonomy of various Cyber Attacks
- Virus, Worm and Trojan, DoS attack, DDOS attack, Phishing attacks, Malware, Ransom
- Vulnerabilities
- Buffer Overflow

Topics to be Covered

- SQL Injection
- Browser Vulnerabilities
- OS vulnerabilities
- Basics Computer Forensics
- Recent Cyber Attacks
- Firewalls and Intrusion Detection Systems
- Important Topics
- Summary

Introduction

- Network security - A term which can be defined in different ways; Securing the hardware present in the network; securing the software/application or most importantly securing the information that is going to be exchanged among devices present in the network.

Cyber Threats, Attacks and Counter Measures

- In 1983, as the internet was introduced to the world, computer crimes also came into the existence. The momentous Morris worm, the first denial of service (DoS) attack came into 1988.
- This worm contains a few dozen lines of code that replicated rapidly and hampered almost 10% of all the computers over the internet.
- Apart from these attacks, flash and browser add on vulnerabilities were introduced in mid-90's by the hackers to control the computers from remote location.
- As the software industry grew rapidly in early 2000s, they were less bothered about the security concerns, and thus the multi- accessed and insecure software were more vulnerable.

Cyber-Threats

- Cyber threat is "the possibility of a malicious attempt to damage or disrupt a computer network or system." also includes an attempt made for unauthorized access of networked systems as well as infiltrating or stealing the data from those systems.
- In other words, threat is defined as a possibility. However, in the cybersecurity community, the threat is more closely identified with the actor or adversary attempting to gain access to a system.

Types of Cyber Threats

- The list of most common threats published by R. A. Grimes in 2012 consists of following unwanted means of attacks in user's system or account.
  - Unpatched Software (such as Java, Adobe Reader,Flash)
  - Phishing
  - Social Engineered Trojans
  - Network traveling worms
  - Advanced Persistent Threats.
  - Various game changing technologies like big data, cloud computing, Internet of things and mobile device usage have contributed to these types of attacks.

Cyber Threats Types of Attacks

- Cyber threats typically consist, but not limited to, one or more of the following types of attacks:
- Advanced Persistent Threats, Malvertising ,
- Rogue Software Phishing ,Trojans , Botnets,
- Ransomware, Distributed Denial of Service (DDoS) ,
- Man in the Middle (MITM) Wiper Attacks, Intellectual Property Theft,
- Data Destruction, Theft of Money, Data Manipulation,
- Spyware/Malware, Man in the Middle (MITM)
- Drive-By Downloads, Malvertising
- Rogue Software.

Sources of Cyber Threats

- There are following major sources of Cyber Threats which can be explored with respect to target being attacked:
- Nation states or national governments
- Terrorists
- Industrial spies
- Organized crime groups
- Hacktivists and hackers
- Business competitors
- Disgruntled insiders

Cyber Attacks

- Cyber Attack can be defined as the threat which has already taken place.
- Although, various definitions can be found in literature, but all of them state the common aim to compromise the data integrity, data confidentiality and availability. e.g. Social engineering, DDoS , Brute force attack, Phishing , Malware.
- There are various attacks given in the literature which results into different set of attacks: For example: Man in the middle attack.

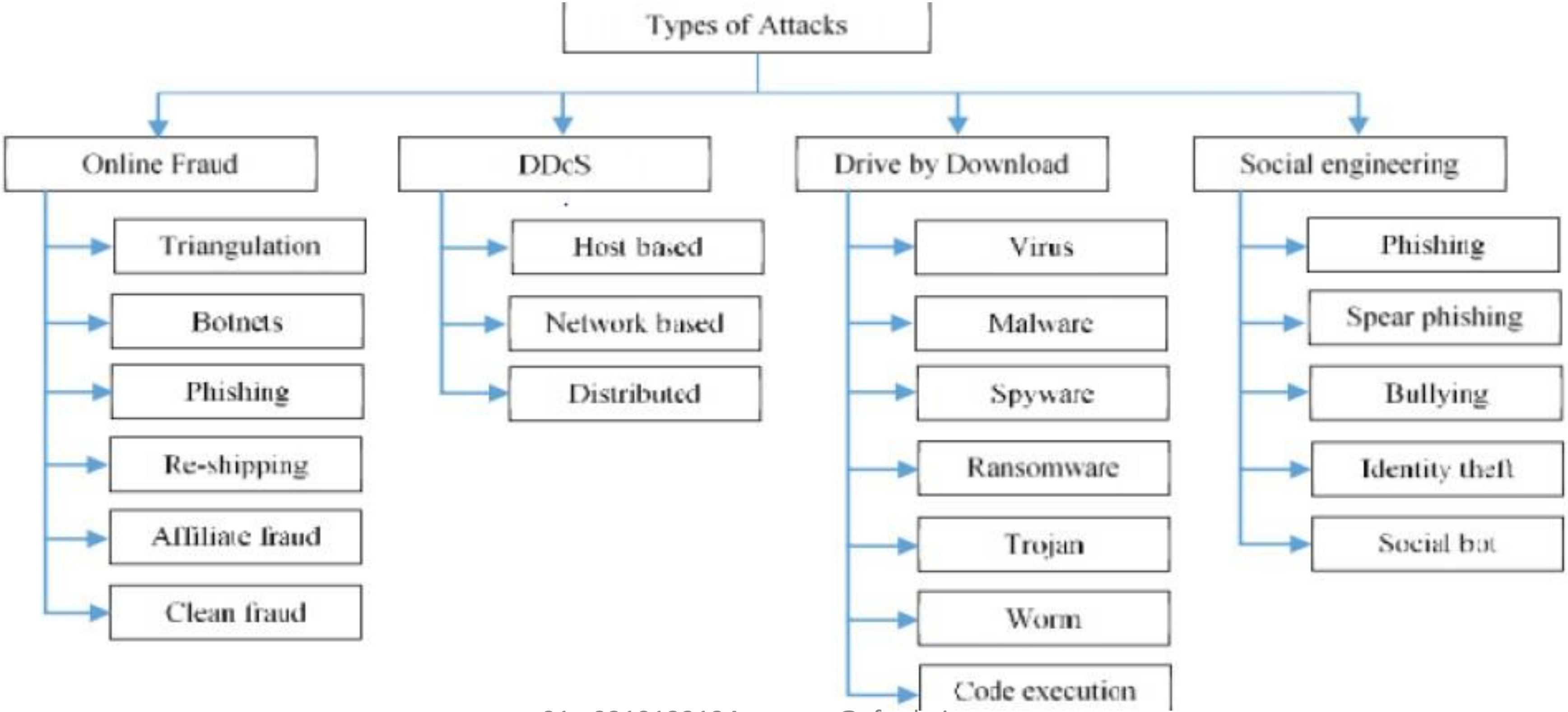
Counter Measures

- Train your Staff members.
- Fully Updated system and software
- Ensuring endpoint protection for remote devices
- Firewall
- Always backup your data
- Access management for your systems
- Enable WiFi Security Options
- Secure Employee personal information
- Passwords

Taxonomy of Various Cyber Attacks

- The term taxonomy can be defined as scientific classification of anything which means arranging things into groups.
- Cyber-attacks came into the existence in 1956, since then a variety of attacks are known in the field of computer security.
- Hence, a proper classification is needed to identify the type of attacks, its behavior and the impact.
- This not only helps in detecting the attacks, it also supports in identifying the counter measures that need to be adopted to mitigate and remediate these cyber vulnerabilities.

Types of Attacks



Cyber Terms

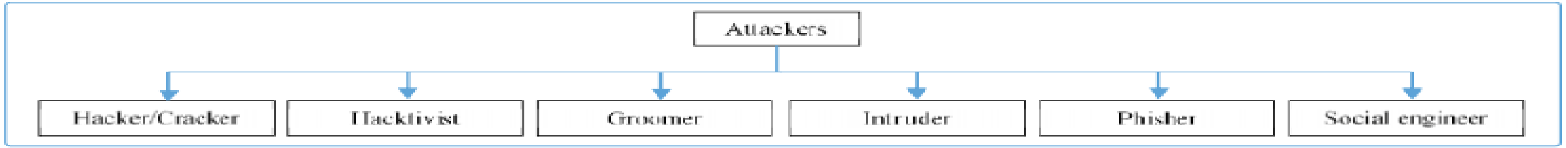


Figure 2: Possible Attackers

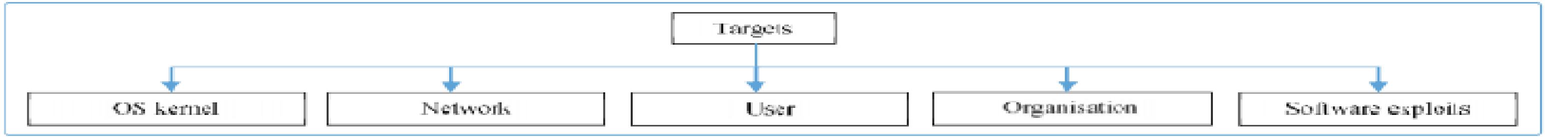


Figure 3: Targets

Figure 4: Motives

Consequences and Defence Mechanism

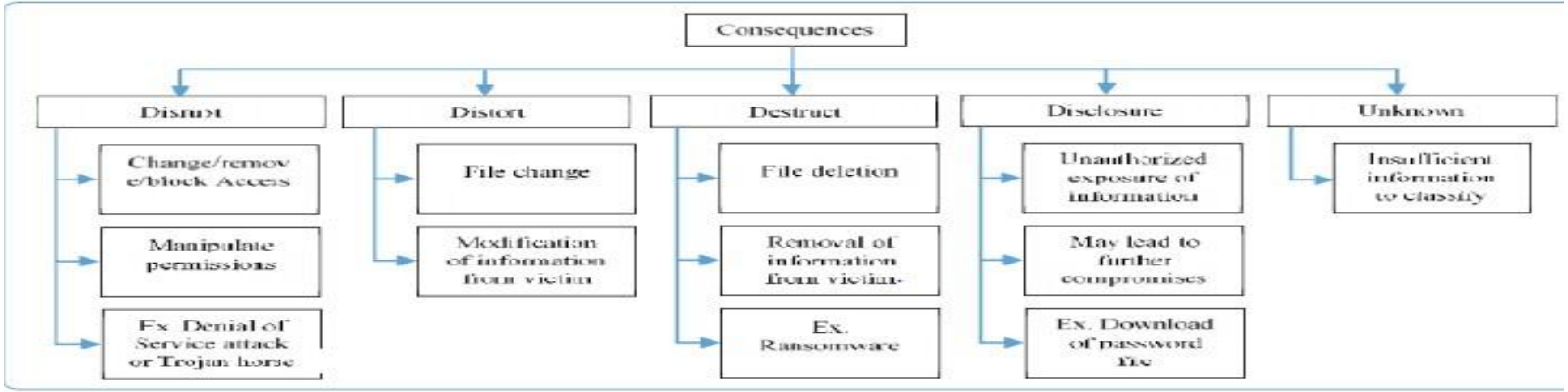


Figure 5: Consequences

Figure 6: Defense Mechanism

Types of Attacks

- **VIRUS:** A self-replicating malicious computer program which affects the other programs in the system/network.
- e.g Boot sector virus, Browser Hijacker, Web Scripting Virus, Resident Virus, Polymorphic Virus, File Infector Virus, Multipartite Virus, Macro Virus.

- **WORM:** It is a type of computer malware. The malwares are self -replicating viruses e.g. Email worms, File sharing worms, **Crypto-Worms,** Internet worms, Instant messaging worms.

Types of Attacks

- **TROJAN:** It can be understood by a simple scenario- Everyone must have noticed, sometimes when you log- in in your system, everything gets popped-up or system behaves inappropriately. e.g. File sharing websites, Email Attachments , Spoofed messages, Unsecured/poorly secured Websites and Hacked WiFi Networks.
- **DOS ATTACK, DDOS ATTACK:** Flooding services e.g. Buffer overflow attacks , ICMP flood, SYN flood , Crashing services.

Types of Attacks

- **PHISHING ATTACKS:** It requires a social engineering techniques to target some specific users
- **MALWARE:** It a malicious software, once installed on victim's device opens up for breaching the cybersecurity and make your device prone to several threats. For gaining unauthorized access or personal information
- **RANSOM :** It is also a malicious software which is used by the cyber criminals to infect the target system. e.g. Locker Ransomware, Crypto Ransomware.

Vulnerabilities

- Cyber criminals are now a days very active to take an advantage of weaknesses present in the systems/networks.
- Though, using these vulnerabilities, objectives of different types of attacks may vary from one to other. Attackers may perform such activities for financial gain, personal gain or due to some political motives. Few major categories of vulnerabilities.
- **eg. Network Vulnerabilities, OS Vulnerabilities, Human Vulnerabilities, Process Vulnerabilities.**

Buffer Overflow

- Buffer overflow is a specific type of process vulnerability which is caused by constrained resources availability in software development phases.
- Specifically, buffer overflow is a situation where temporary space supported by any software consumed to its defined storage capacity.
- In such a case, the excess data take place into already occupied memory locations which leads to corrupting or overwriting the data present earlier at those locations.
    - **Impacts that may occur in your system:**
- **Complete System Crash, Access control, Additional Security concerns**

SQL Injection

- SQL stands for structured query language which is used for querying and modifying the databases.
- In the internet world, any web service hosted over the internet uses database to store the information it's users and other details.
- Cyber criminals mainly try to intrude in database to capture the information from a web service.

Browser Vulnerabilities

- Revolution in world of Internet came with the introduction of web browsers.
- Every employee, be it from small or large organization, use these browsers to access internet to perform their jobs.
- However, it has dramatically increased the productivity, but
- also exposes the organization’s security breach points.

OS Vulnerabilities

- **OS vulnerabilities are classified as follows**
  - **Client side vulnerabilities:** an attacker tries to infiltrate via different applications installed/supported by operating system.
  - E.g Any web-browsers, different office software, email client.
  - **Server side vulnerabilities:** These vulnerabilities generally found on server side computer system or in between the client and server, in which an attacker can attack thorough SQL injection method on any web application.

Basic Computer Forensics

- Computer forensics is an art which describes the present state of a digital artifact such as mobile phones/devices, computer systems, hard drives, pen drives.
- Moreover, it also includes investigating and linking the data present in the computer systems, email applications etc. for describing any event.
- The field of computer forensics is comparatively new than cyber security, however recent advancements done in the past twenty years have overwhelmed this field to contribute in various legal proceedings in the court.

Major Techniques Being Used in Investigations

- **Major techniques being used in investigations by forensics team:**
- Cross-Drive Analysis:
- Live Analysis:
- Analysis by recovery:
- Stochastic forensics :
- Stenography analysis:

Recent Cyber Attacks

- In 21st century, cyber security is essential for every individual and organization due to vast use of Internet. Although, various self-learning/experience based solutions and software solutions such as antiviruses exists, but cyber criminals are in continuous search for getting success in breaching out these solutions.

- Attack on Popular Social media website Facebook
- Attack on graphic design website CANVA.
- Attack on Servers of MGM hotel.
- Ransomware attack on California university.
- Attack on World Health Organization.
- Attack on Zoom video conferencing service.

Firewalls and Intrusion Detection Systems

- Preventive measures that can be considered to protect any individual or an organization from cyber-attacks.
- A firewall and Intrusion detection system both works for protecting you from various cyber-attacks at different levels.
- However, there are some similarities and differences in the working of both firewall and an IDS.

Important Topics

- Cyber Threats, Attacks and Counter Measures
- Virus, Worm and Trojan, DoS attack, DDOS attack, Phishing attacks, Malware, Ransom
- Buffer Overflow
- SQL Injection
- Browser Vulnerabilities
- OS vulnerabilities
- Firewalls and Intrusion Detection Systems

Summary

- In this session, best efforts are made to present available knowledge in the domain of network security.
- Currently, extreme use of computers in distributed manner require networking strategies to make them connected with each other.
- This implementation further poses a lot of security concerns for the systems involved in such networks.
- So, this session, gives a detailed overview of security concerns like threats which can result into many cyber-attacks. After that discussion move towards defining the major reasons due to which one may be exposed to these attacks are known as vulnerabilities.

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